CSTE4103 — Cloud Computing • Lab 3 Tutorial

1

Part 1 of 6

Download and Verify the Ubuntu Server ISO

Check your PC, download the installer image, and prove it was not corrupted

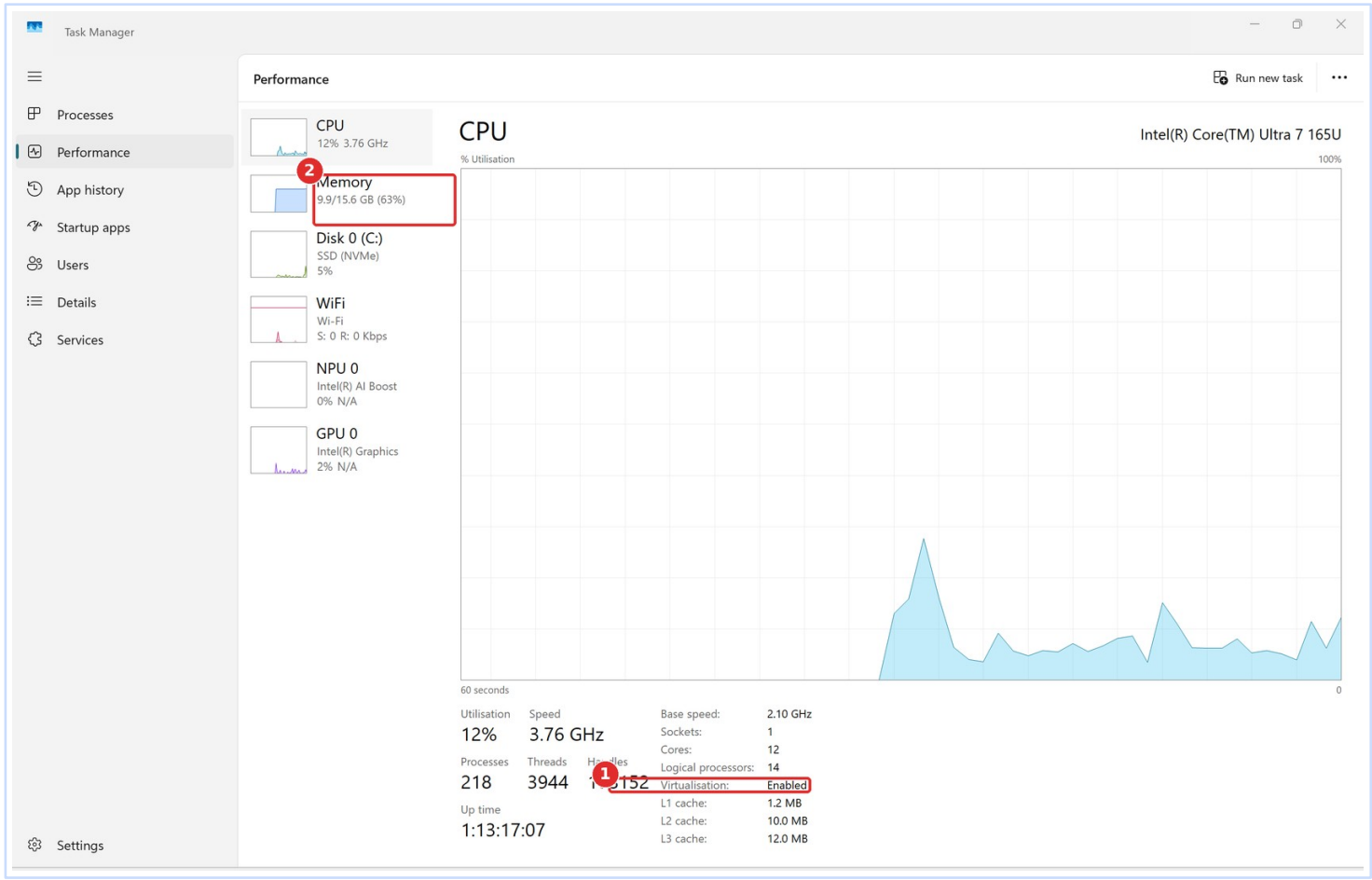
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PART 1 OF 6

Before you start

- Windows 10/11 PC with virtualization enabled in BIOS/UEFI
- At least 8 GB RAM and about 30 GB free disk space
- A stable internet connection: the ISO is about 3 GB
- About 25 minutes, most of it waiting for the download
- Every screenshot in this series is from a real PC. Personal details were masked on purpose.

Check that virtualization is enabled

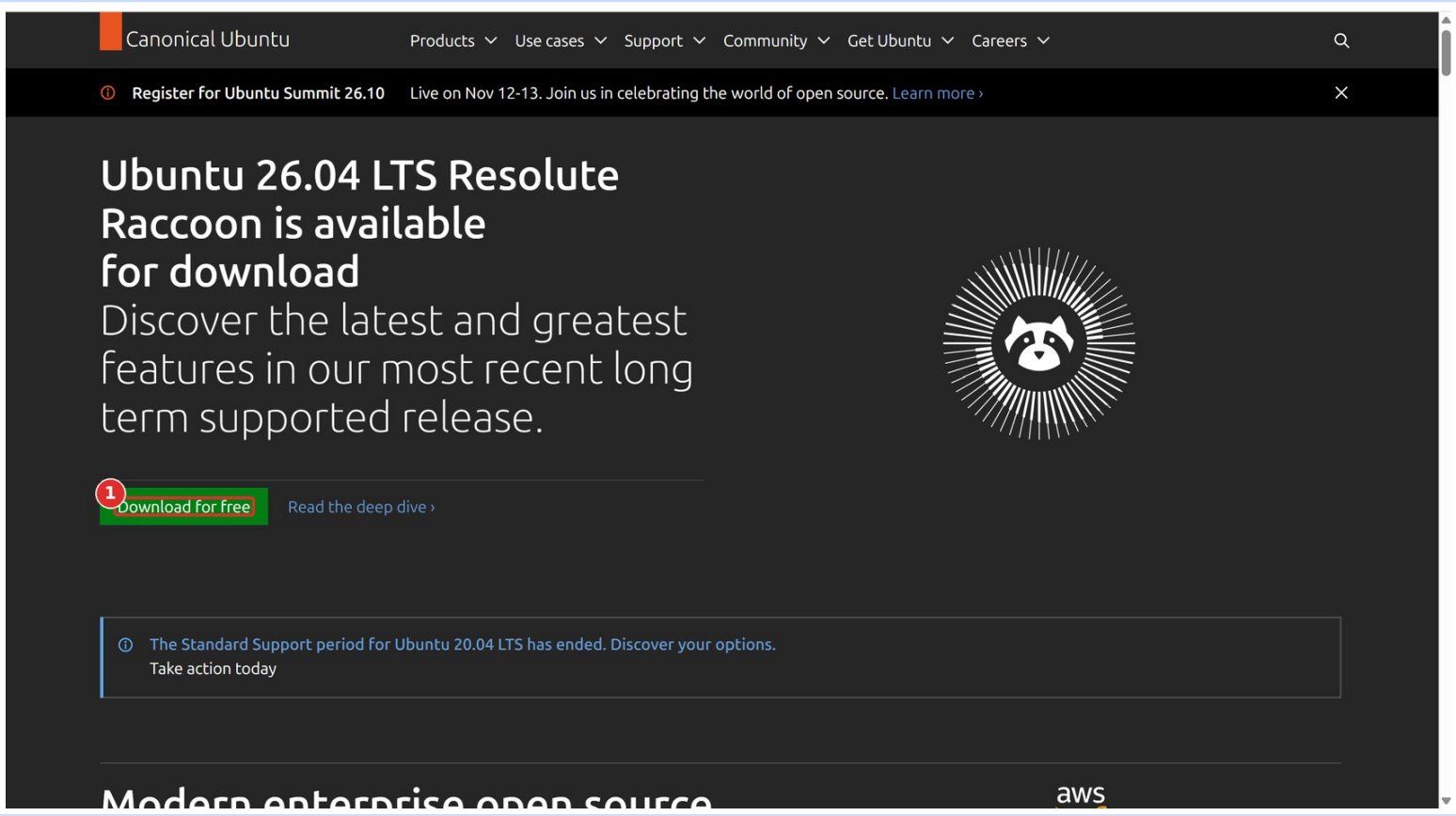


1. Open Task Manager (Ctrl+Shift+Esc), Performance tab, CPU. Look for “Virtualization: Enabled”.

2. Check the installed memory in the Memory tab. 8 GB or more is comfortable for a host plus one VM.

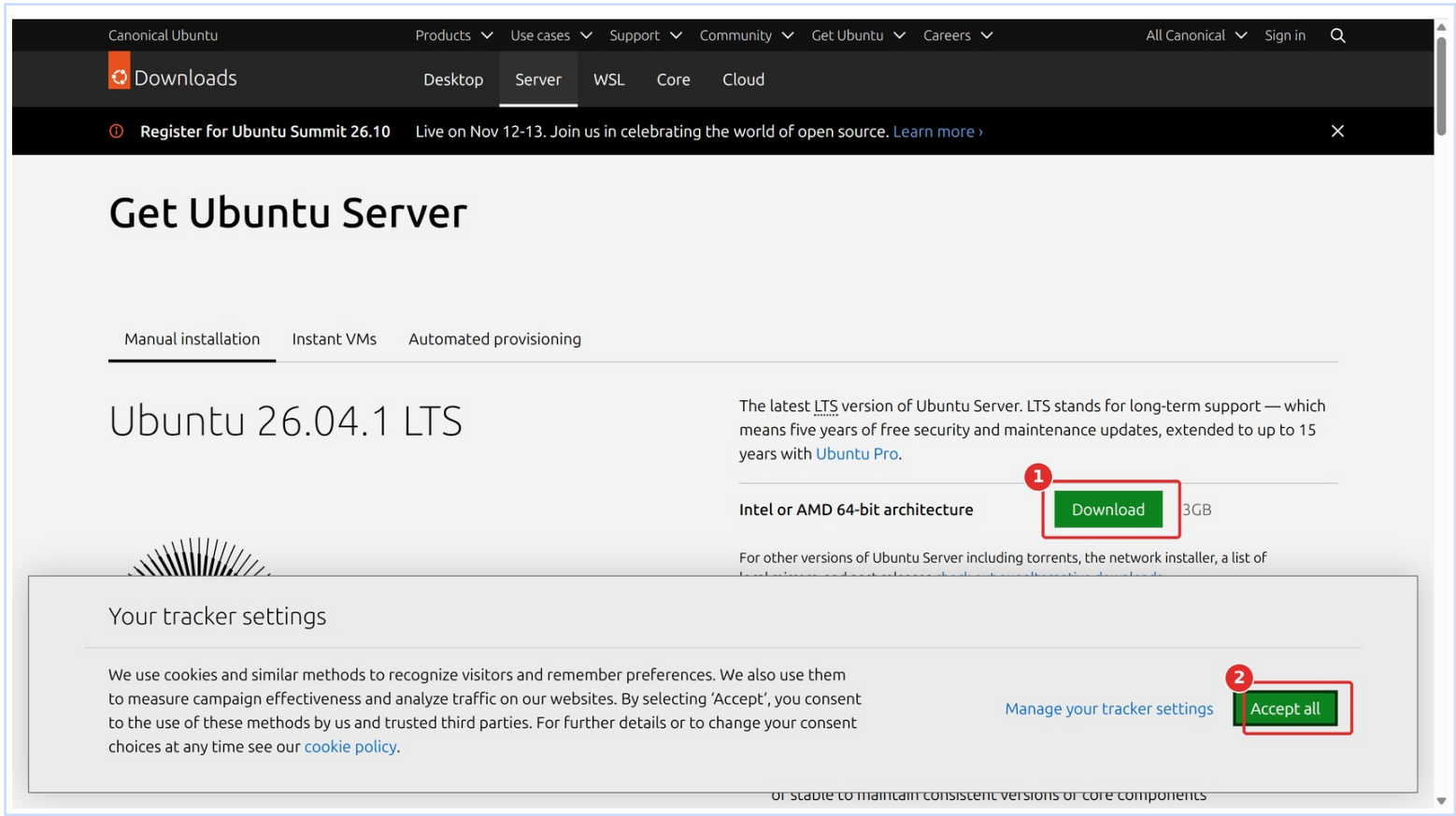
💡 If it says Disabled, enable Intel VT-x or AMD-V in BIOS/UEFI first.

Open the Ubuntu Server download page



1 Go to ubuntu.com/download/server and press “Download for free”. Ubuntu Server LTS is the long-term support release.

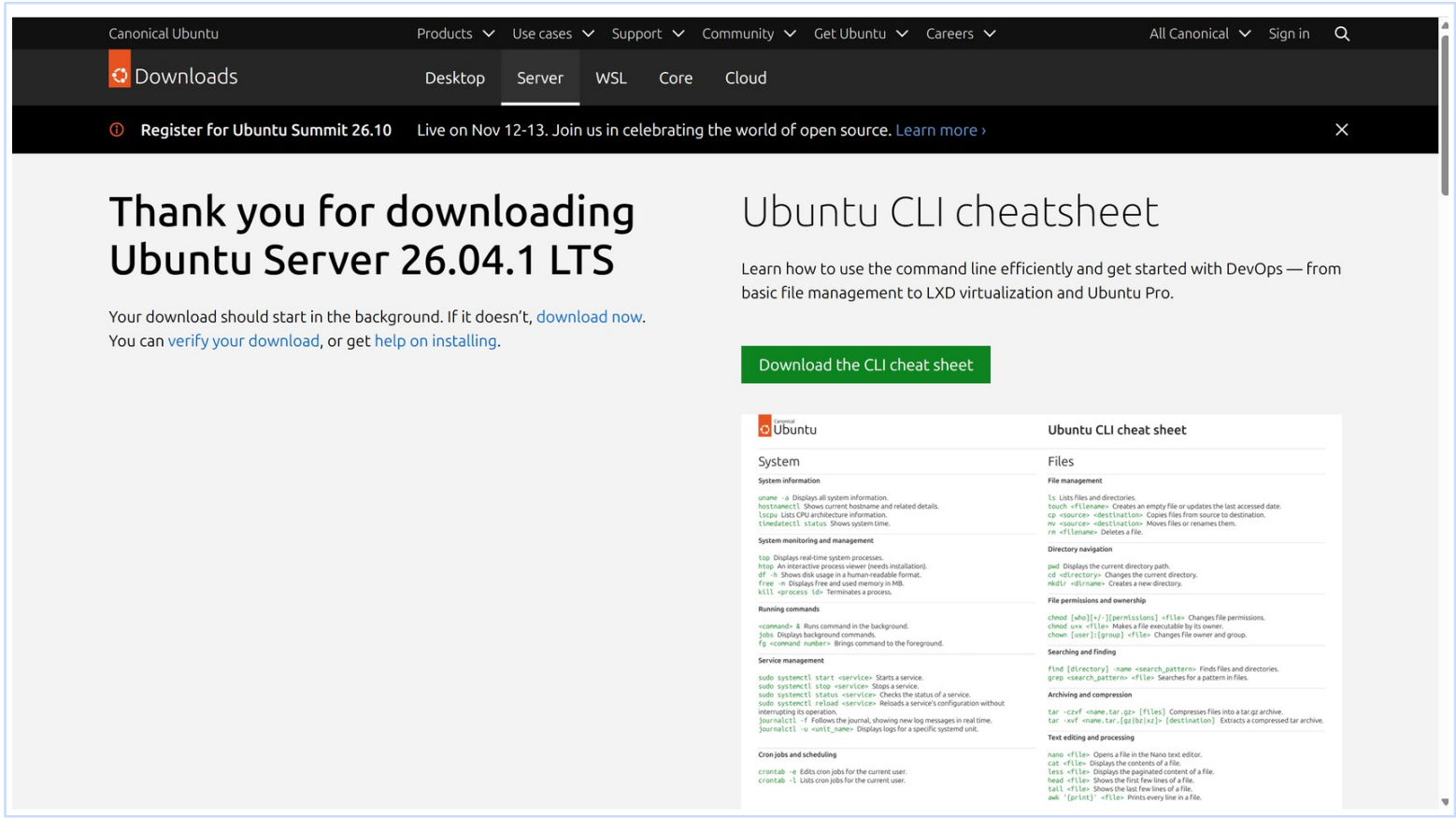
Choose the 64-bit installer



1. Press the green Download button for "Intel or AMD 64-bit architecture" (about 3 GB).

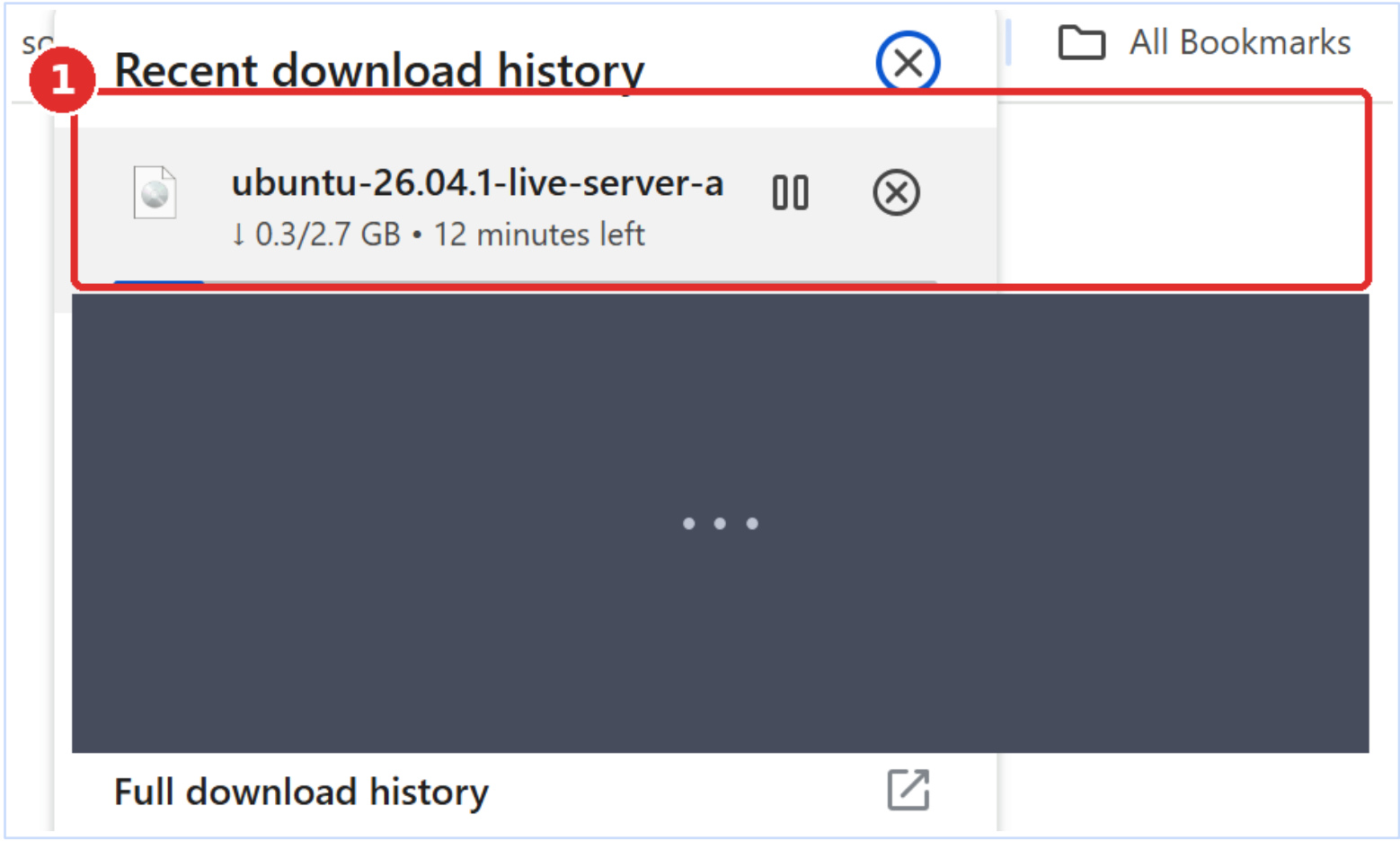
2. If the cookie banner appears, accept or manage it. It does not affect the download.

The download starts



1 The "Thank you for downloading" page confirms that the browser started the download. Do not close the browser yet.

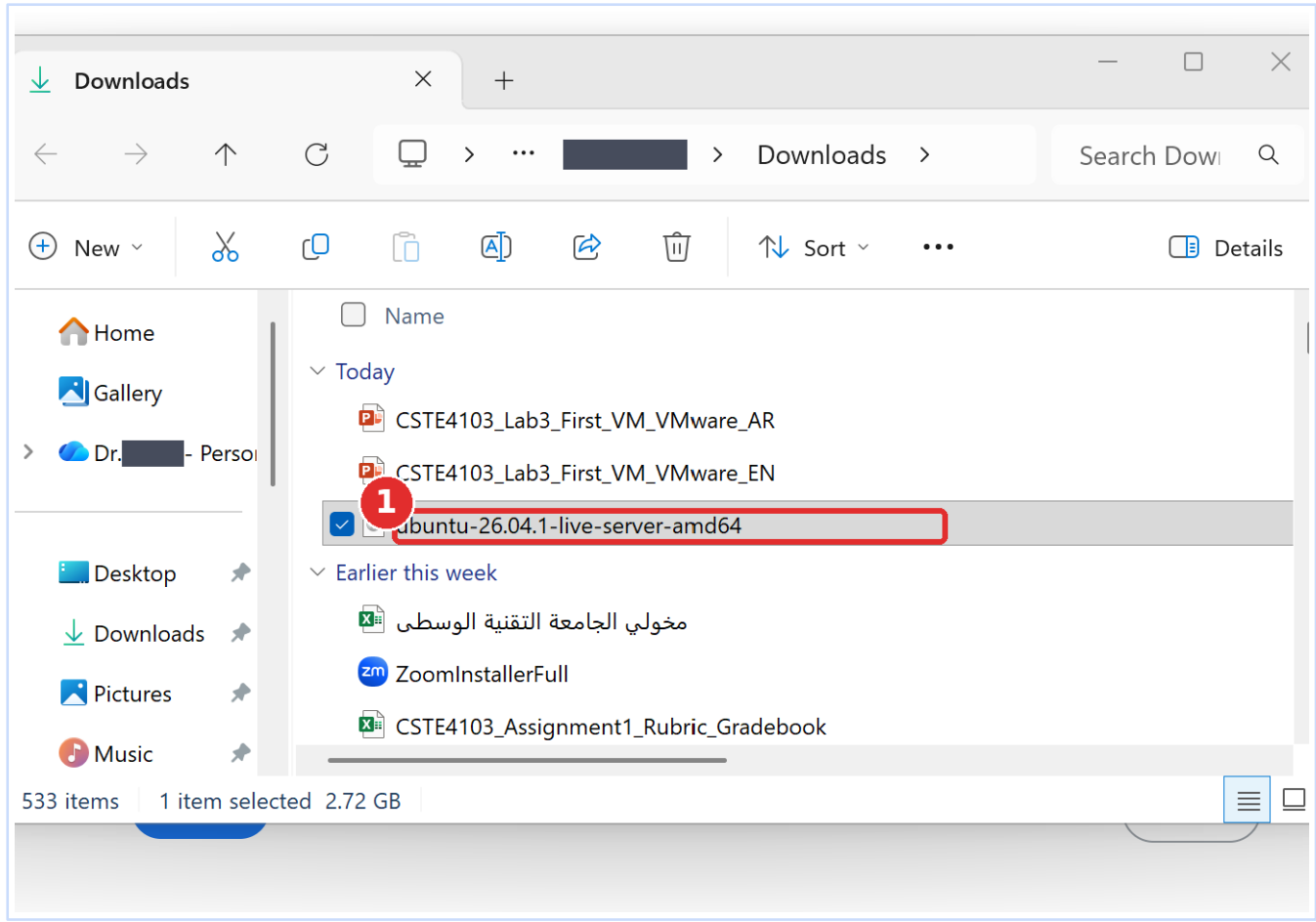
Watch the progress



1 Open the browser download list. The ISO is about 2.7 GB, so it may take a while depending on your connection.

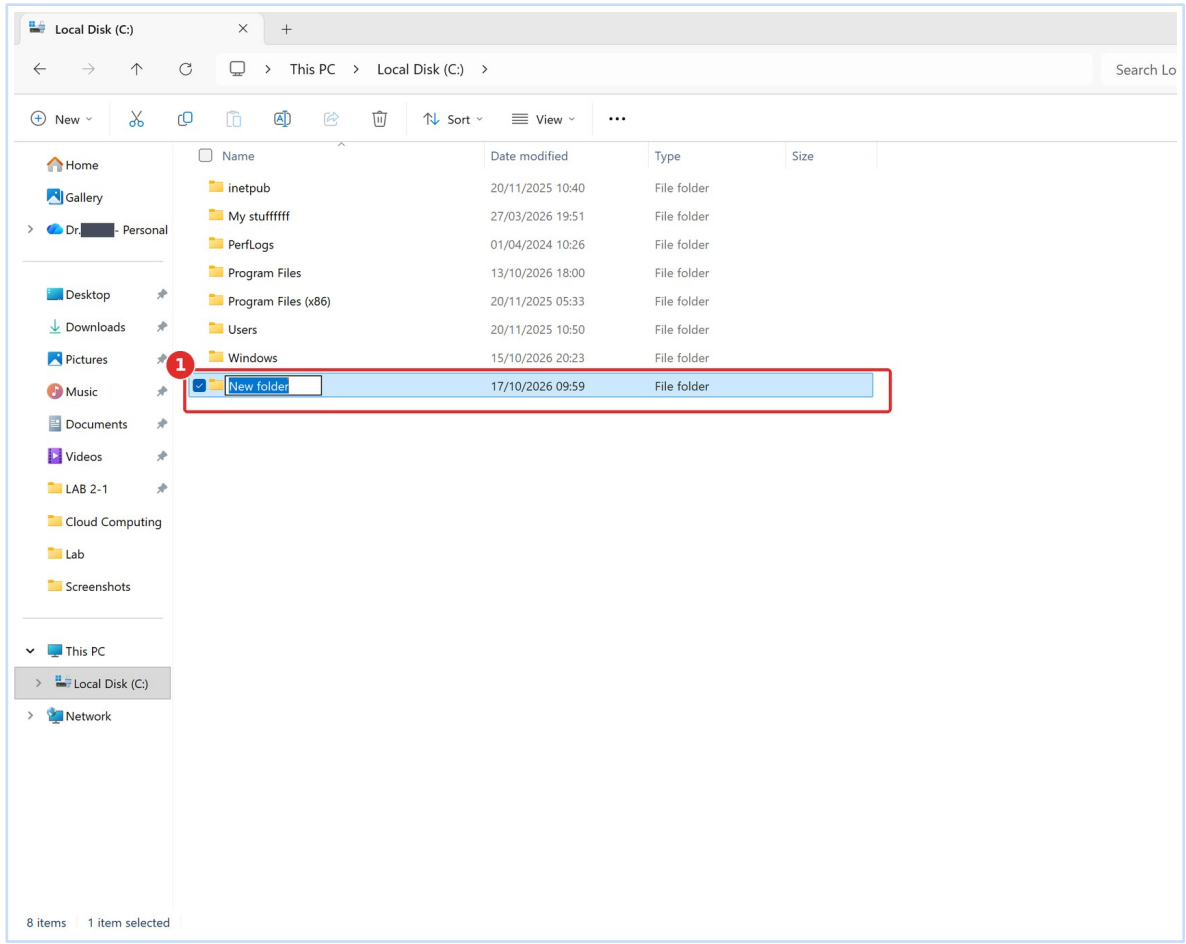
💡 Other entries in the list were hidden to protect privacy.

Find the file in Downloads



1 Open File Explorer, then Downloads. The file is named ubuntu-26.04.1-live-server-amd64. Do not rename it.

Create a folder for ISO files



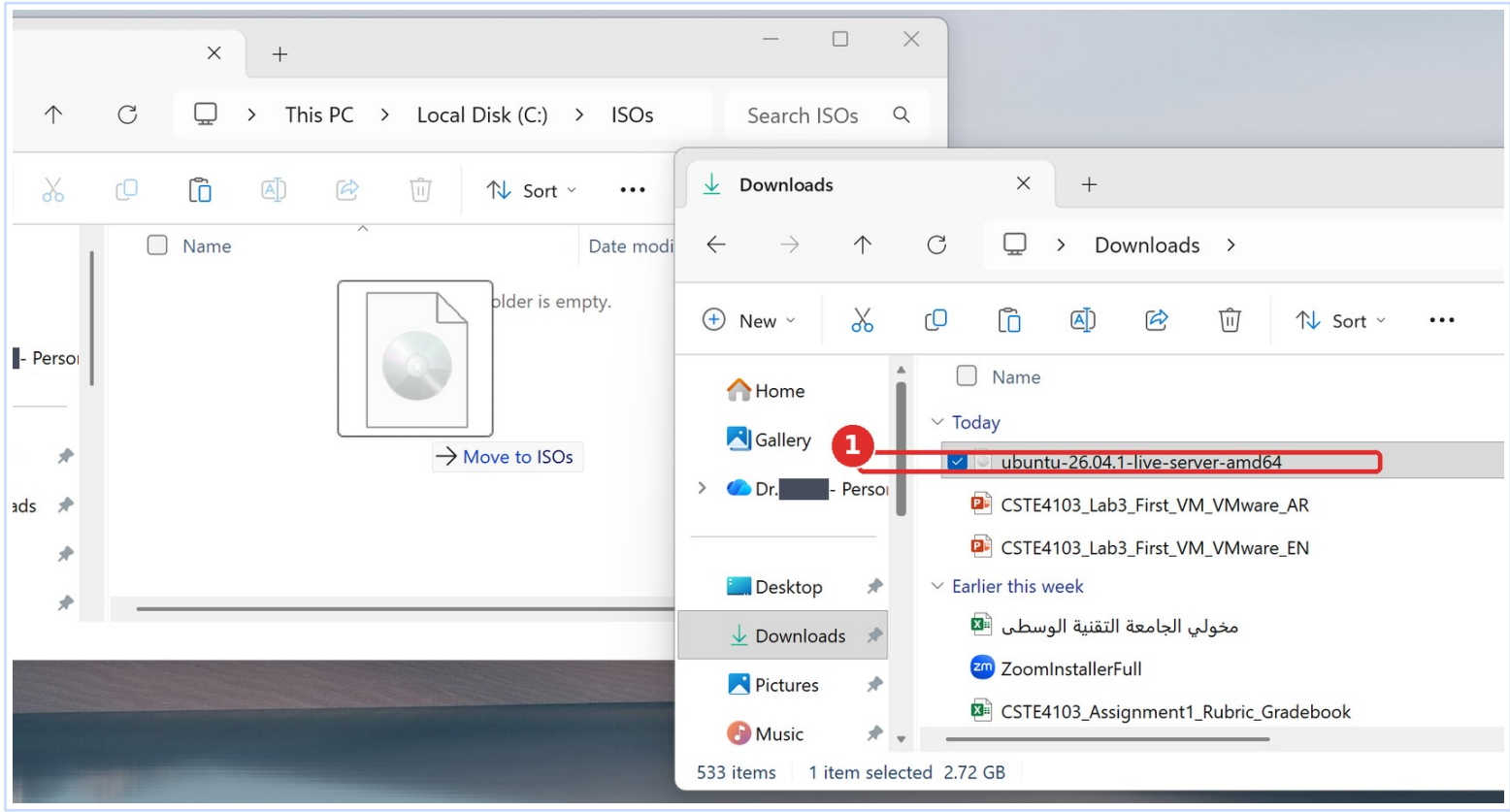
1. Open This PC, Local Disk (C:), then New folder. A short path without spaces avoids trouble later.

Name it ISOs

1 Type ISOs and press Enter. You will point VMware to this folder in Part 4.

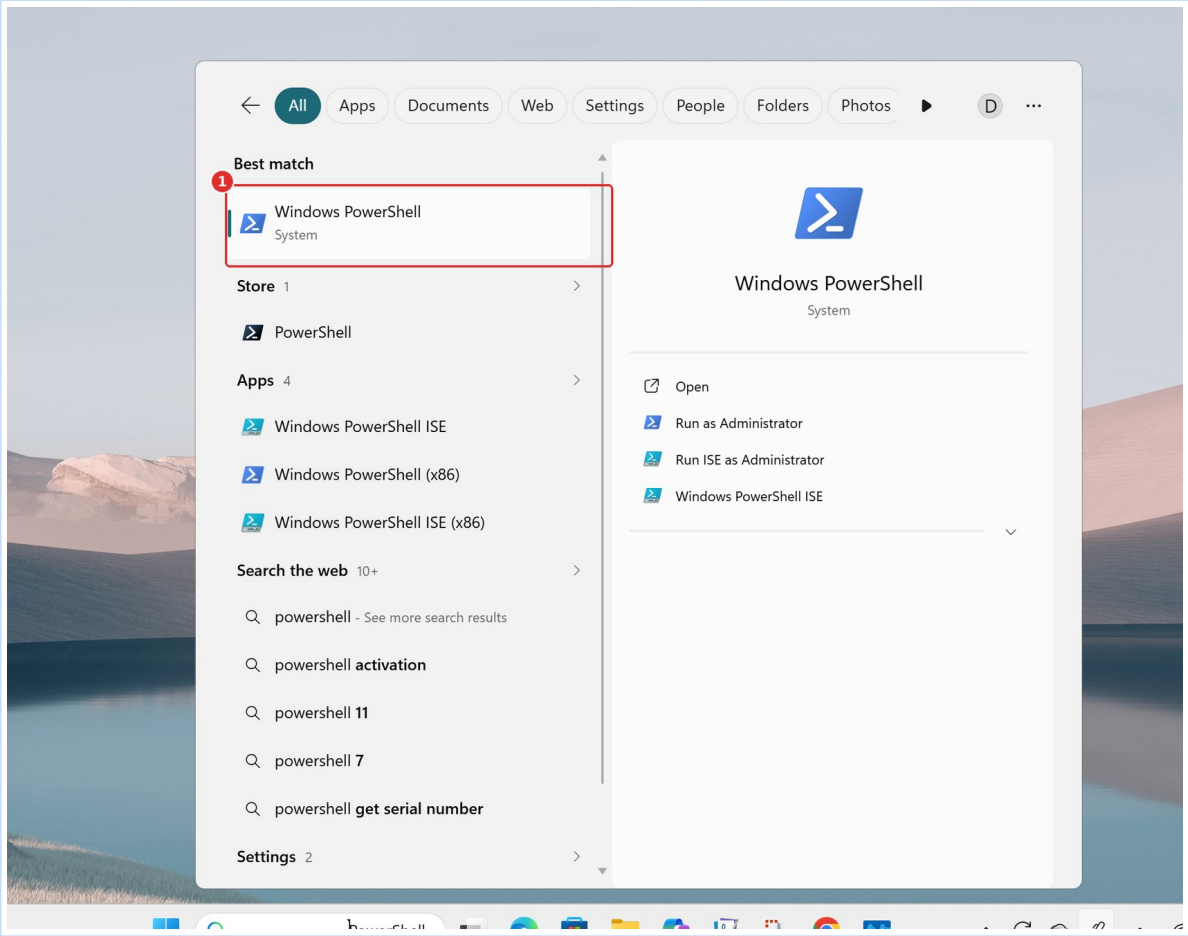


Move the ISO into C:\ISOs



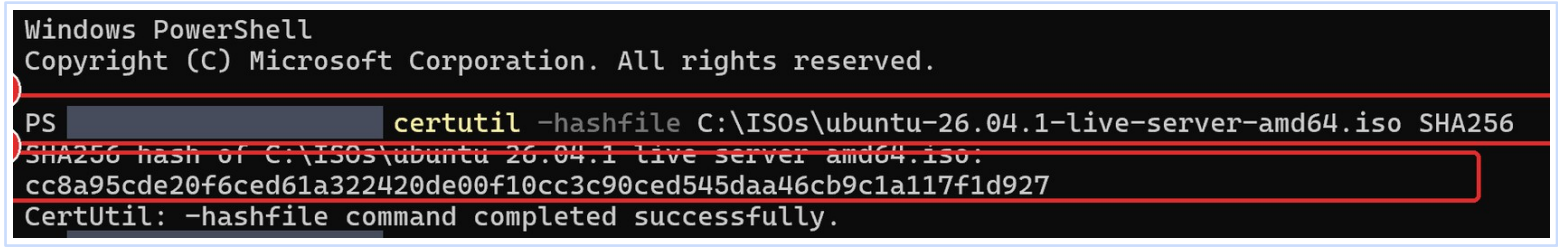
1 Drag the ISO from Downloads into the ISOs folder (“Move to ISOs”). Moving it, rather than copying, saves 3 GB of disk space.

Open Windows PowerShell



1. Press the Windows key, type powershell, and open “Windows PowerShell”. Administrator rights are not needed for this step.

Calculate the SHA-256 checksum



1. Run: certutil -hashfile C:\ISOs\ubuntu-26.04.1-live-server-amd64.iso SHA256

2. The 64-character result is the file's fingerprint. Keep it for the next step.

💡 Tip: type part of the path and press Tab to auto-complete it.

Compare with the official value

1. Open releases.ubuntu.com/26.04.1/SHA256SUMS in the browser. Find the line ending with live-server-amd64.iso.

```
487f87faaf547ea30e0aba4d5b53346292571256b25333a978db1692bcee9dd2 *ubuntu-26.04-desktop-amd64.iso
dec49008a71f6098d0bcfc822021f4d042d5f2db279e4d75bdd981304f1ca5d9 *ubuntu-26.04-live-server-amd64.iso
96c7f5fb28a7fe28245331f9bfbe4375f18dd29a4850116ad3c4f60f6700c55c *ubuntu-26.04-wsl-amd64.wsl
601e30fbf5d97759367c632e2c33630665039b7e2158fd068403da3ccf1bda1f *ubuntu-26.04.1-desktop-amd64.iso
cc8a95cde20f6ced61a322420de00f10cc3c90ced545daa46cb9c1a117f1d927 *ubuntu-26.04.1-live-server-amd64.iso
48d56724b5c8e60f24893e83e73bbb58c60b3ca22fba3da97707542Oacd54104 *ubuntu-26.04.1-wsl-amd64.wsl
```

💡 The two values must match exactly. If not, delete the file and download again.

PART 1 OF 6

Why verify a download?

- A checksum proves the file is complete and unmodified. A damaged ISO can fail halfway through installation.
- It is also a security habit: attackers sometimes distribute tampered installers.
- Always compare against the value published on the official site, not one sent by a third party.
- If releases.ubuntu.com times out on your network, retry later or from another connection.

Checkpoint

- ✓ Virtualization shows Enabled
- ✓ ISO is in C:\ISOs
- ✓ Both SHA-256 values match

Next: Part 2 — Broadcom account and VMware Workstation download

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Part 2 of 6

Broadcom Account and VMware Workstation Download

Create the free account, find Workstation Pro in the free downloads, and verify the installer

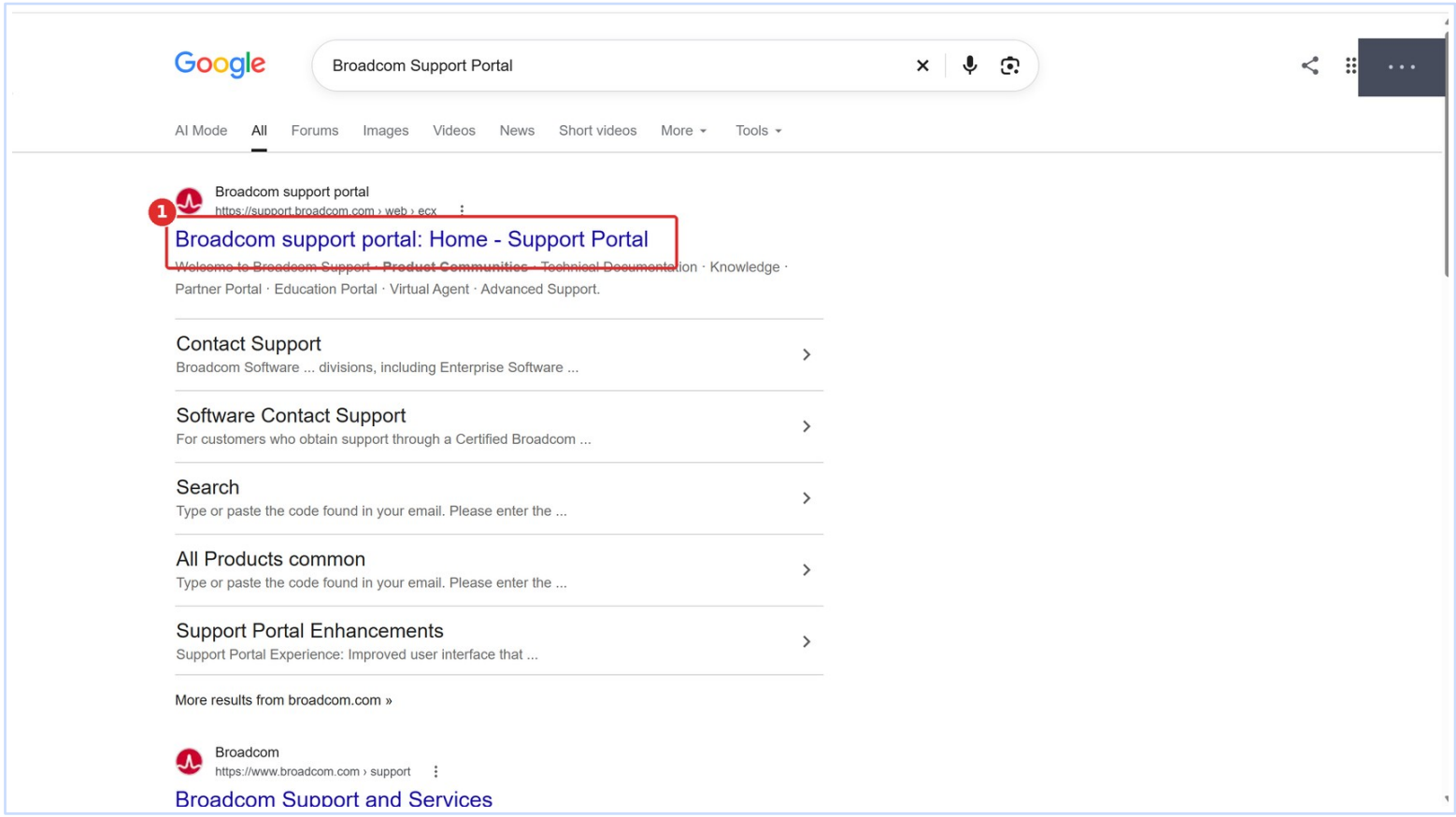
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PART 2 OF 6

What this part does

- VMware Workstation Pro is now free for personal use, but the download sits behind a free Broadcom account.
- You need an email address you can open right now: a verification code will be sent to it.
- Use your real name, country and address. The trade-compliance form can block wrong or inconsistent data.
- In every screenshot, names, emails, phone numbers and addresses are hidden with dark boxes.
- Allow about 30 minutes. Broadcom pages change often, so wording may differ slightly.

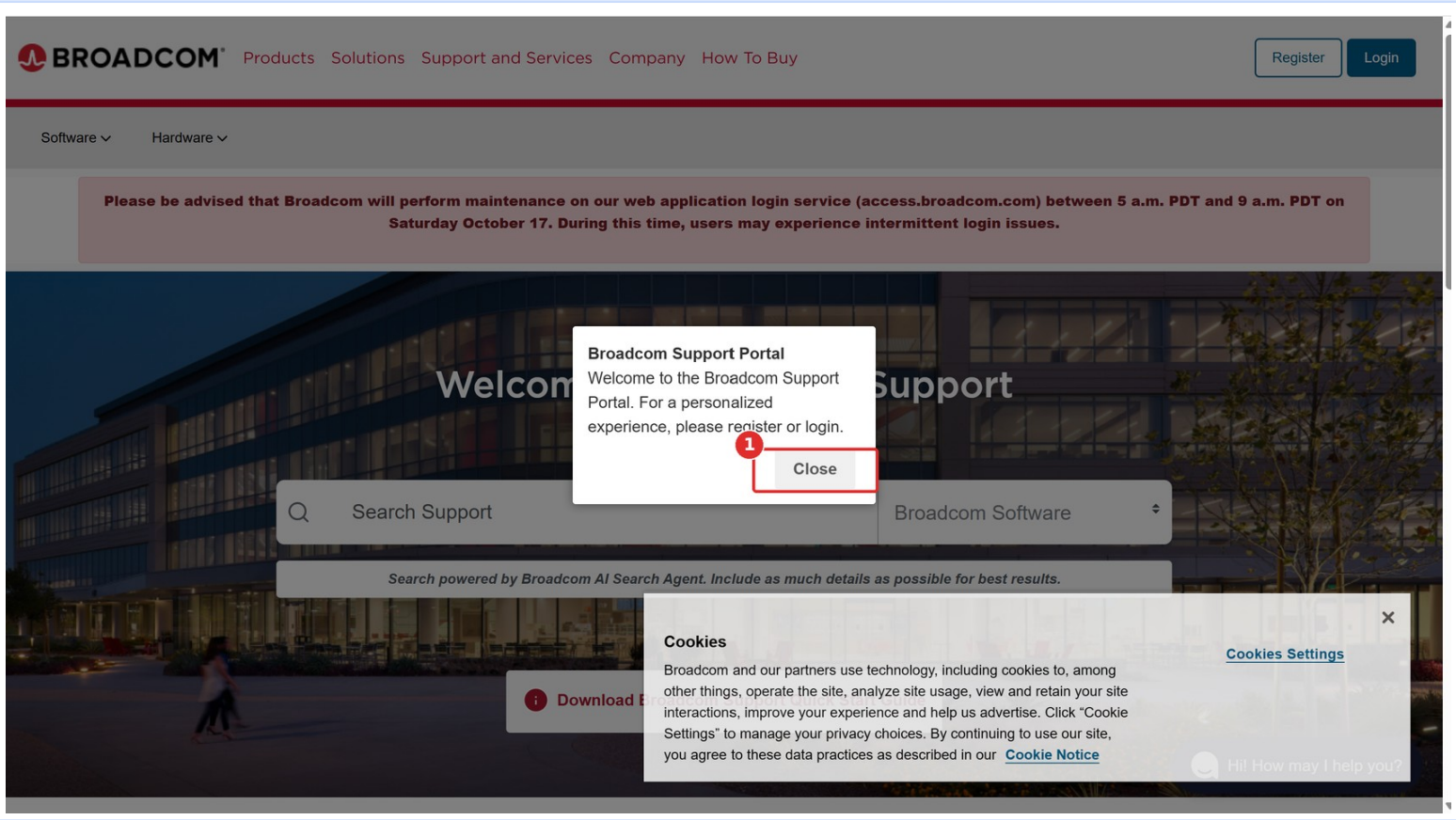
Find the Broadcom Support Portal



1. Search Google for “Broadcom Support Portal” and open the official result: support.broadcom.com.

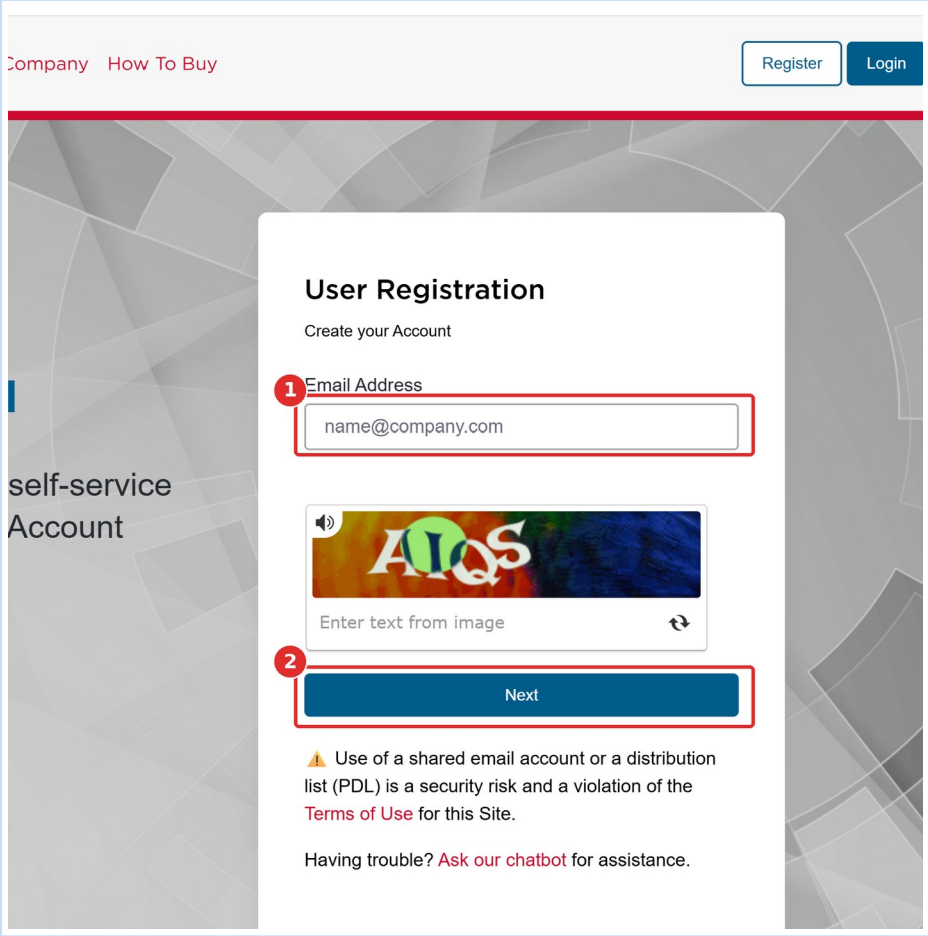
💡 Check the address bar says support.broadcom.com before typing any data.

Close the welcome pop-up



1 Press Close on the "Welcome to the Broadcom Support Portal" message. The cookie notice can also be accepted or dismissed.

Start the registration

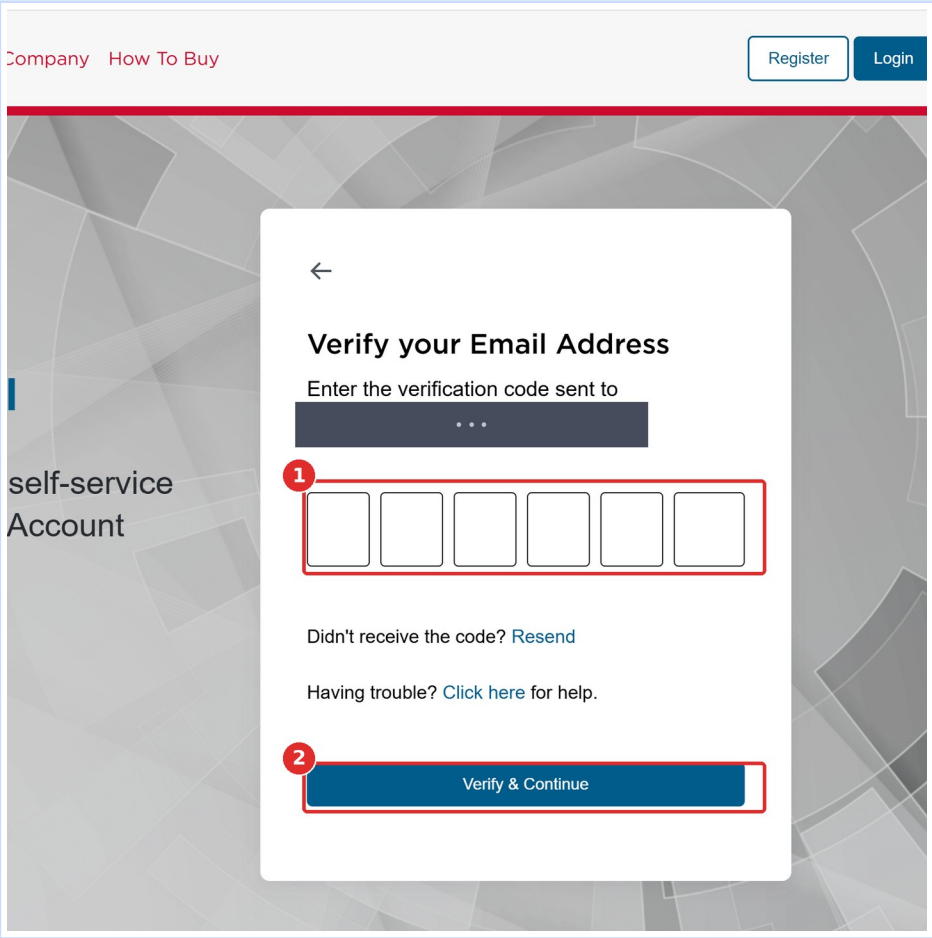


1. Press Register, then type the email address you will use for the account.

2. Type the characters shown in the image (the captcha), then press Next.

💡 Use a personal or university mailbox, not a shared or group address.

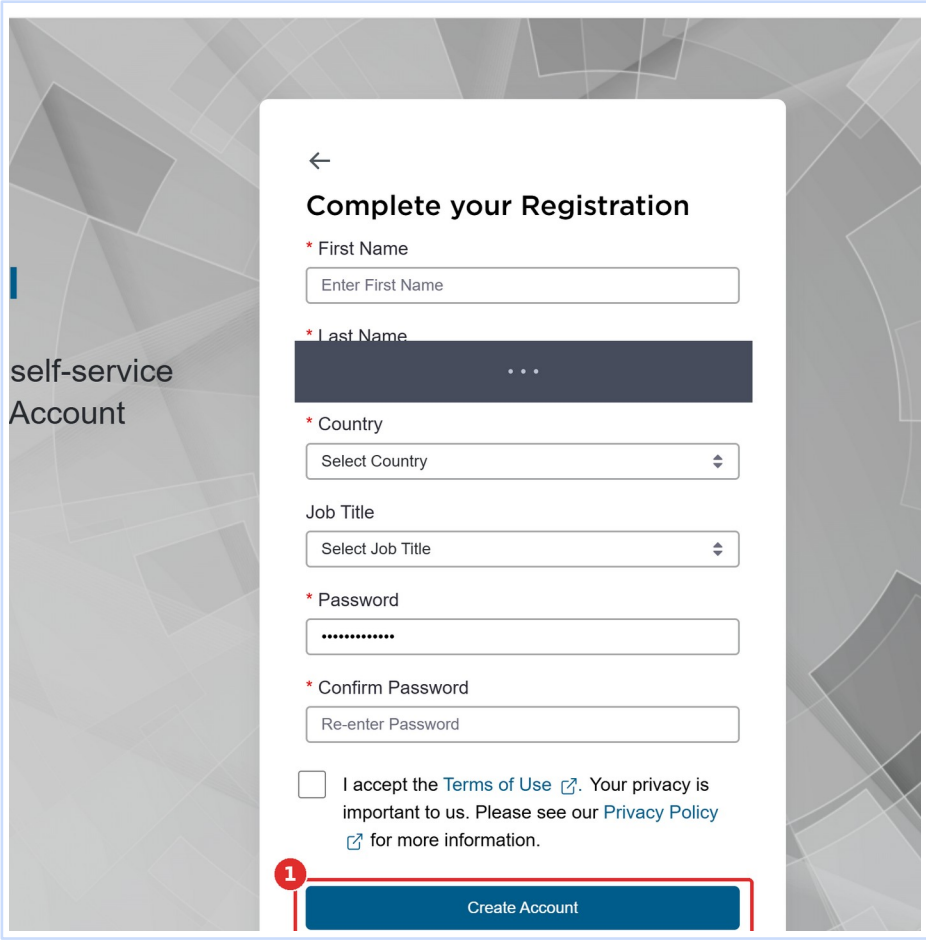
Verify your email address



1. Broadcom sends a 6-digit code to your mailbox. Type it in the six boxes.

2. Press Verify & Continue. If the code does not arrive, press Resend and check Spam.

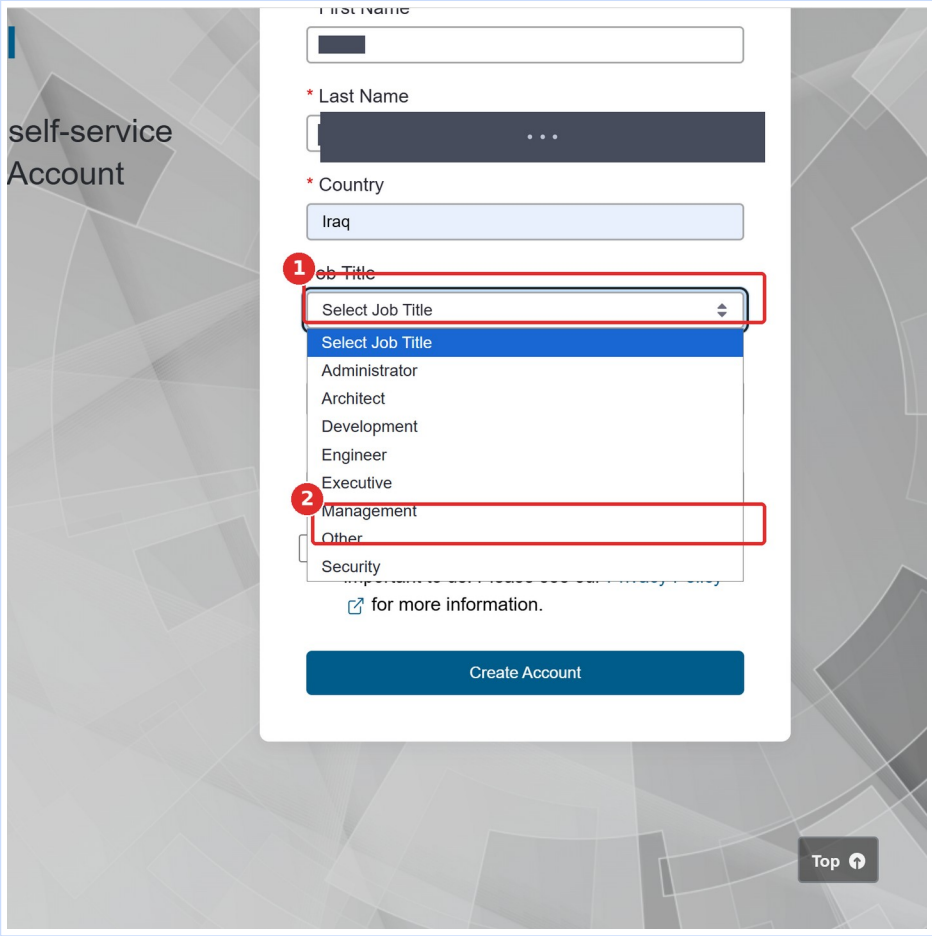
Complete your registration



1. Fill First Name, Last Name, Country, Job Title, Password and Confirm Password. The name fields are hidden in this screenshot. Then press Create Account.

💡 Choose a long, unique password and keep it in a password manager.

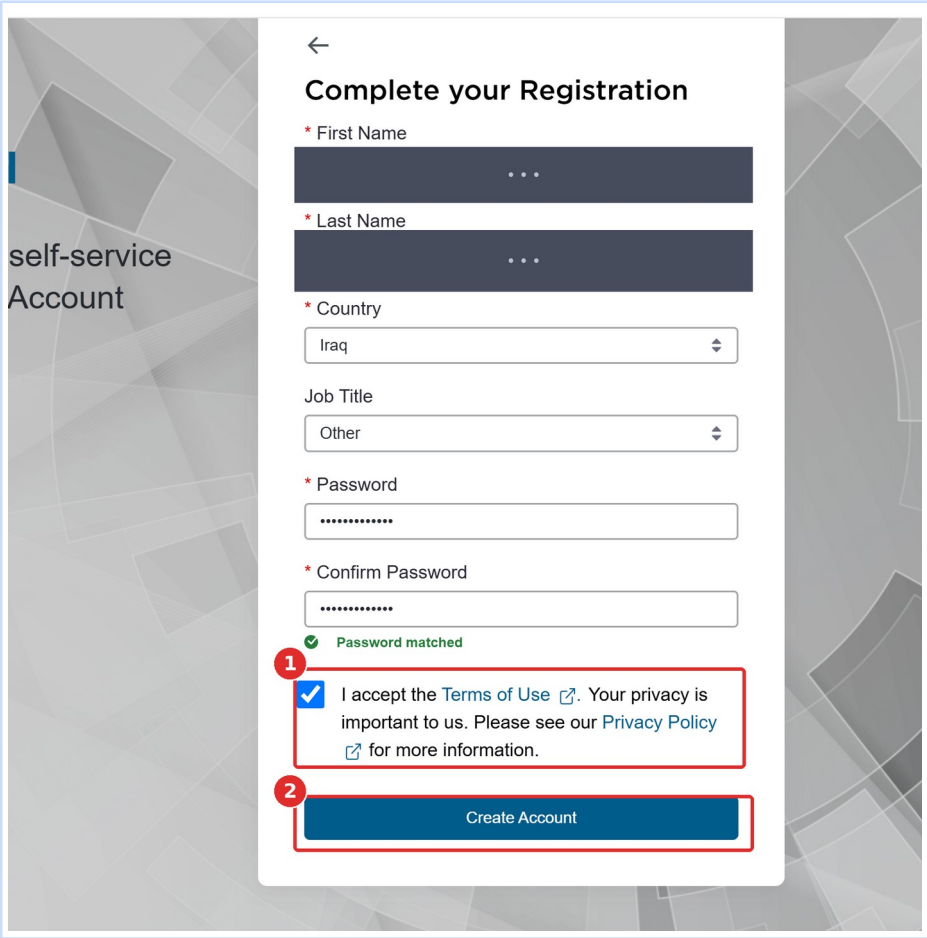
Pick a job title



1. Open the Job Title list and choose the closest entry. Students can choose “Other”.

2. “Other” is a valid choice here.

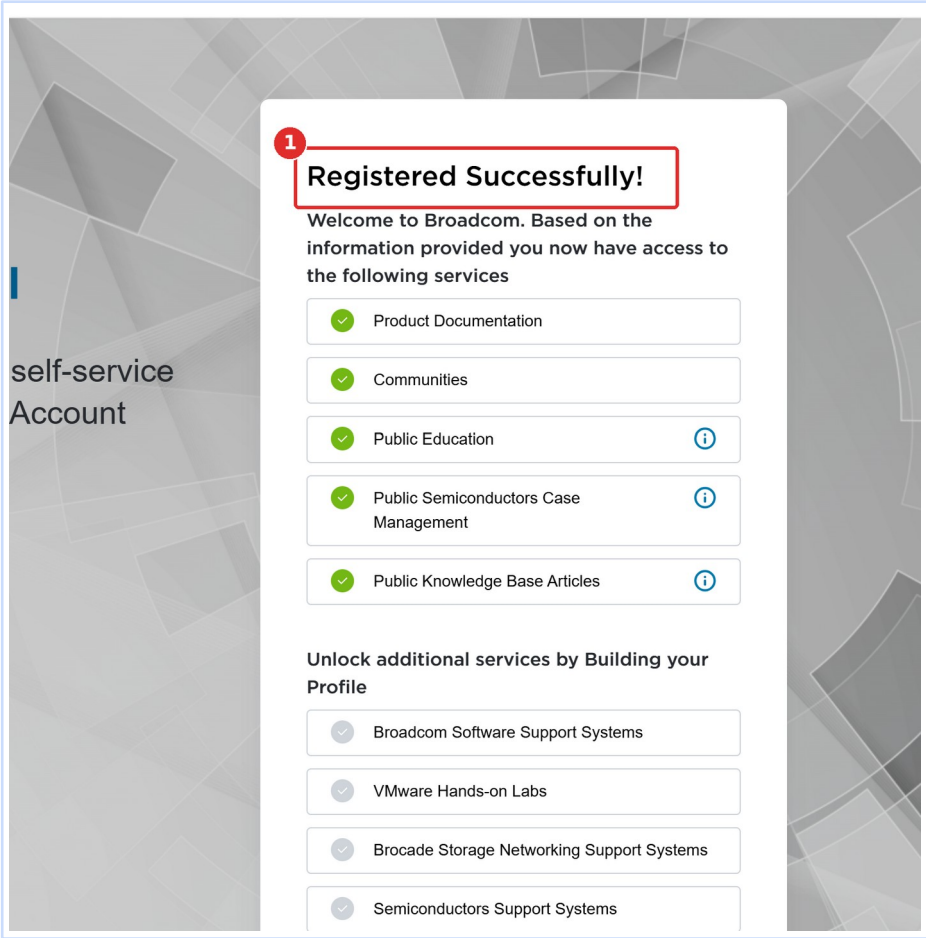
Accept the terms and create the account



1. Tick “I accept the Terms of Use”. Wait for “Password matched” to appear.

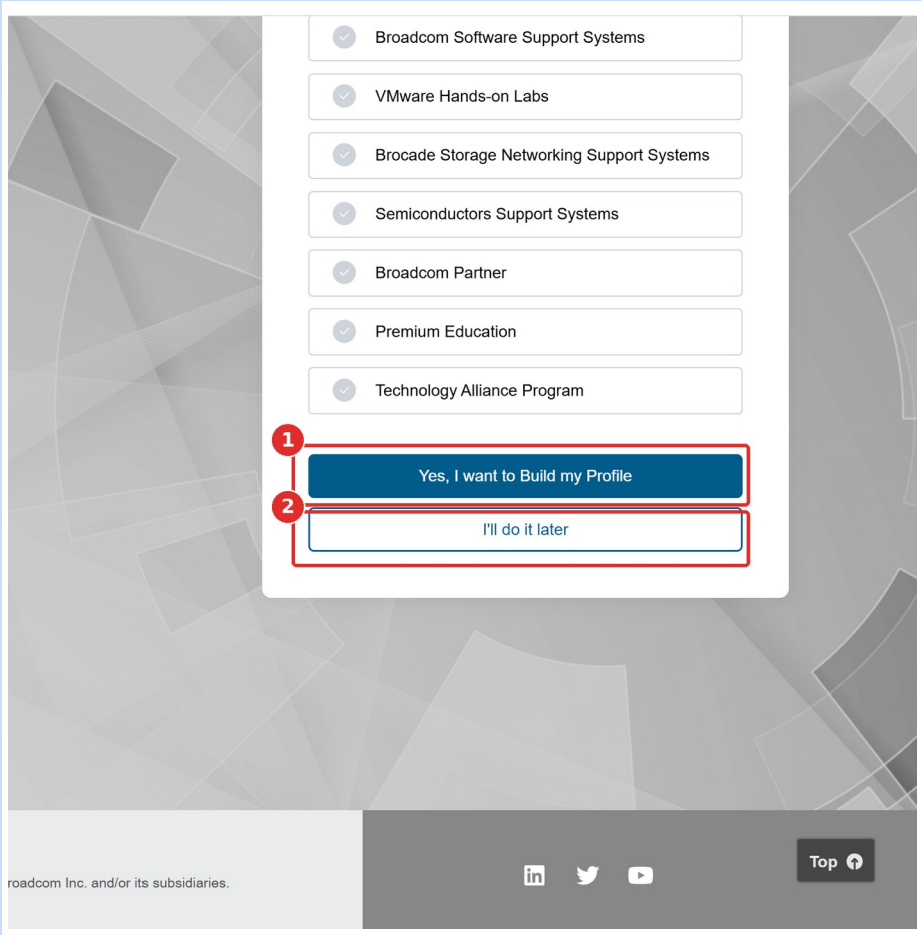
2. Press Create Account.

Registered successfully



1 Broadcom confirms the account and lists the services it unlocks. Scroll down to see the next choice.

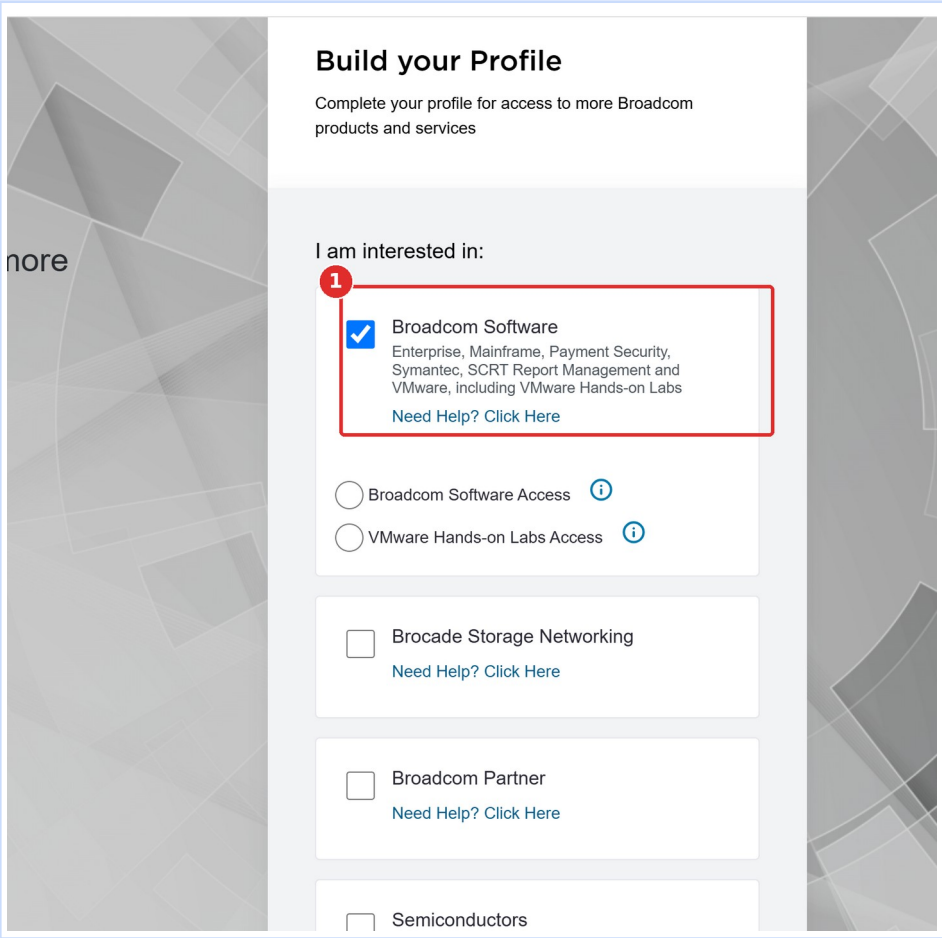
Build your profile now or later



1. "Yes, I want to Build my Profile" asks for company details such as a Site ID.

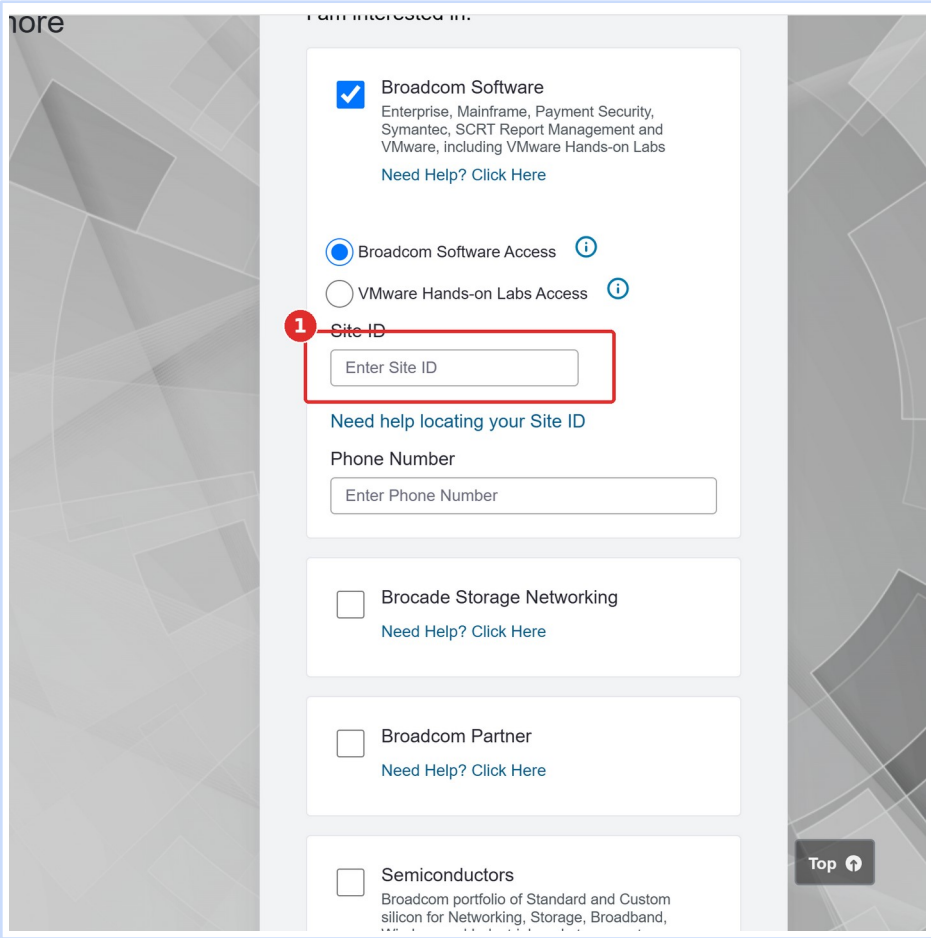
2. "I'll do it later" skips it. Free Workstation downloads do not need a Site ID.

The profile form



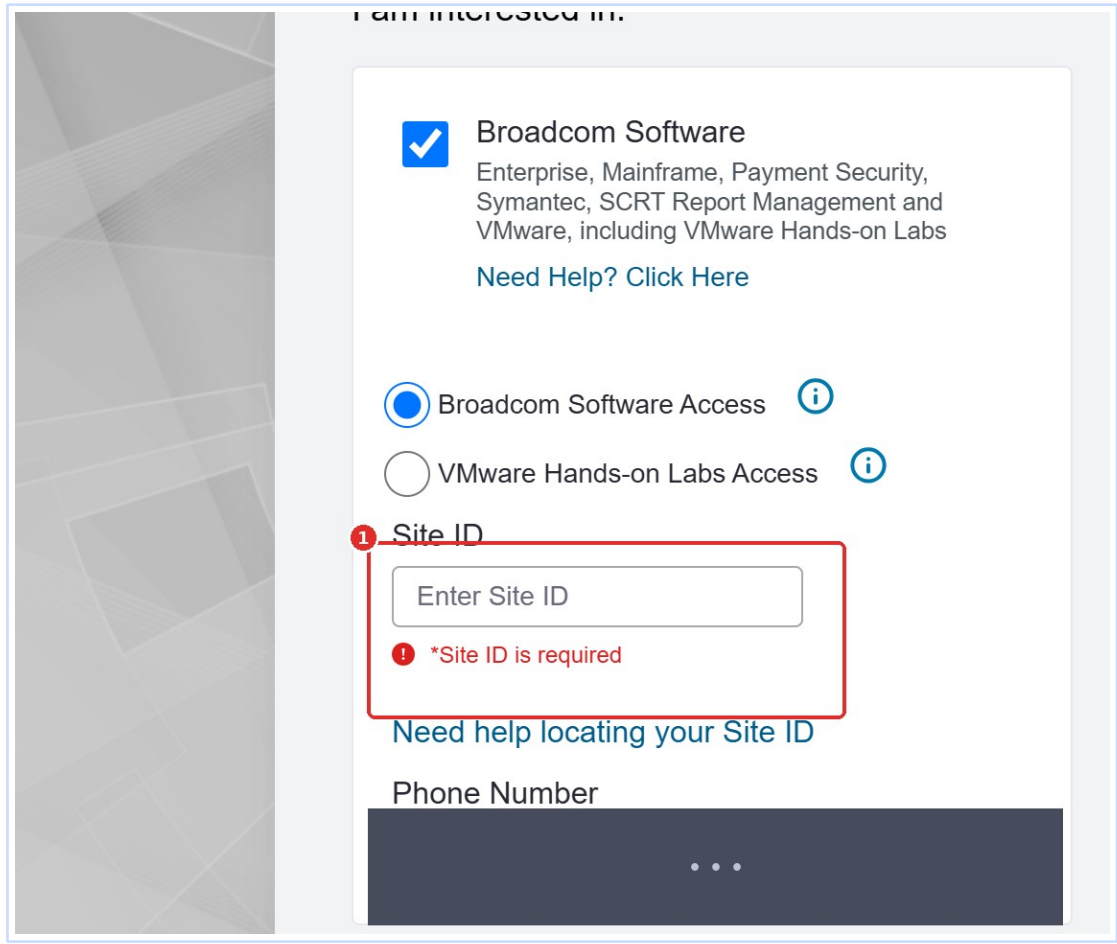
1 This form is for customers with paid products. Students do not have to complete it.

Why the Site ID field is optional for us



1. Site ID is issued to organizations that bought Broadcom products. A student account does not have one.

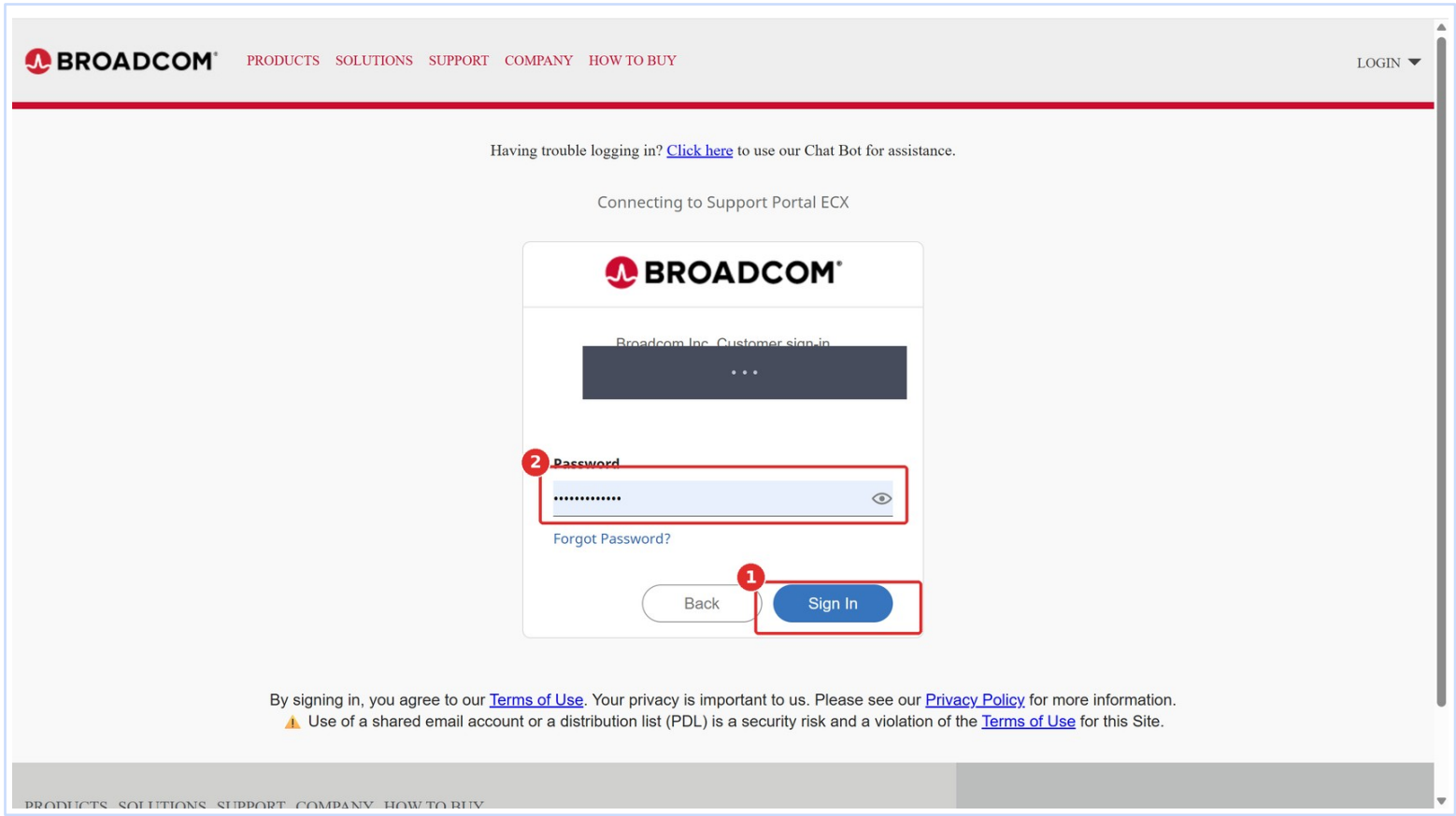
If you see “Site ID is required”



1 The error appears when the profile is saved without a Site ID. The phone number box below it is hidden here.

💡 Do not invent a Site ID. Close this form and sign in instead.

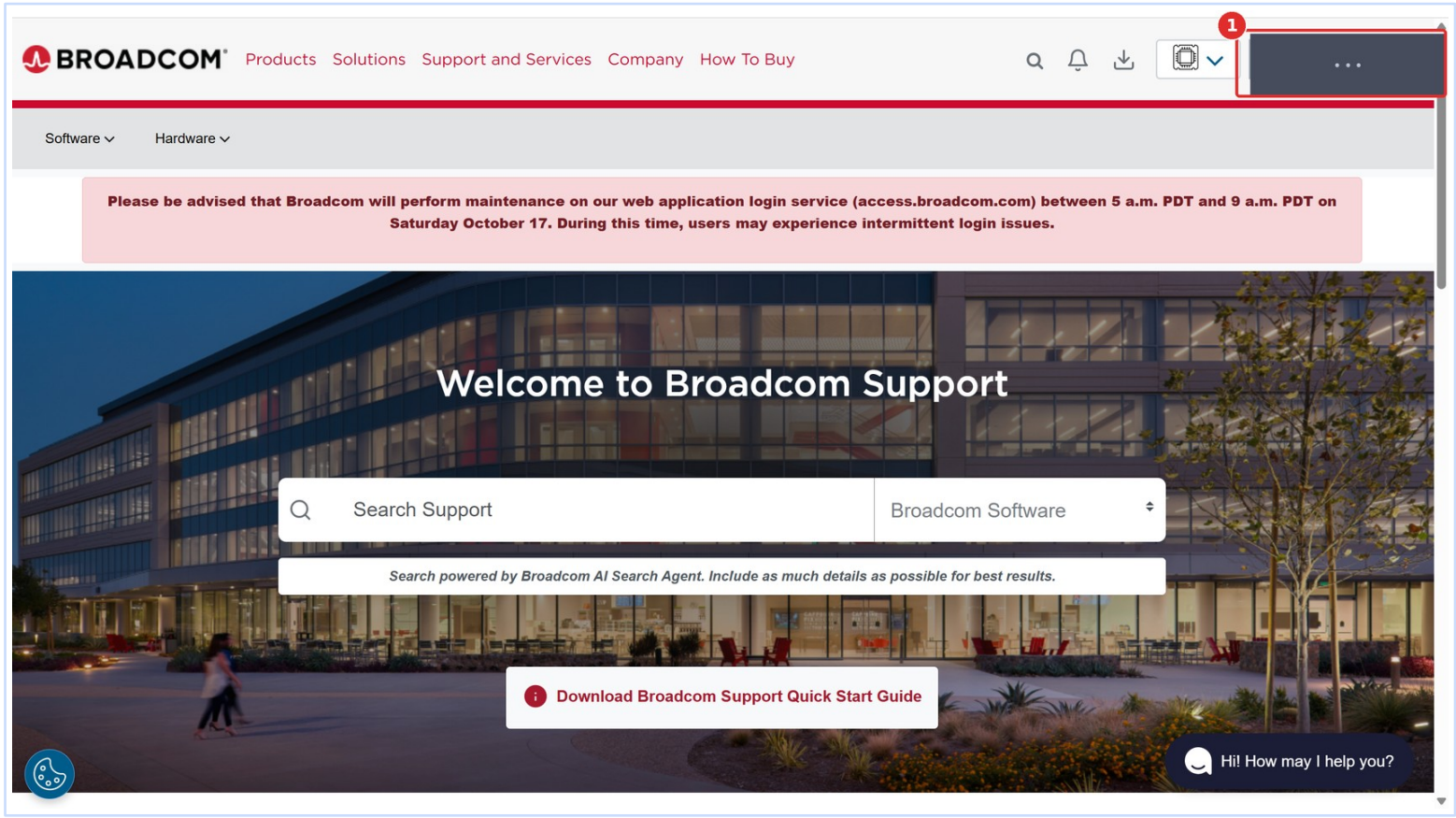
Sign in



1. Type your password. The email address shown above it is hidden here.

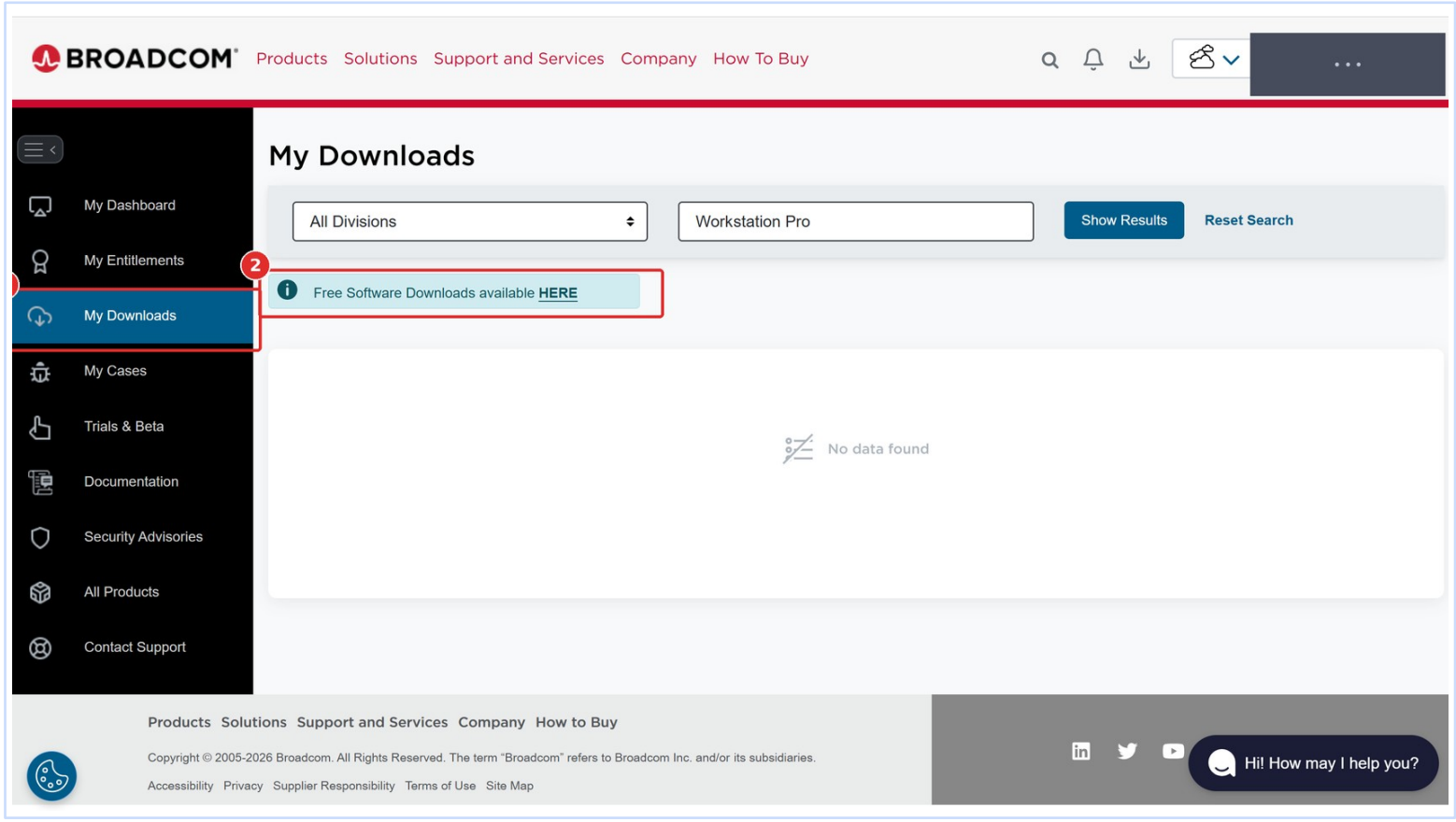
2. Press Sign In.

You are signed in



1 Your name appears at the top right of the Broadcom Support home page (hidden here).

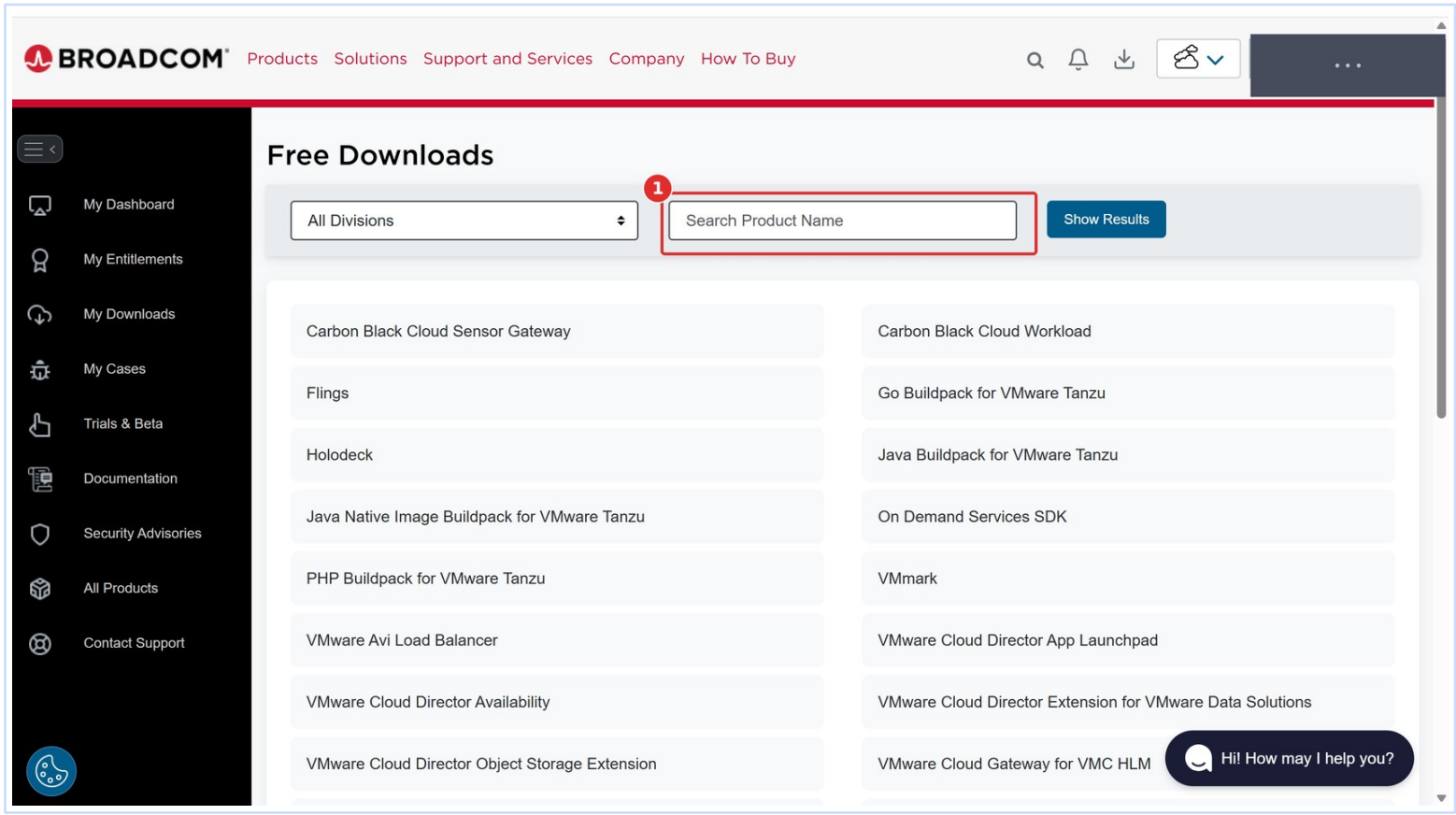
Open My Downloads



1. Choose My Downloads in the left menu.

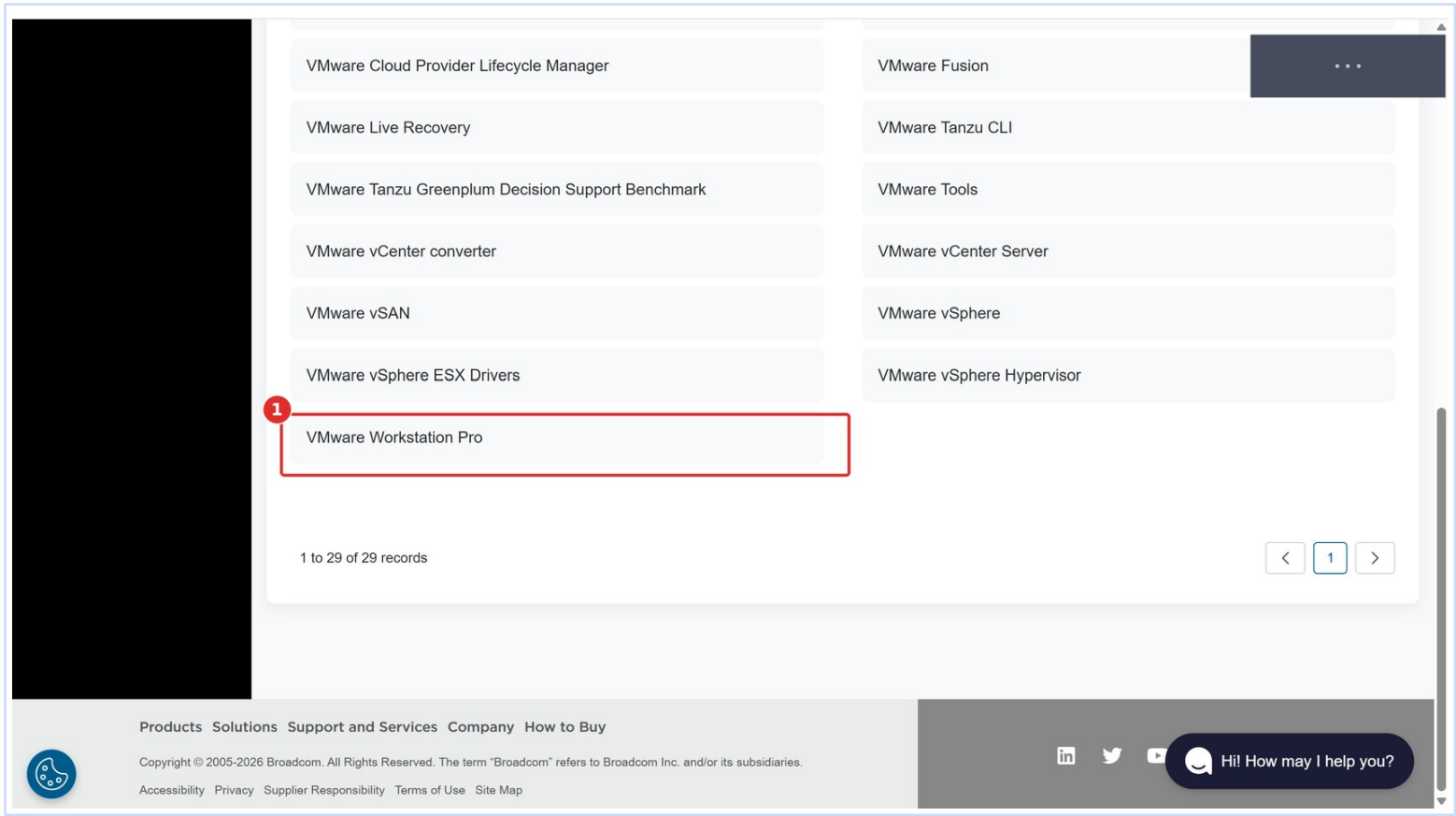
2. Press the link "Free Software Downloads available HERE".

The free downloads list



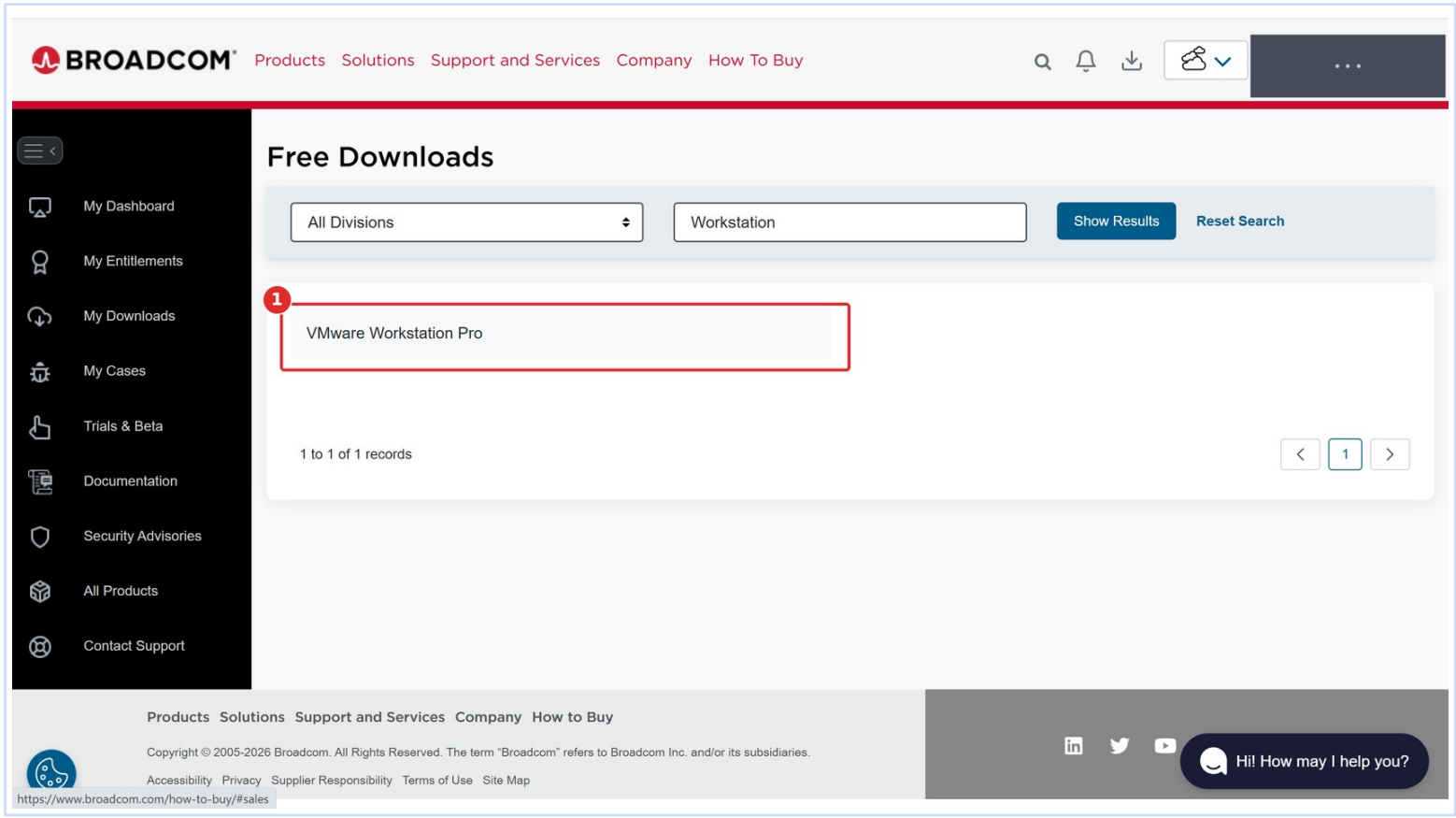
1 Type “Workstation” in Search Product Name and press Show Results. The list is long, but you can also scroll.

Scroll to VMware Workstation Pro



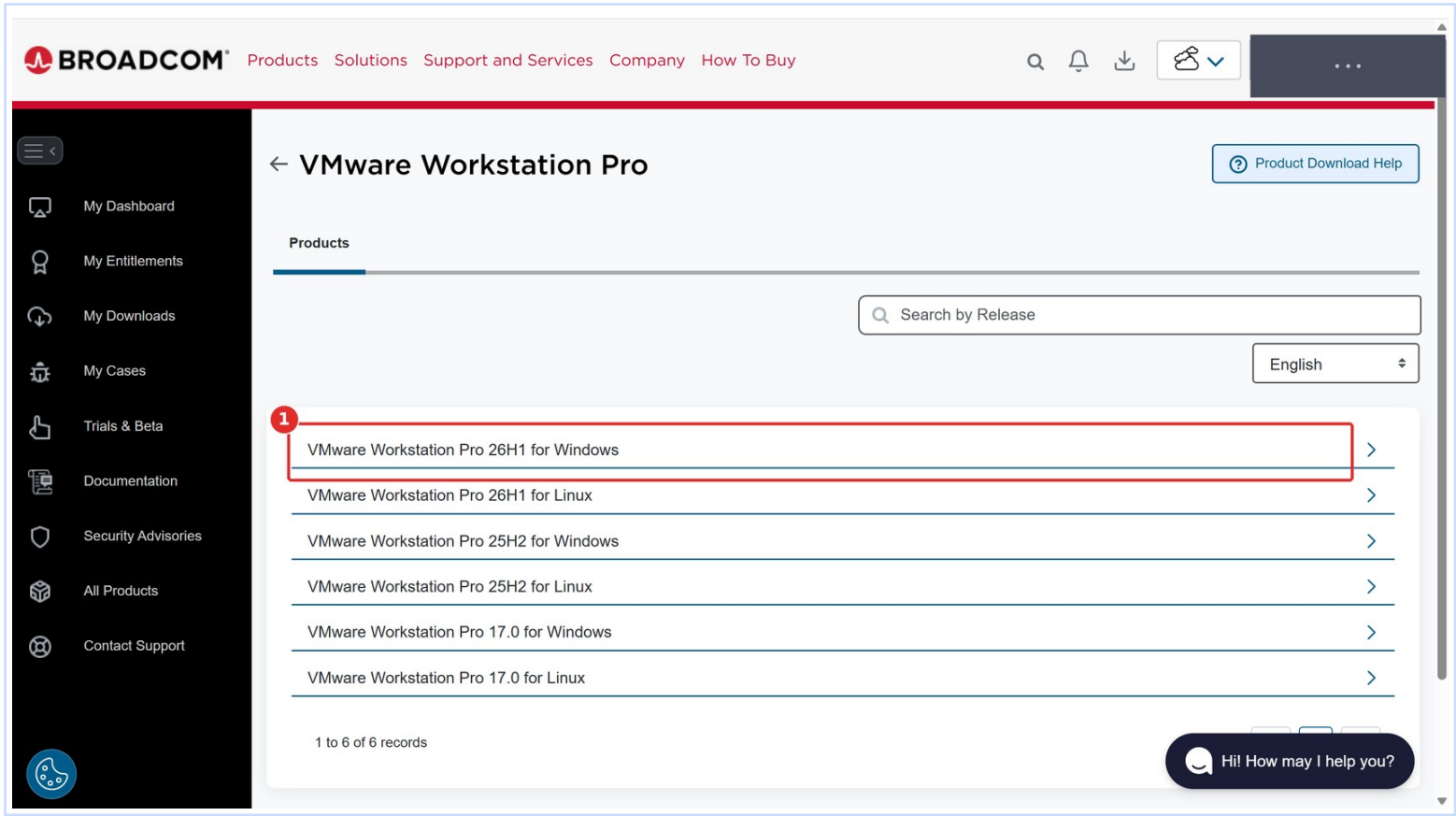
1 Near the end of the list, click "VMware Workstation Pro".

Or search for it



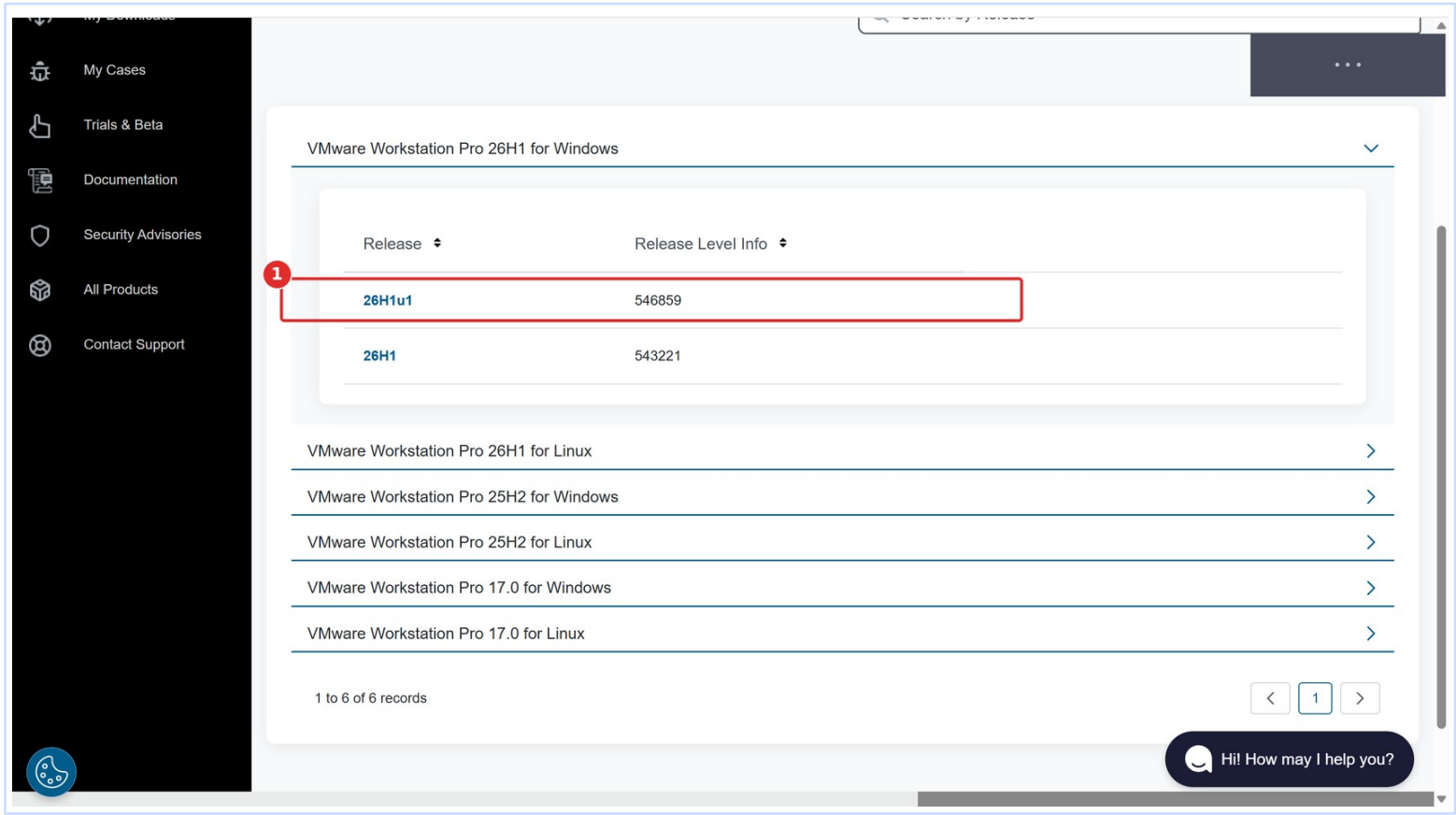
1 After searching, only one result remains. Click VMware Workstation Pro.

Choose the Windows edition



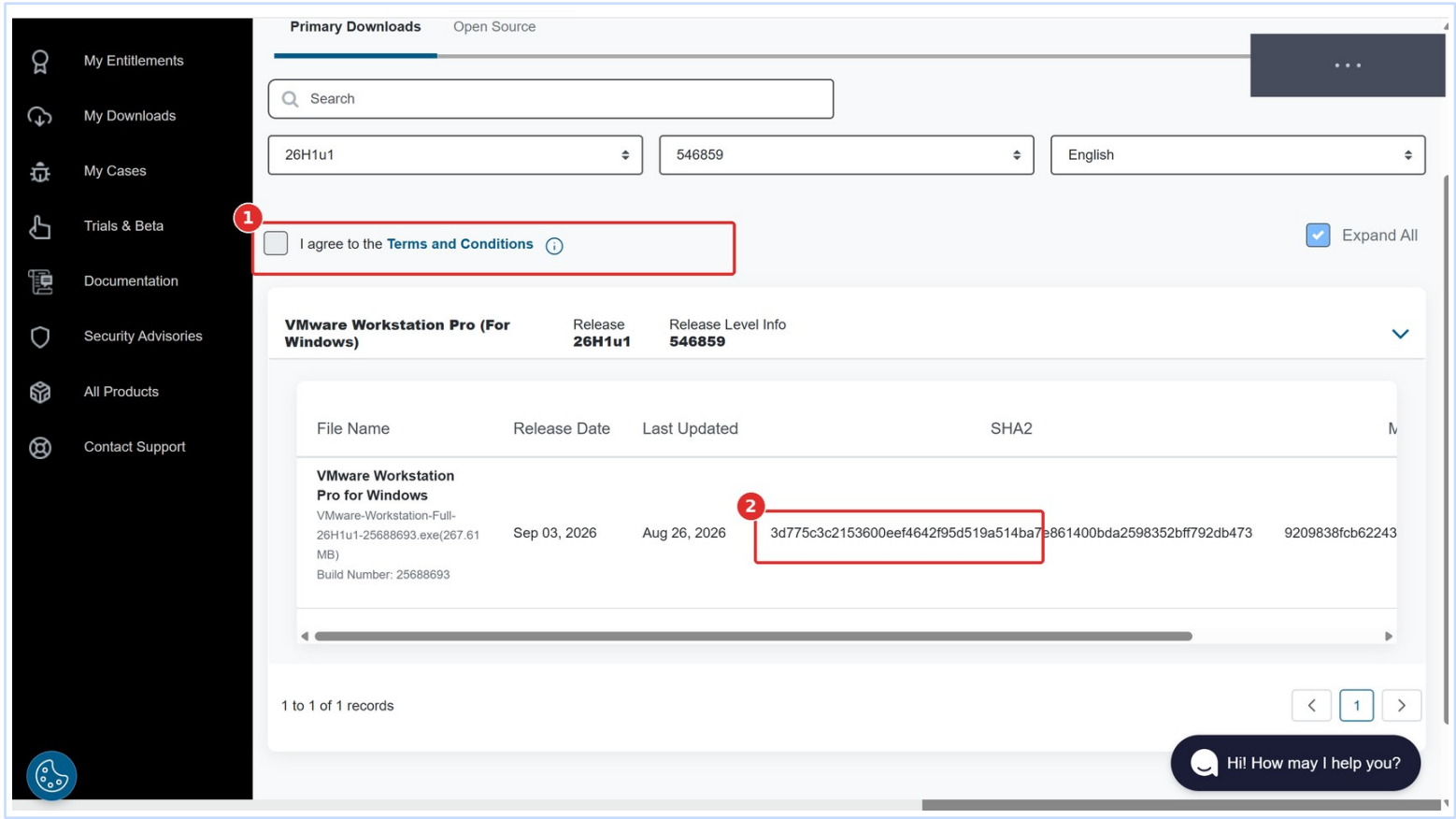
1 Click “VMware Workstation Pro 26H1 for Windows”. Linux versions are listed too, but we run on Windows.

Choose the release



1. Pick the newest update, 26H1u1 (build 546859). Newer builds contain security fixes.

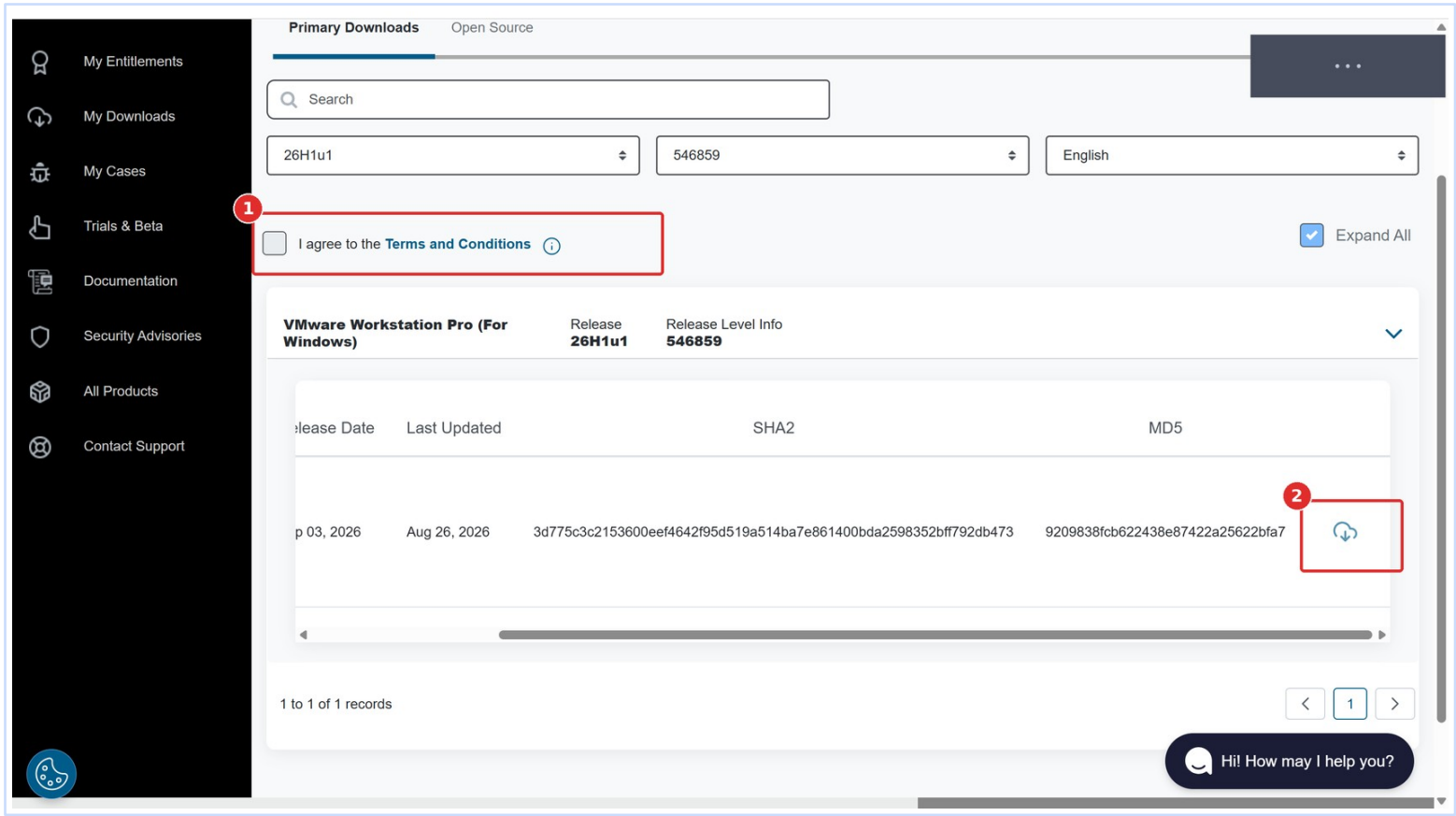
Accept the terms



1. Tick “I agree to the Terms and Conditions”. The download button stays disabled until you do.

2. The SHA2 column shows the official checksum of the installer. Note it for later.

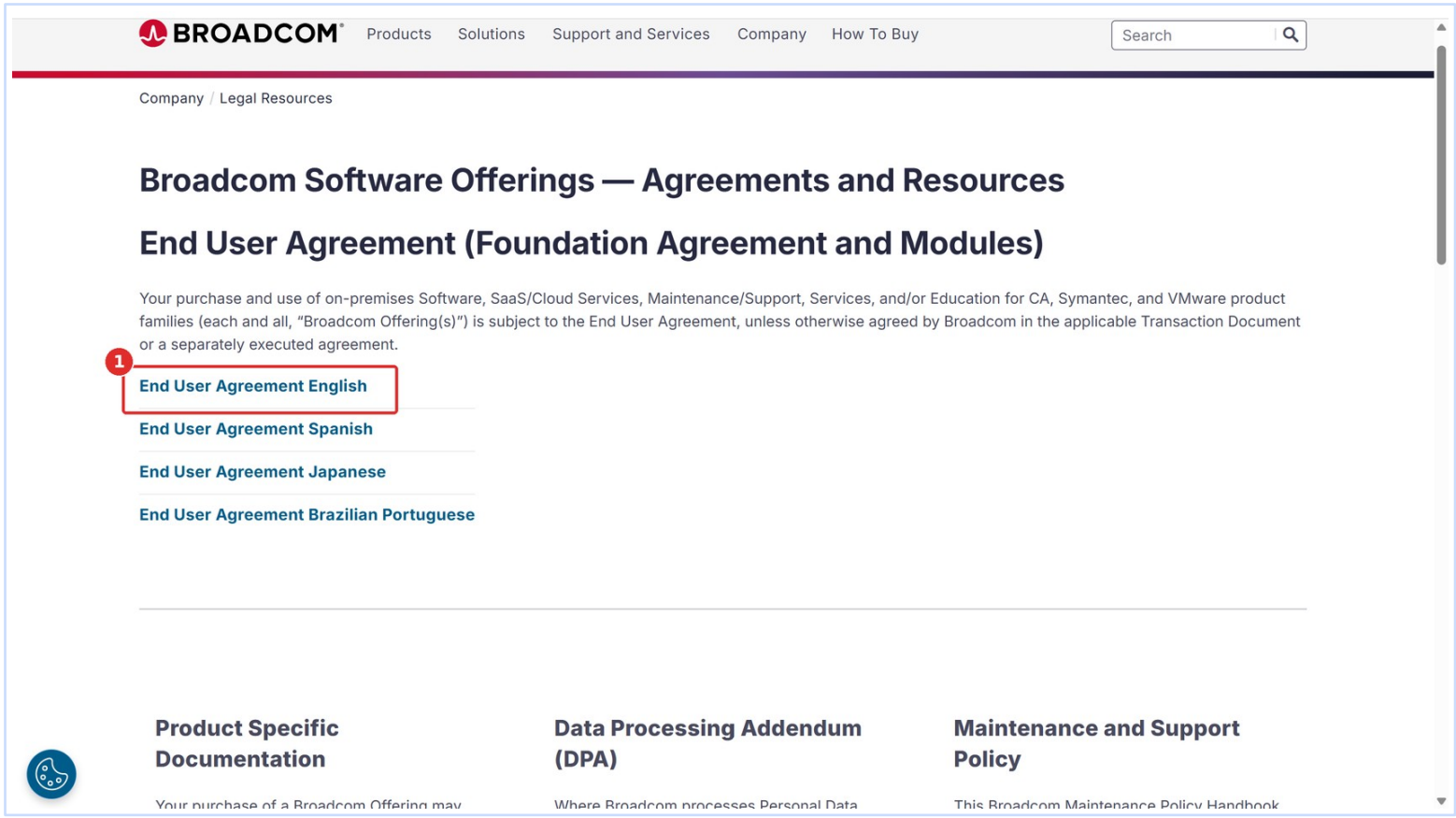
Press the download icon



1. The box is now ticked.

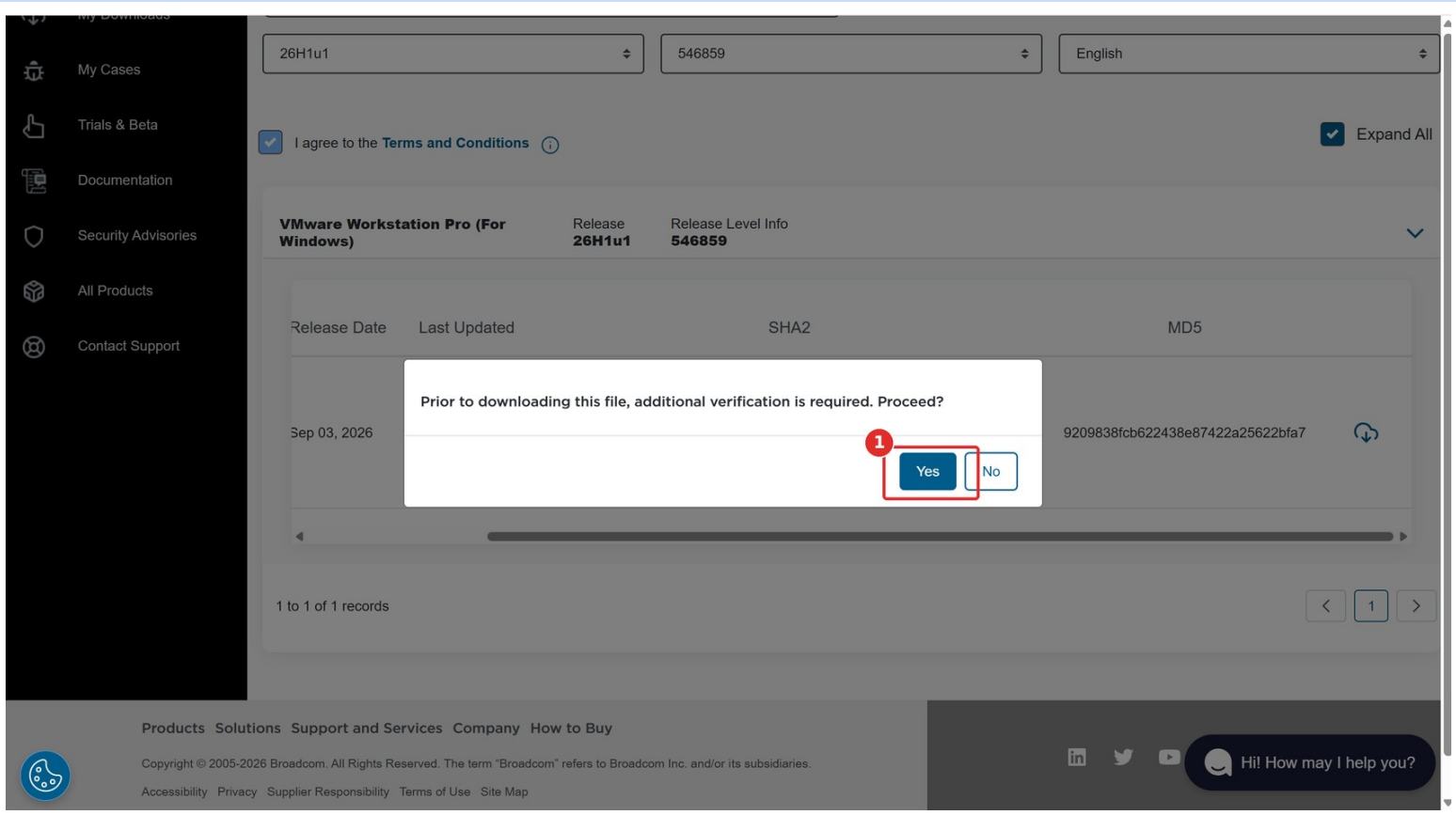
2. Press the cloud download icon at the end of the row.

Optional: read the license



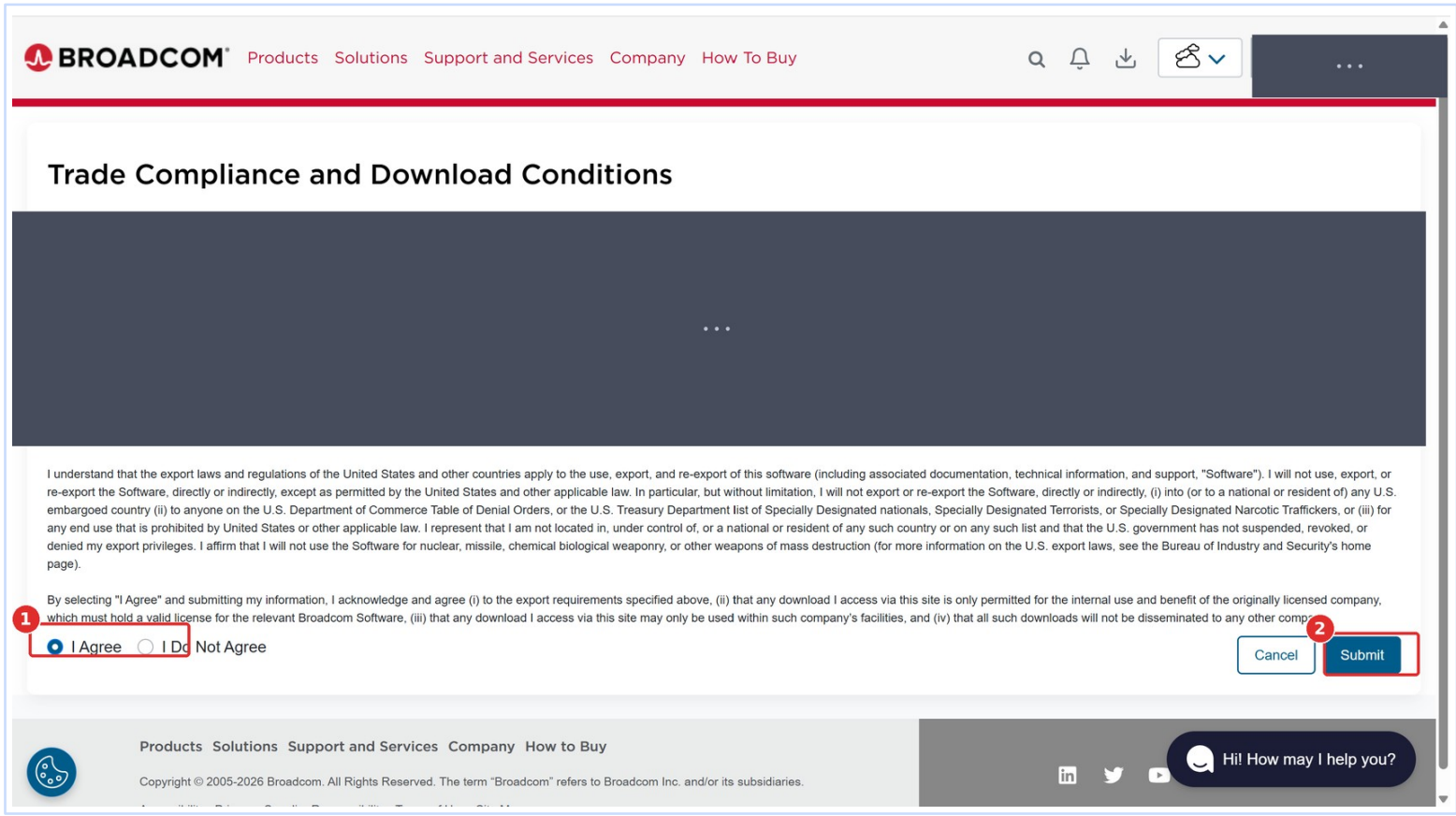
1 The links open Broadcom's End User Agreement pages. Read them: they decide where you may use and share the software.

Confirm the extra verification



1 "Prior to downloading this file, additional verification is required." Press Yes.

Trade compliance form

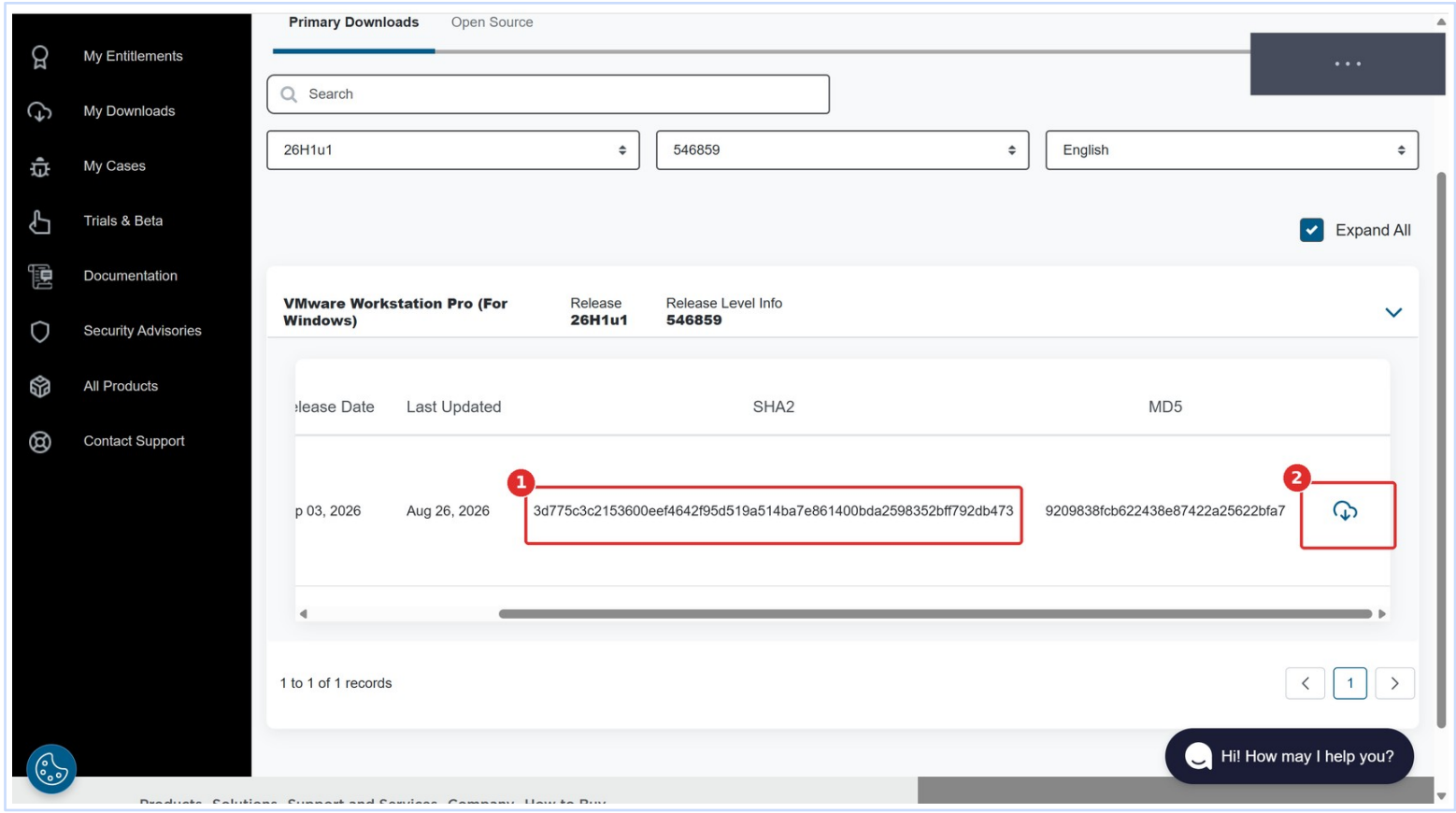


1. Fill name, company, address, country and postal code truthfully. All of these boxes are hidden in this screenshot.

2. Choose “I Agree” and press Submit.

💡 Wrong data can block the download. Re-enter the correct values and submit again.

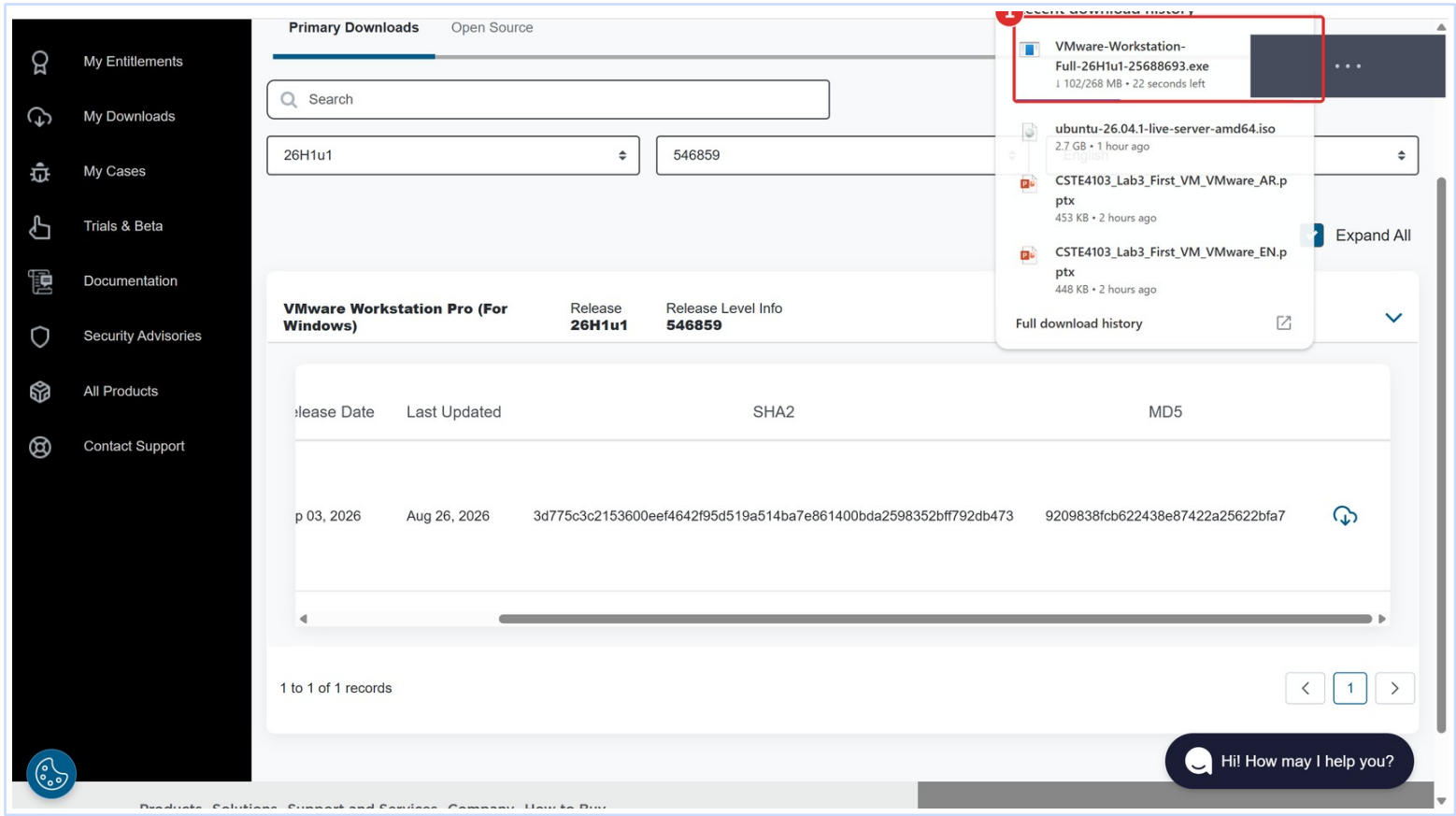
Back to the file list



1. Compare this SHA2 value with your own calculation later.

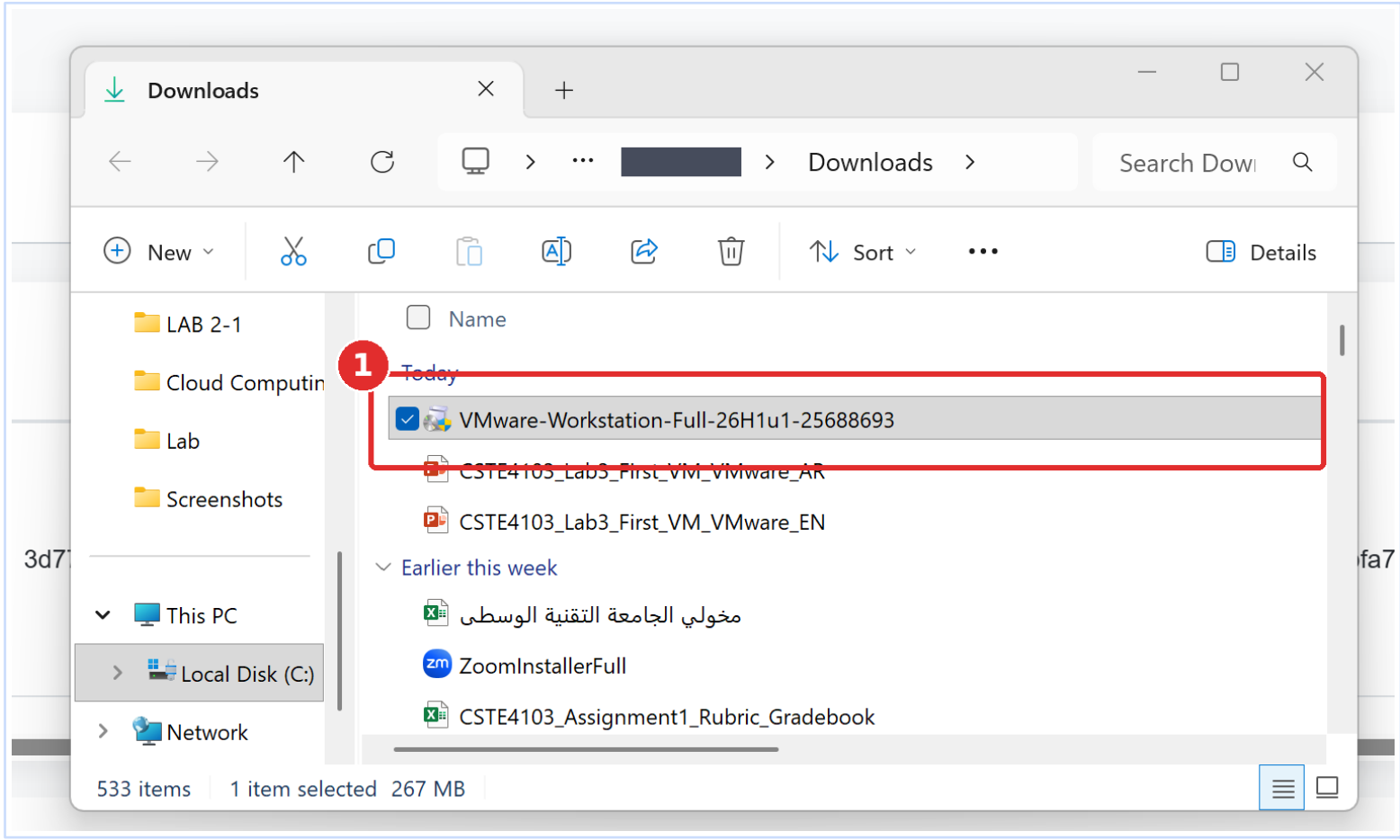
2. Press the download icon again. The installer is about 270 MB.

The installer downloads



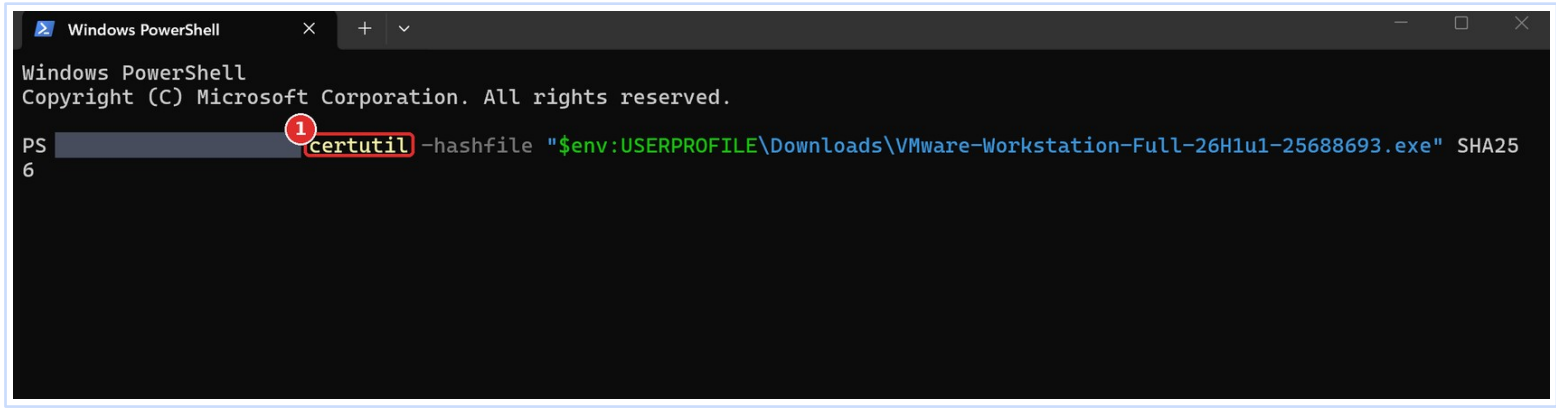
1 VMware-Workstation-Full-26H1u1-25688693.exe appears in the browser download list.

Locate the file



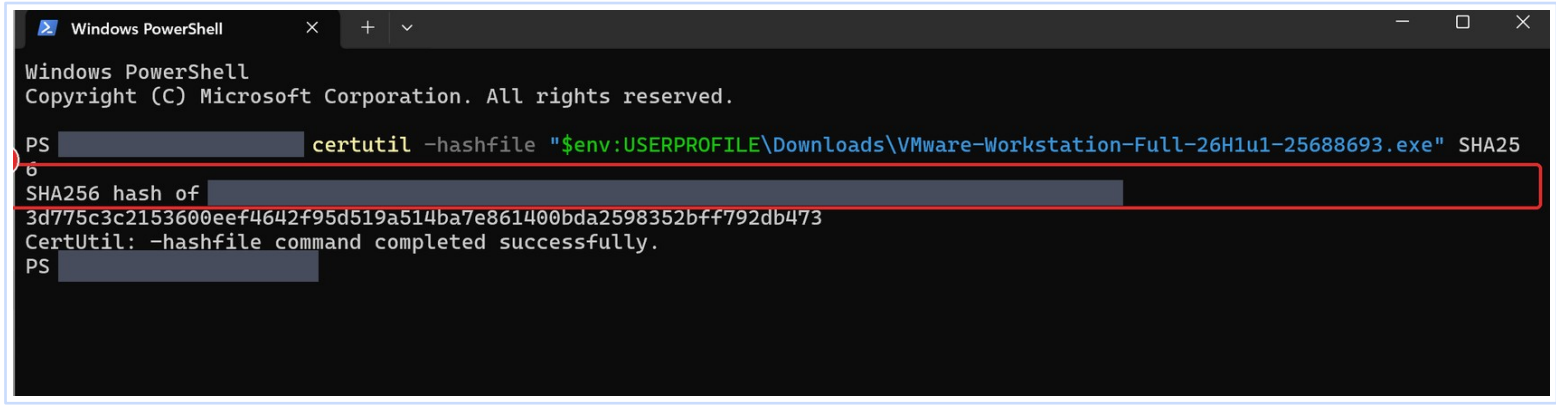
1 Open Downloads. The installer is selected, about 267 MB.

Calculate its checksum



1. In PowerShell run: certutil -hashfile "$env:USERPROFILE\Downloads\VMware-Workstation-Full-26H1u1-25688693.exe" SHA256

Compare with Broadcom's value



1. The result must equal the SHA2 value shown on the Broadcom page (starts with 3d775c3c…). If different, download again.

Common problems

- No verification email: wait a minute, press Resend, and check Spam.
- “Site ID is required”: skip the profile form; free downloads do not need it.
- Trade compliance rejected: re-enter name, address and country exactly and consistently.
- The portal changes often: the same steps may sit under slightly different menus.
- Before sharing the installer with others, check Broadcom’s current license terms.

PART 2 OF 6

Checkpoint

- ✓ Broadcom account created and signed in
- ✓ VMware Workstation Pro 26H1u1 installer downloaded
- ✓ Installer SHA-256 matches the Broadcom value

Next: Part 3 — Install VMware Workstation Pro

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Part 3 of 6

Install VMware Workstation Pro

Run the installer as administrator and walk through every wizard screen

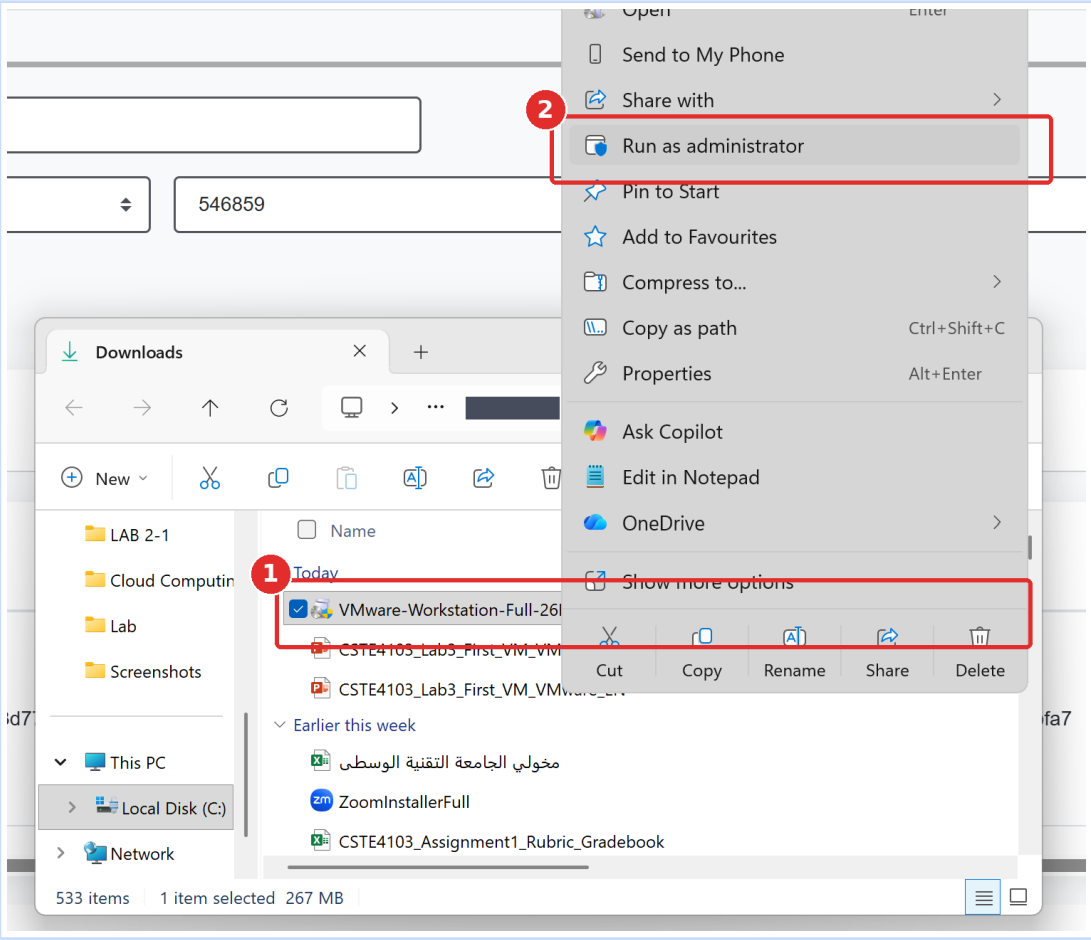
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PART 3 OF 6

Before you install

- You need the installer from Part 2 and administrator rights on this PC.
- Close other programs. The install may ask for a restart.
- Allow 10 to 15 minutes and about 1.5 GB of free disk space.
- Dark boxes in the screenshots hide your Windows user name and other personal details.

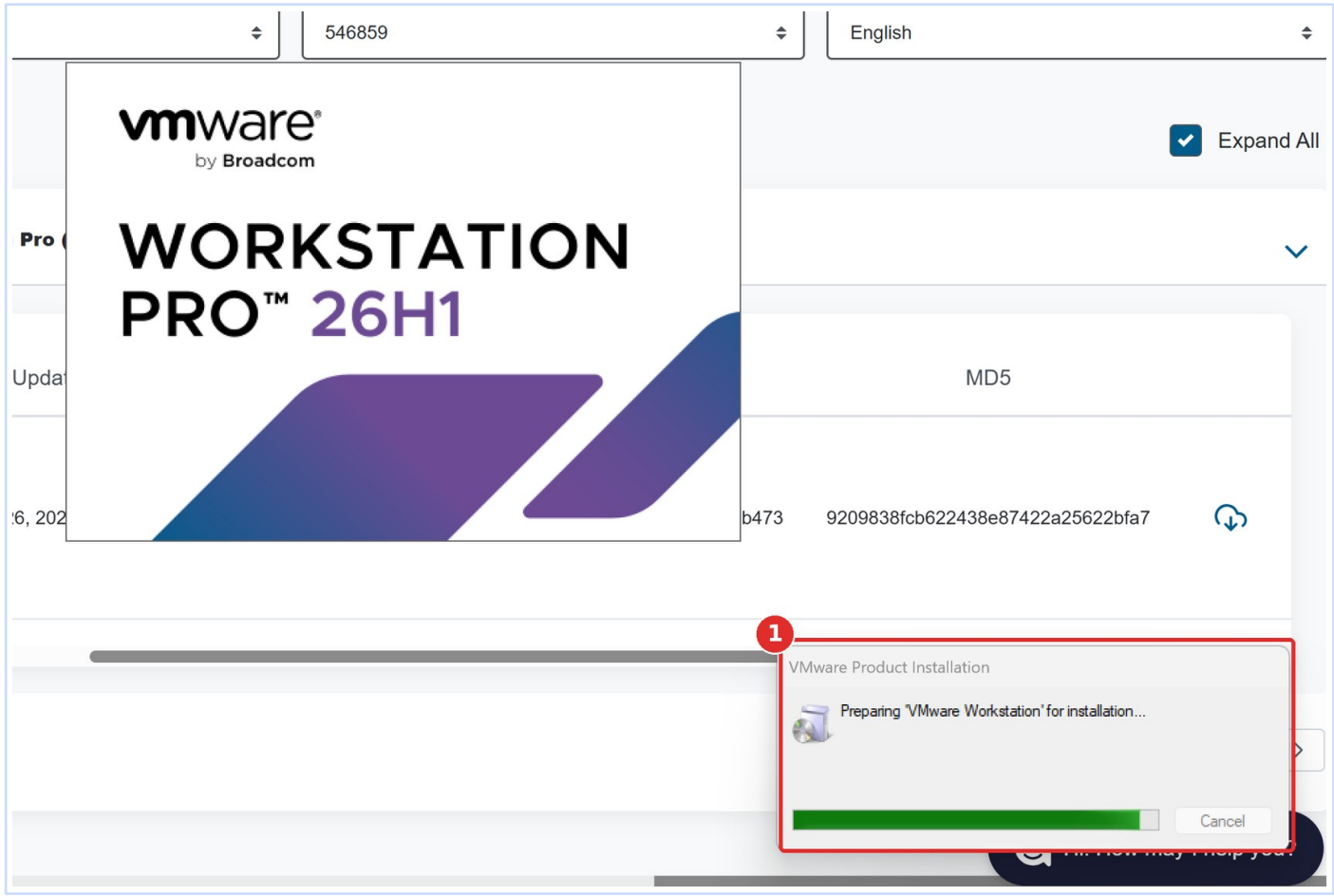
Run the installer as administrator



1. Find VMware-Workstation-Full-26H1u1 in Downloads.

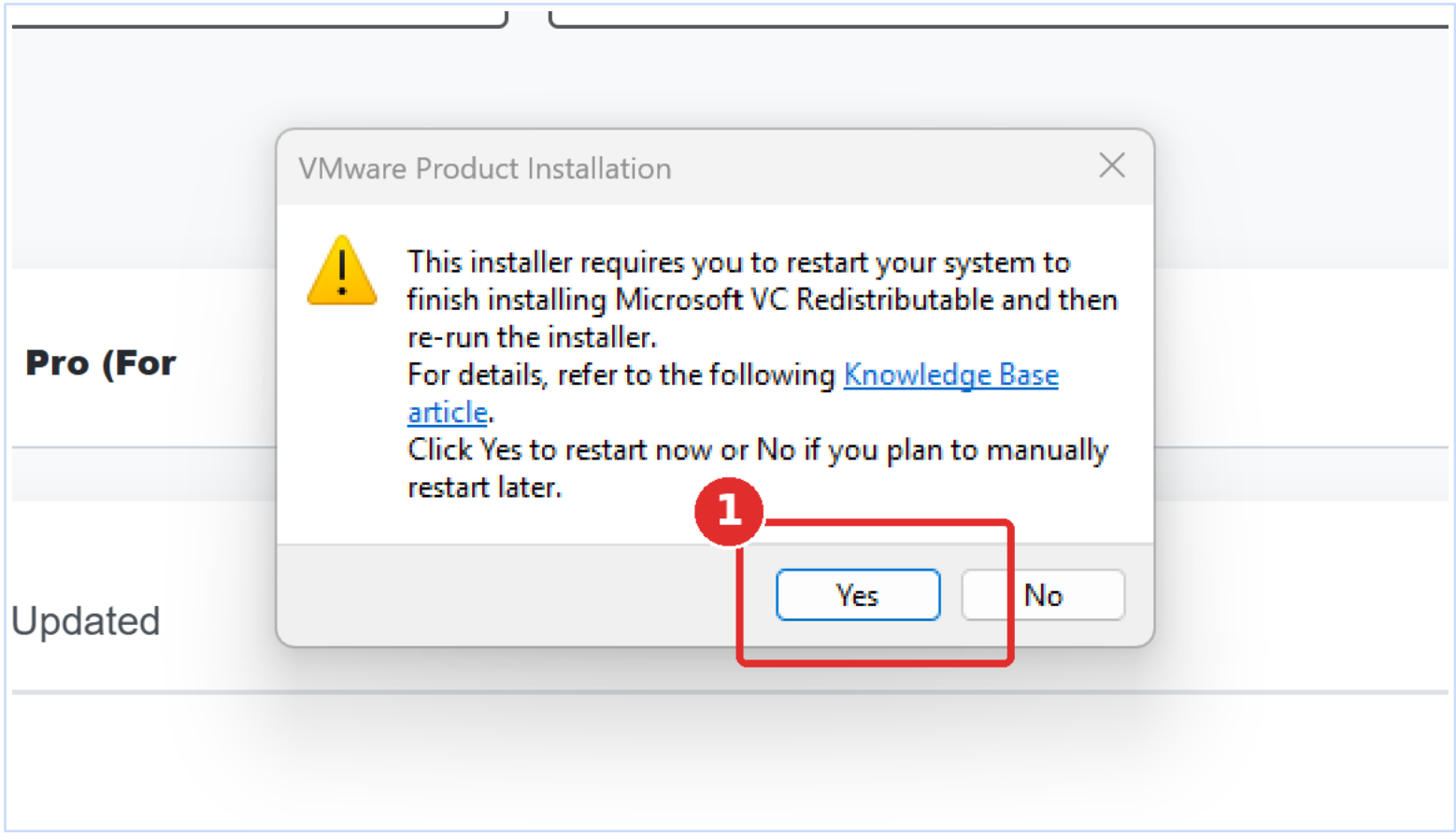
2. Right-click it and choose Run as administrator (in Windows 11 it may be under “Show more options”).

The installer prepares itself



1. The VMware splash screen appears and “Preparing VMware Workstation for installation” runs. Wait for it.

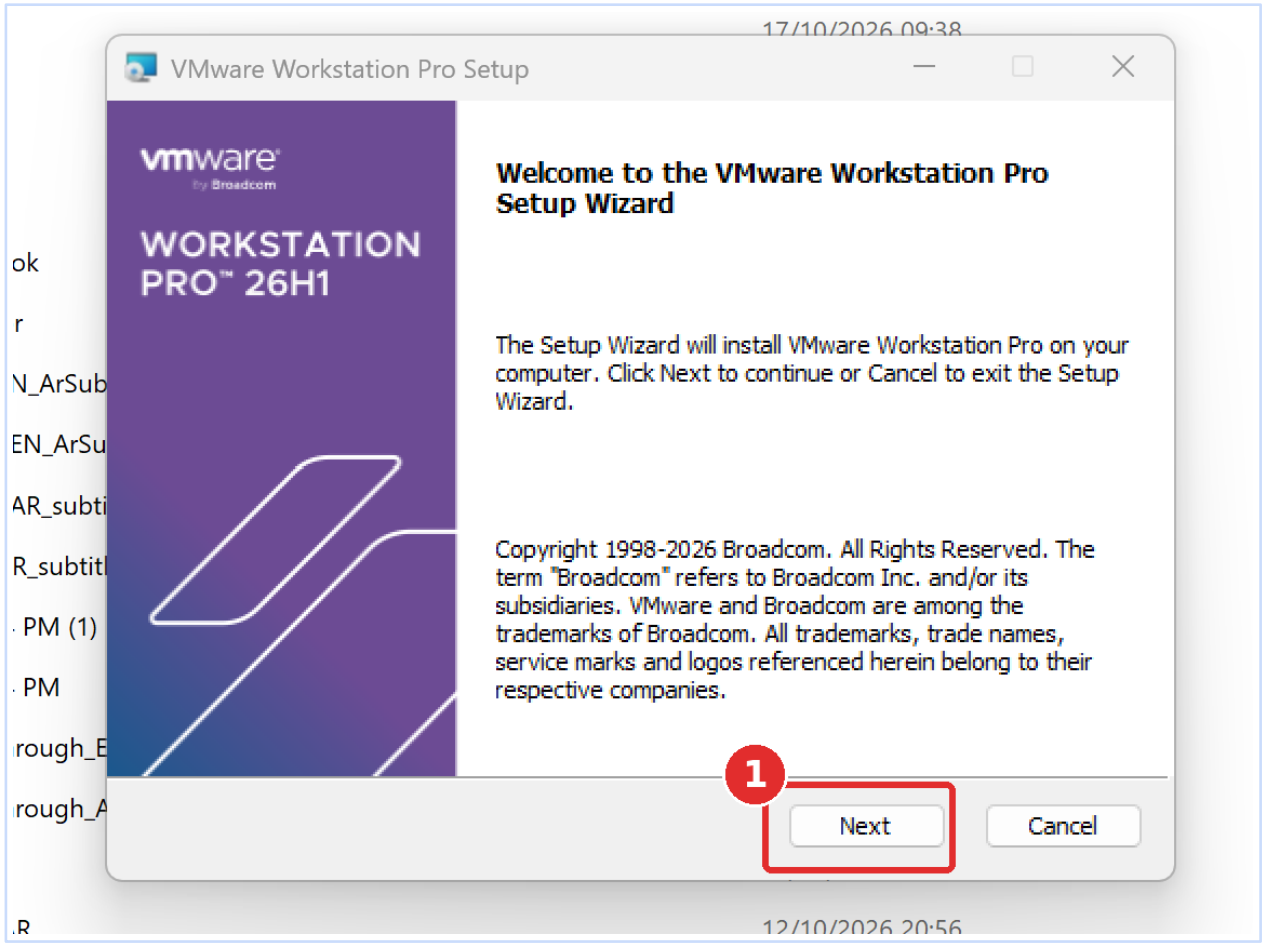
Restart if asked



1 This message appears when Microsoft Visual C++ Redistributable was installed first. Press Yes to restart.

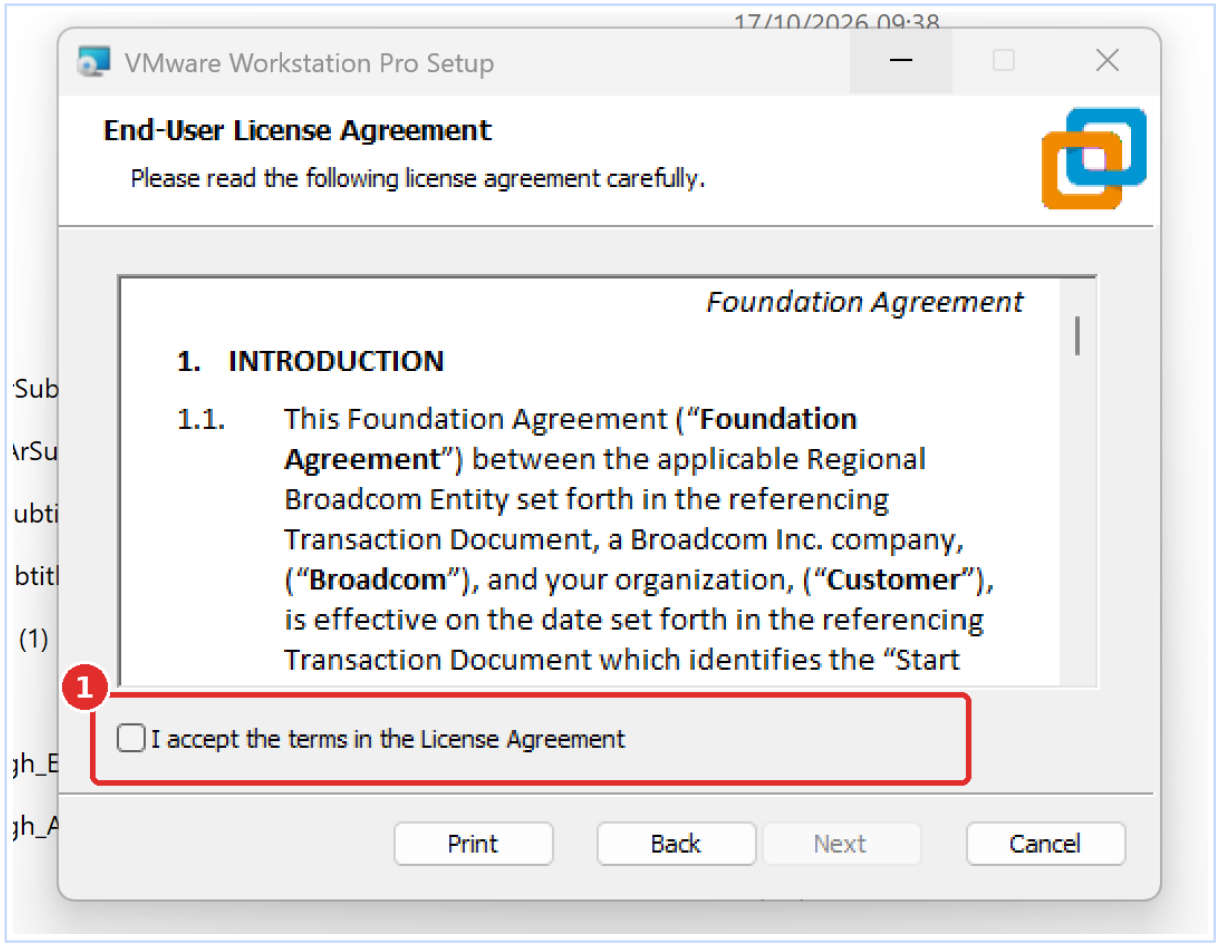
💡 After the restart, run the installer again as administrator.

Welcome screen



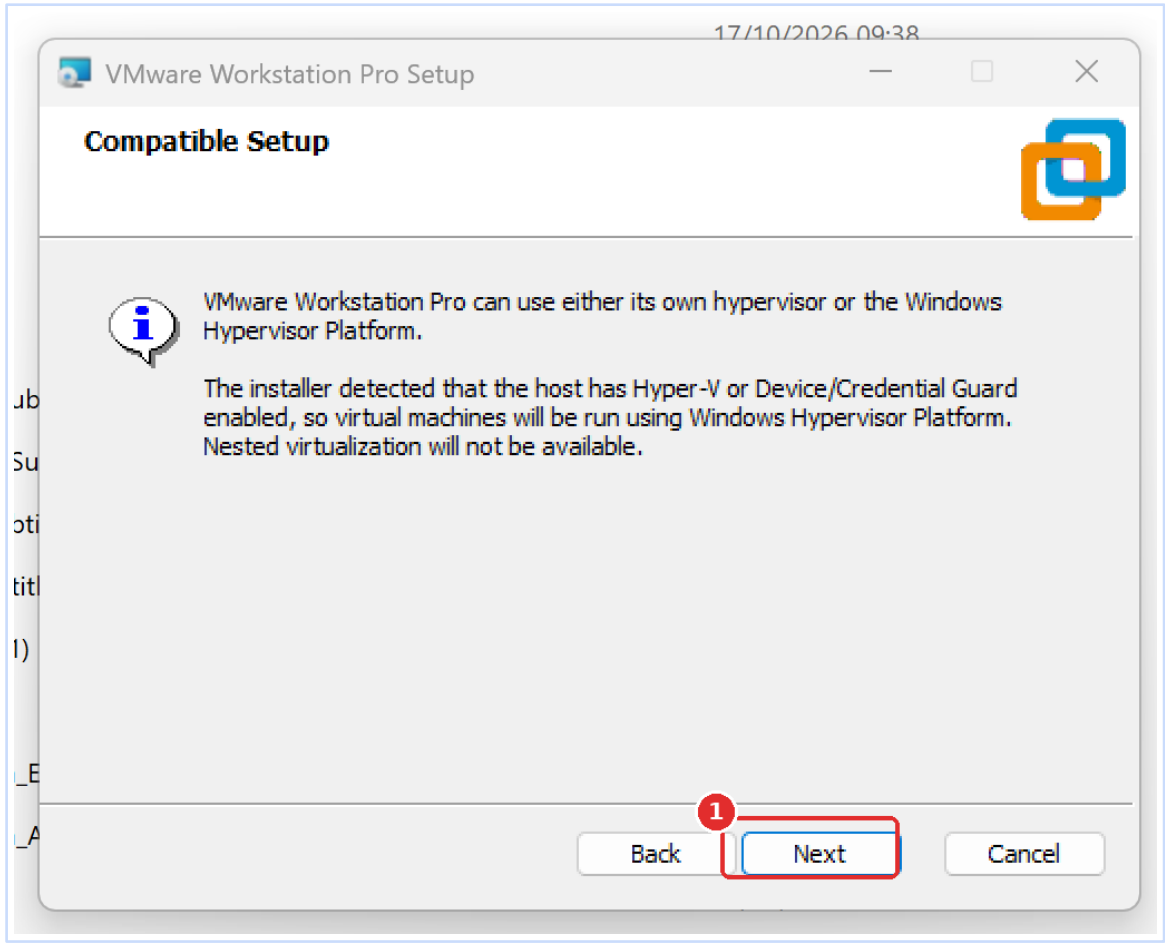
1. Press Next on “Welcome to the VMware Workstation Pro Setup Wizard”.

Accept the license agreement



1. Read the End-User License Agreement, tick "I accept the terms", and Next becomes available.

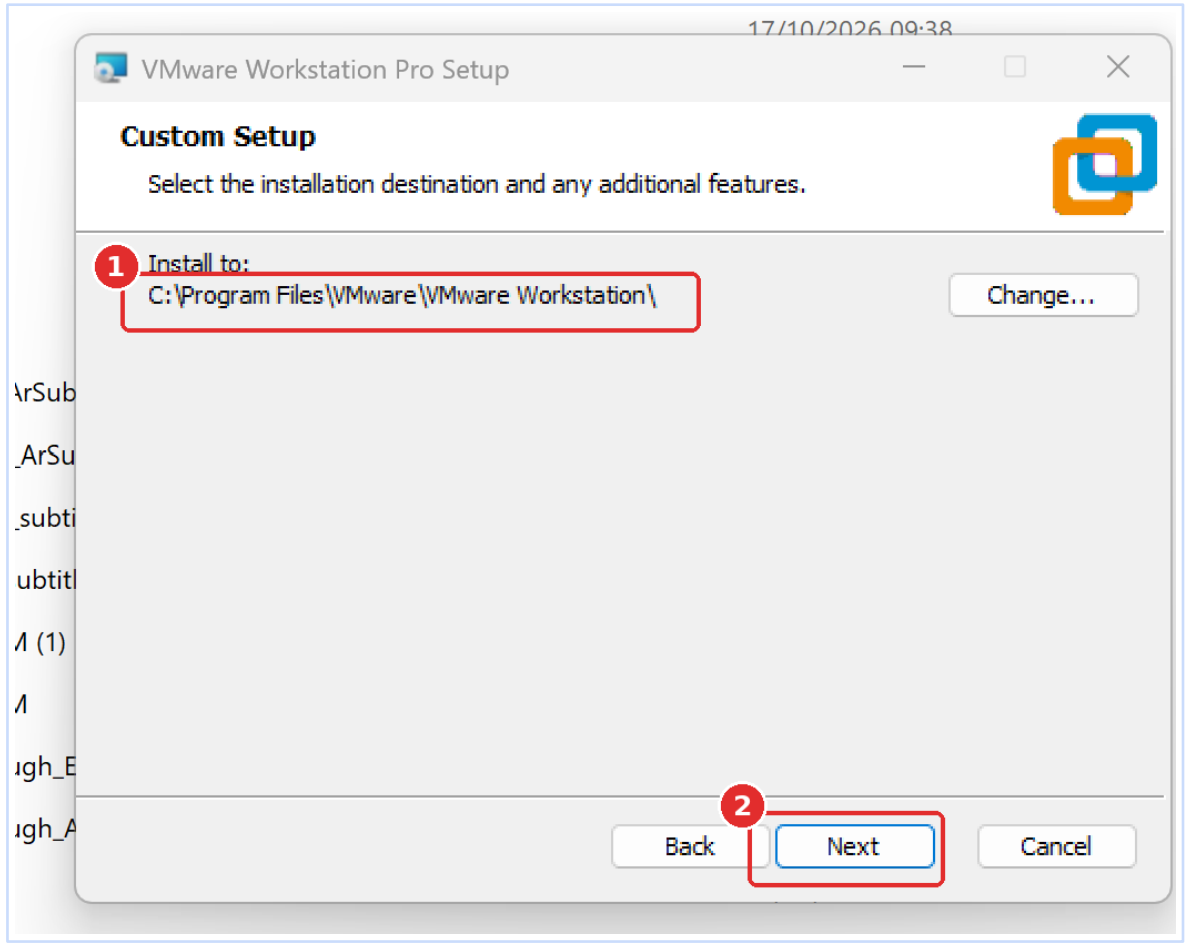
Compatible Setup message



1. Windows on this PC runs Hyper-V or Device Guard, so VMware runs on the Windows Hypervisor Platform and nested virtualization is not available. This is normal. Press Next.

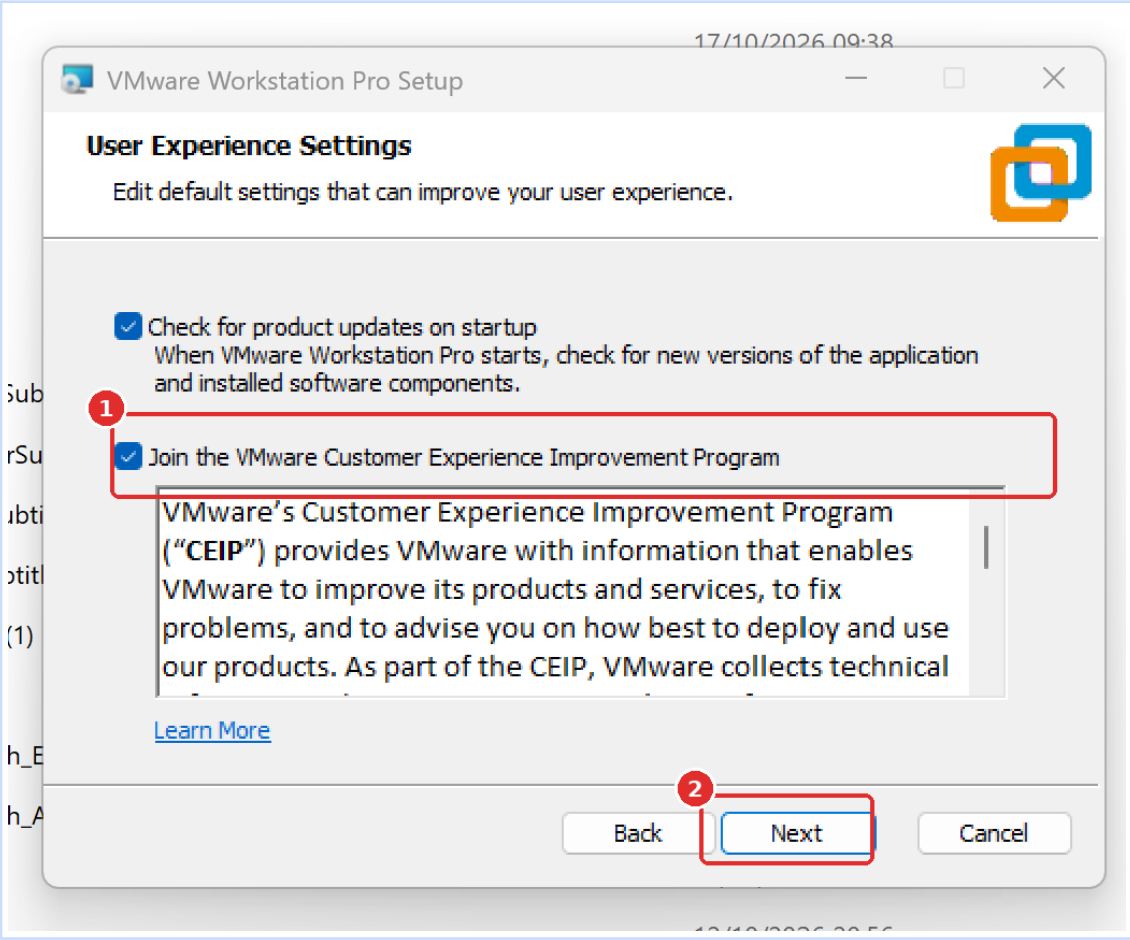
💡 If your PC does not show this screen, VMware uses its own hypervisor. Both work for this lab.

Choose the install folder



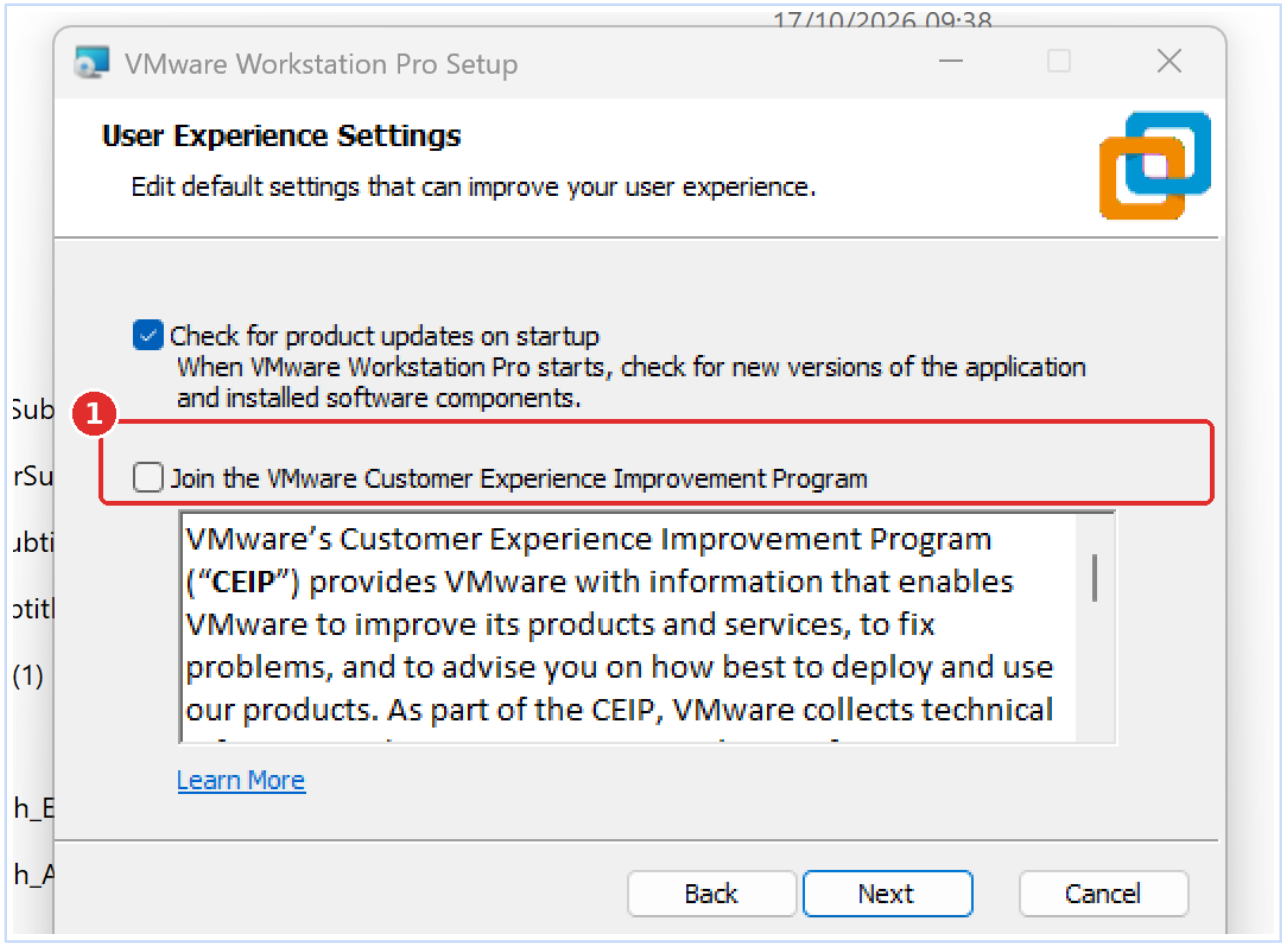
1. Keep the default folder C:\Program Files\VMware\VMware Workstation\.
2. Press Next.

User experience settings



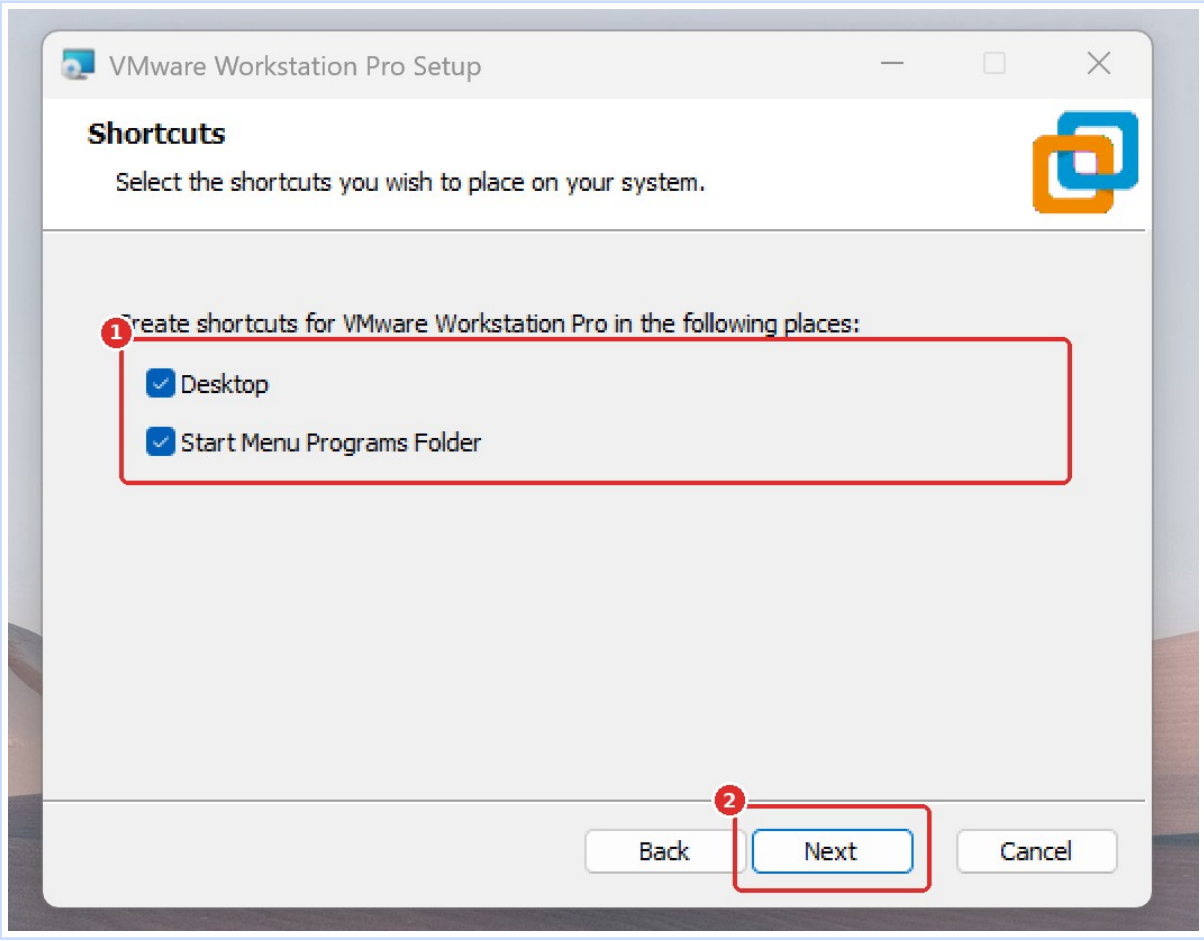
1. "Check for product updates on startup" is on by default, and so is "Join the VMware Customer Experience Improvement Program" (CEIP).
2. Press Next after deciding.

We turned CEIP off



1 CEIP sends anonymous usage data to VMware. We untick it for privacy; this is optional and does not affect the lab.

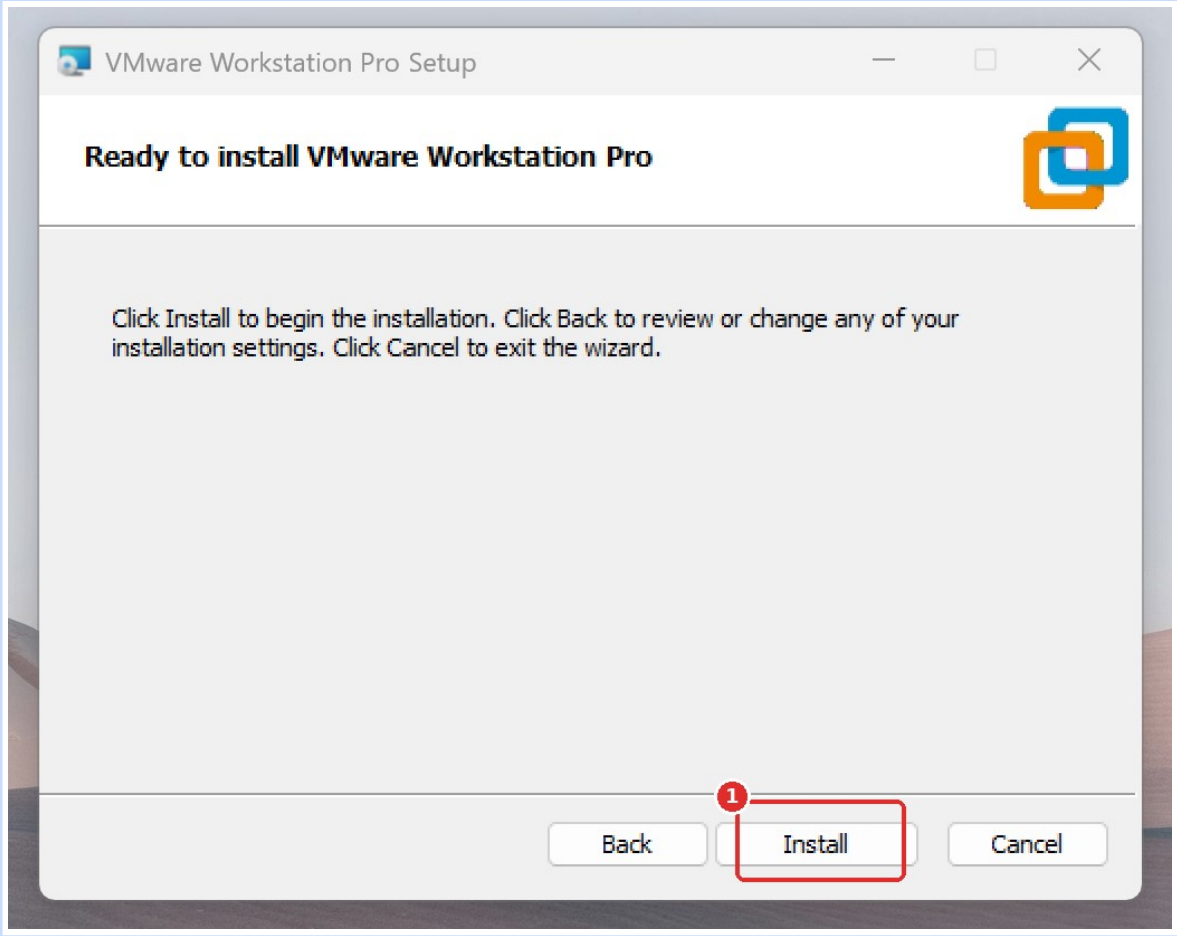
Shortcuts



1. Keep Desktop and Start Menu Programs Folder ticked for easy access.

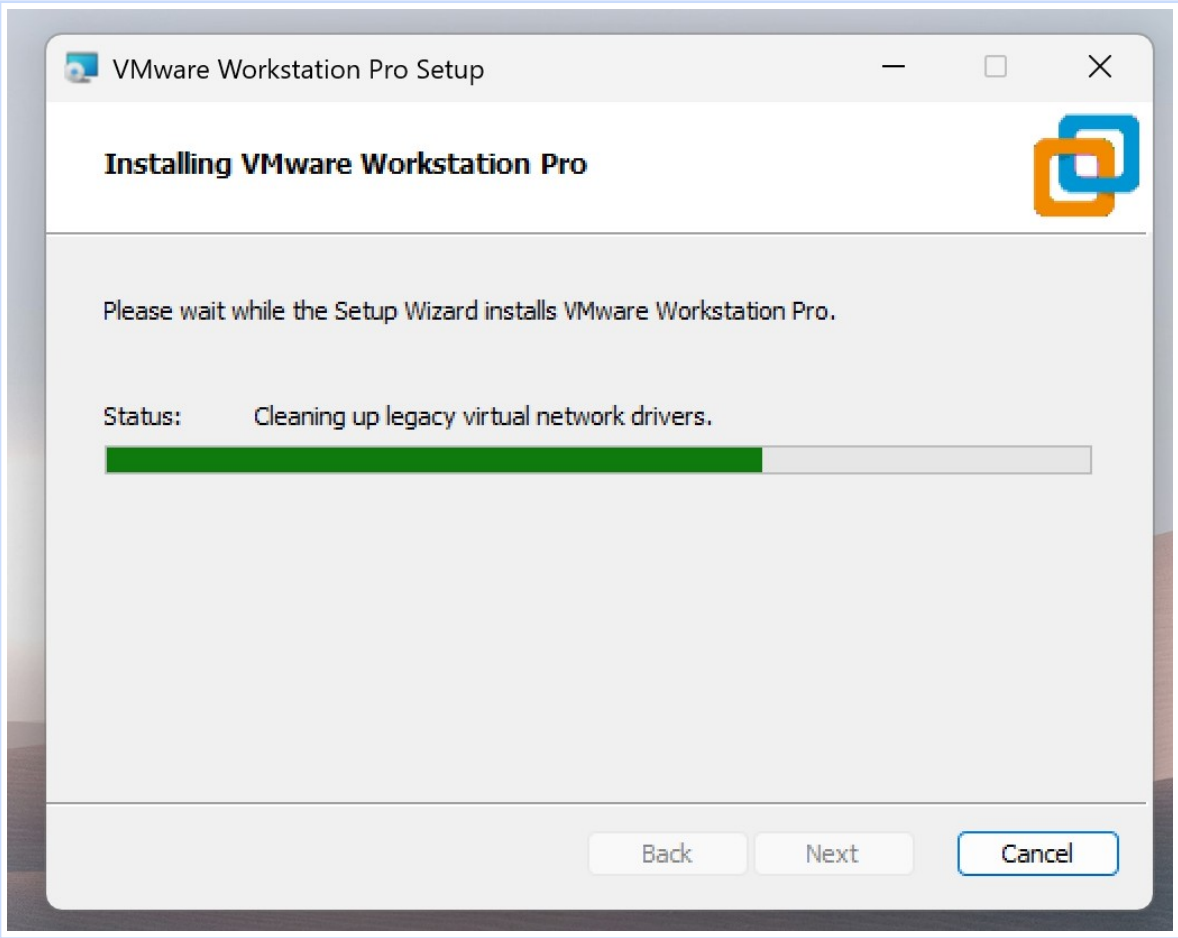
2. Press Next.

Ready to install



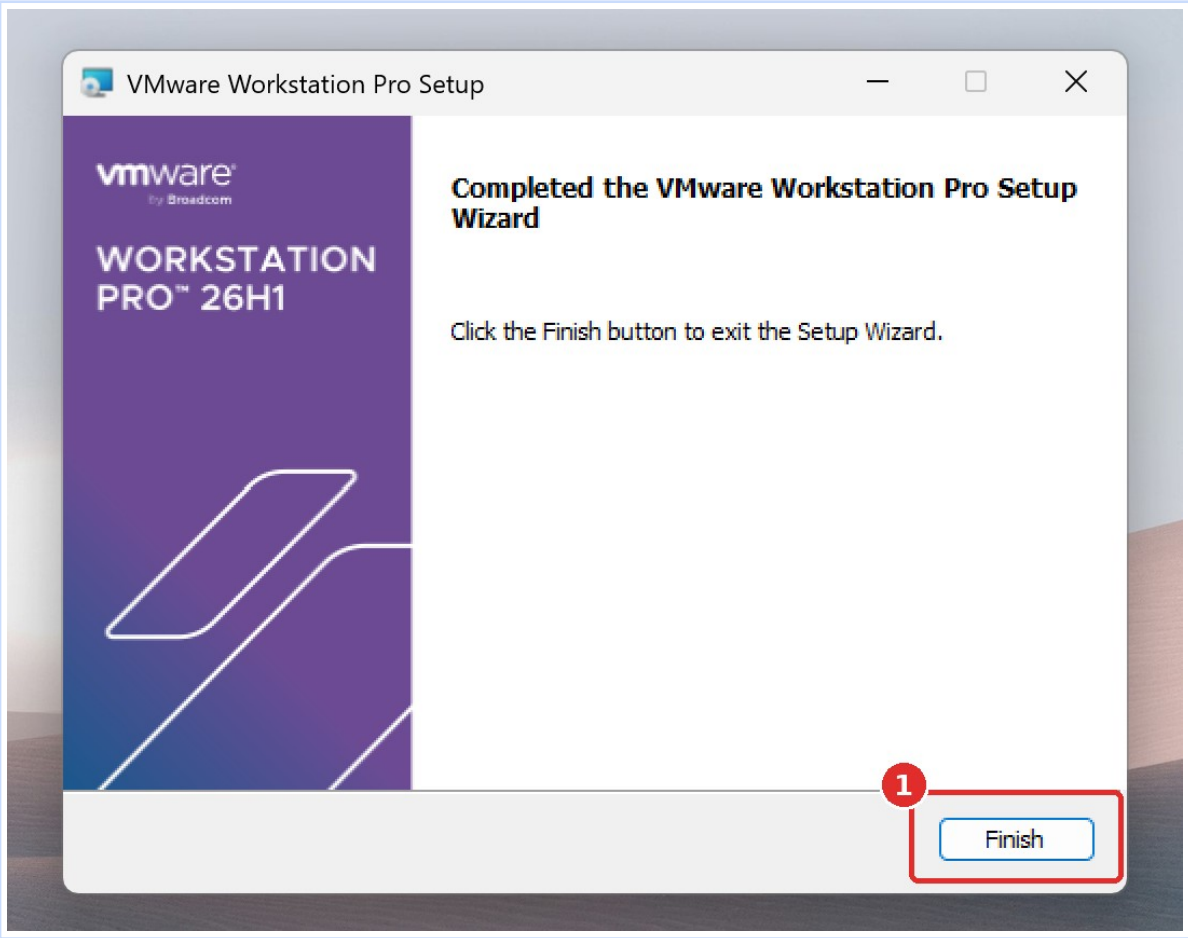
1. Review the choices and press Install. Press Back if you want to change anything.

Installing



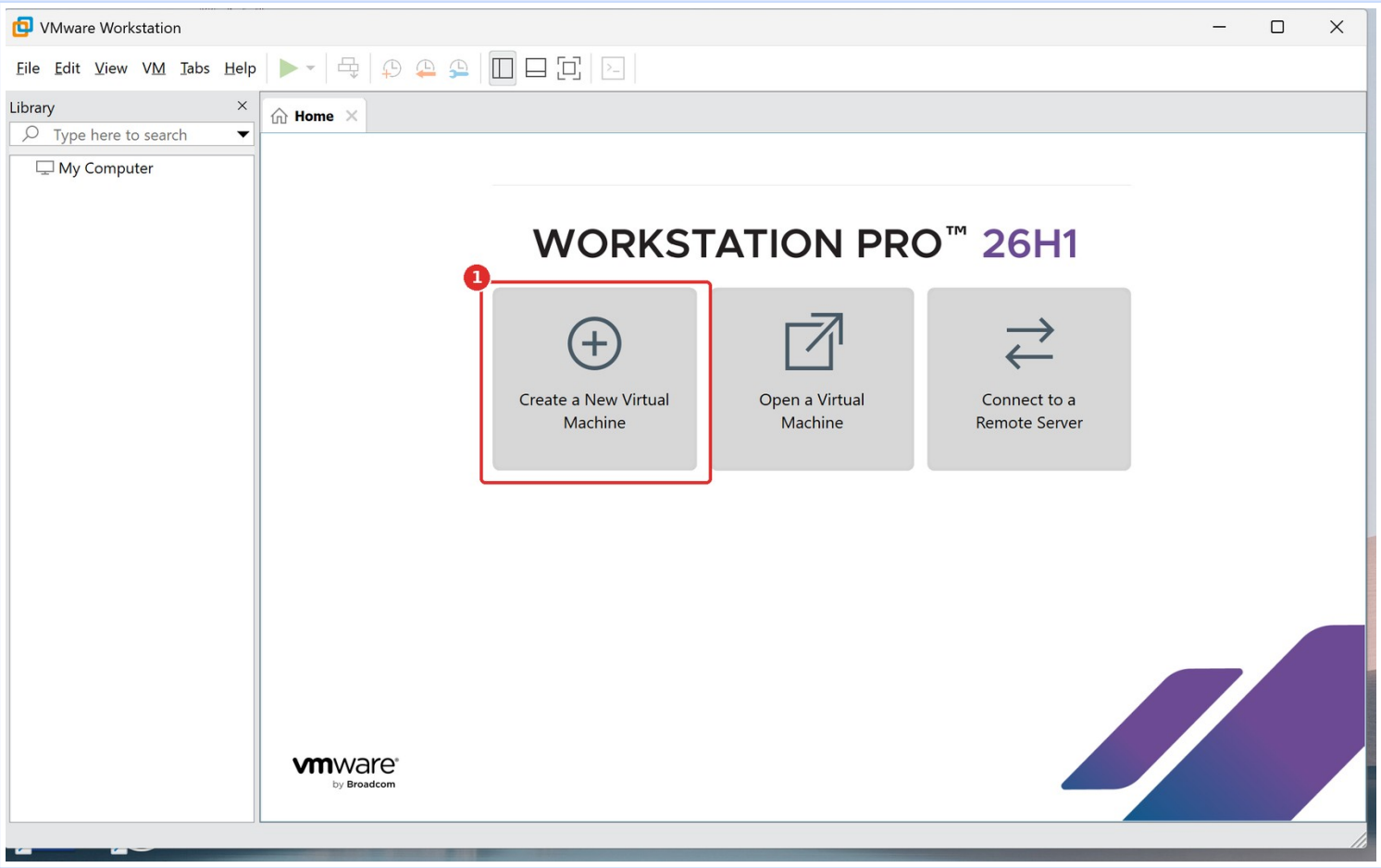
- The wizard copies files and installs virtual network drivers. Windows may ask permission; allow it. This takes a few minutes.

Setup finished



1. Press Finish. VMware Workstation Pro is now installed.

First launch



1. Open VMware Workstation from the desktop shortcut. The Home tab shows three buttons. “Create a New Virtual Machine” starts Part 4.

Checkpoint

- ✓ VMware Workstation Pro opens without errors
- ✓ Home tab shows “Create a New Virtual Machine”
- ✓ Shortcut available on the desktop

Next: Part 4 — Create the virtual machine

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Part 4 of 6

Create the Virtual Machine

Name it, size it, and choose the virtual hardware for the Ubuntu Server guest

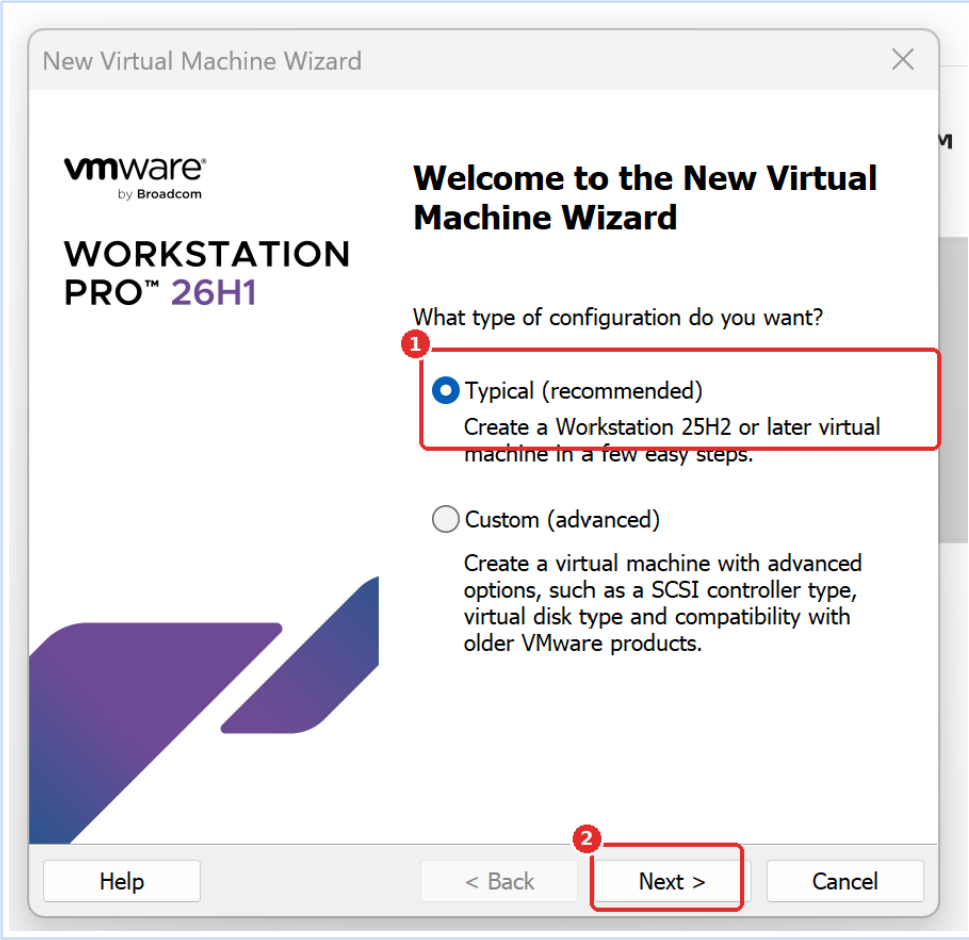
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PART 4 OF 6

Our virtual machine plan

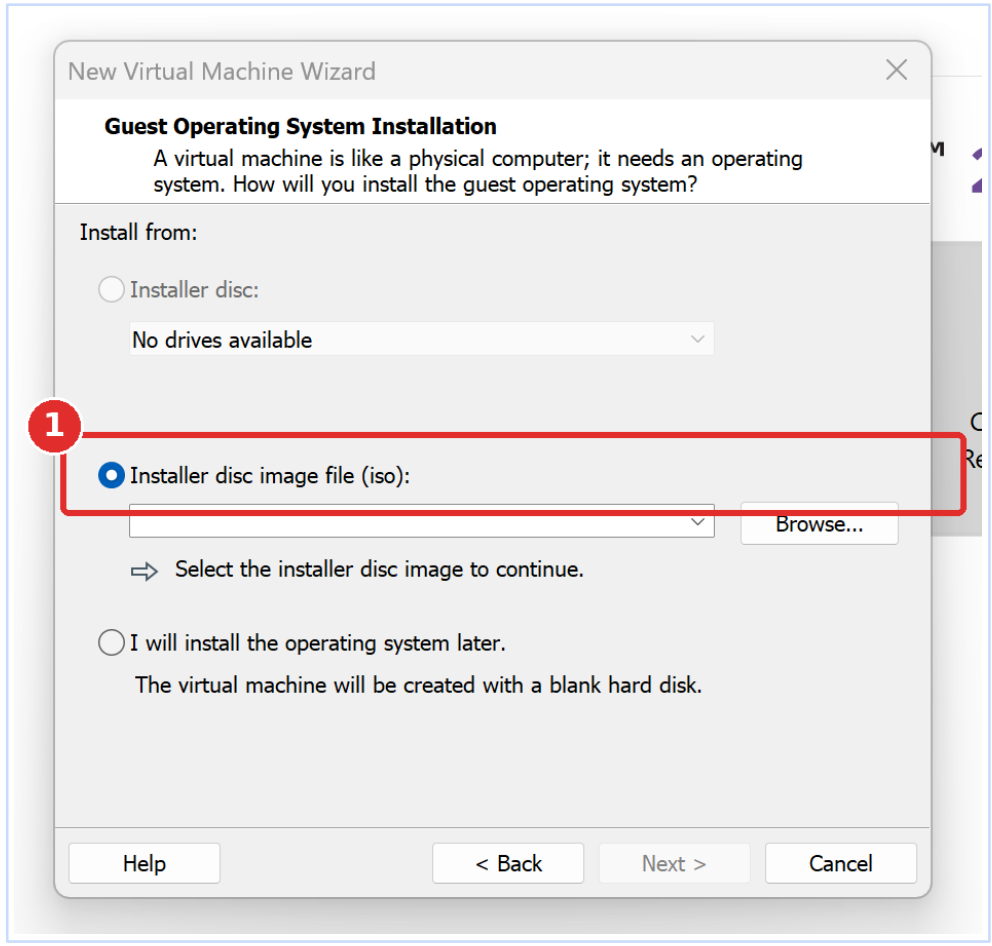
- Name: ubuntu-lab, stored in C:\VMs\ubuntu-lab
- Disk: 20 GB, split into several files
- Memory: 4096 MB (4 GB). VMware suggests more, but our PC has to share its RAM.
- Processors: 2 virtual CPUs with 1 core each
- Network: NAT. 3D graphics acceleration: off.

Start the New Virtual Machine Wizard



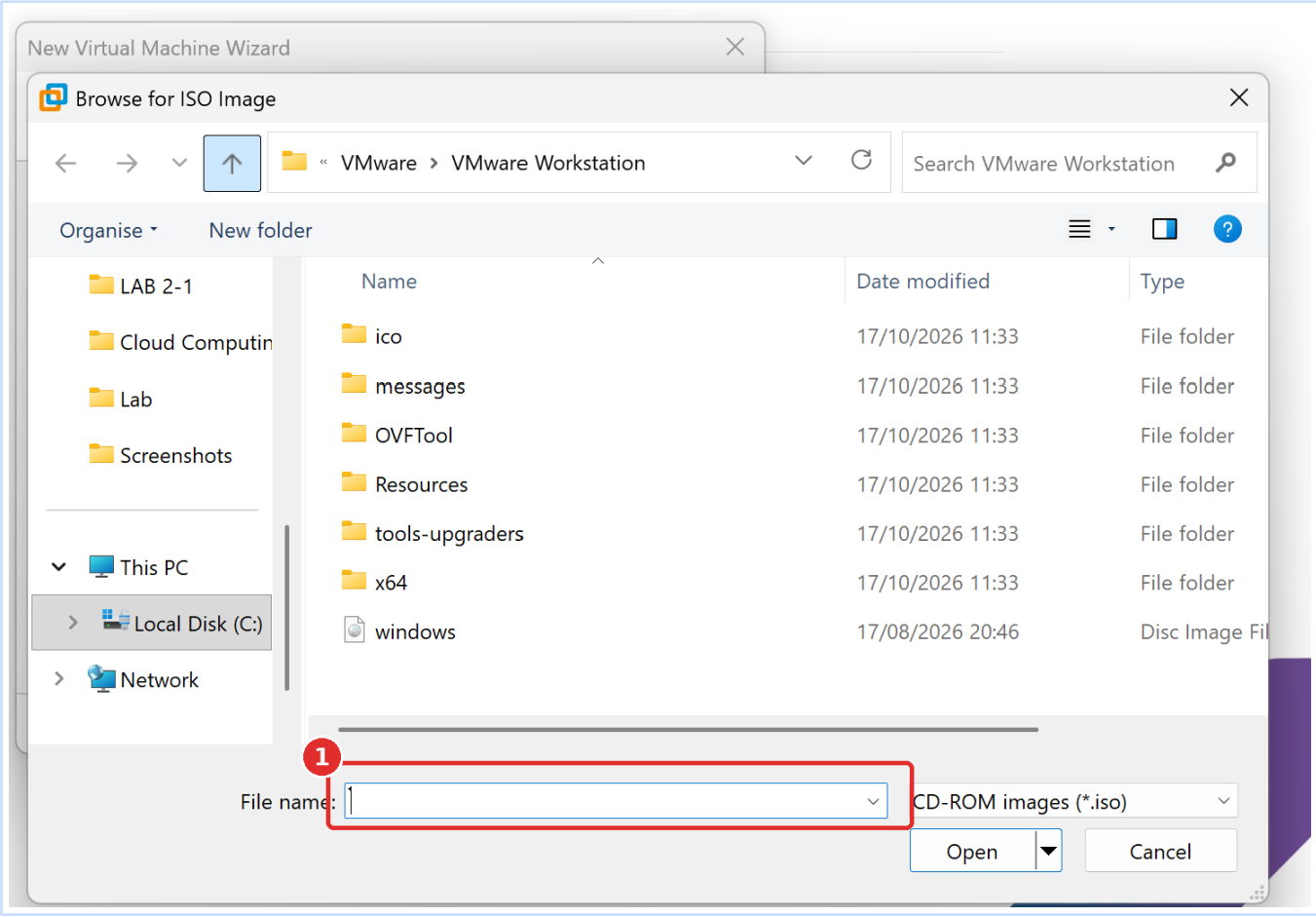
1. Choose “Typical (recommended)”.
2. Press Next.

Choose how to install the guest OS



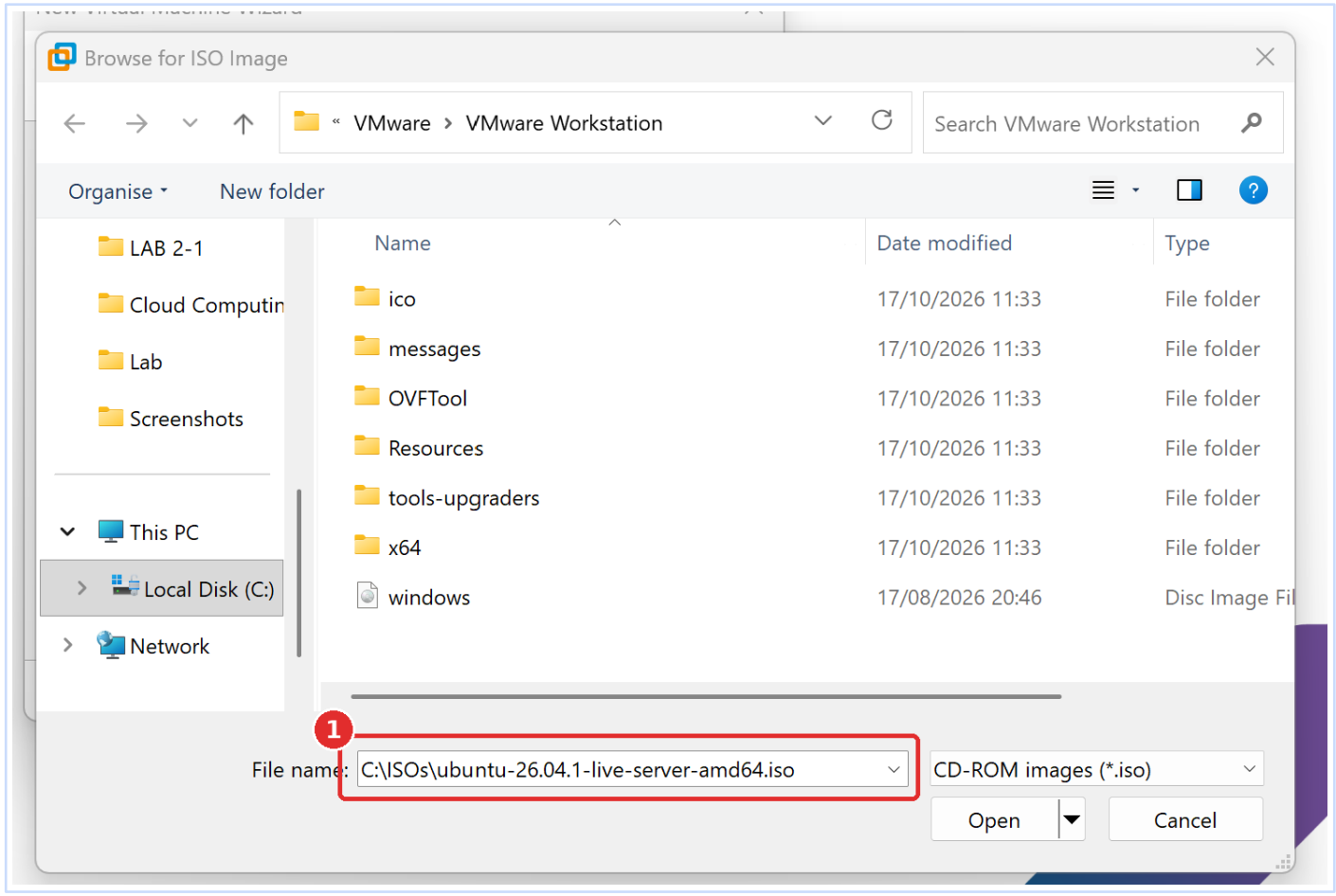
1. Select “Installer disc image file (iso)”.

Browse to the ISO



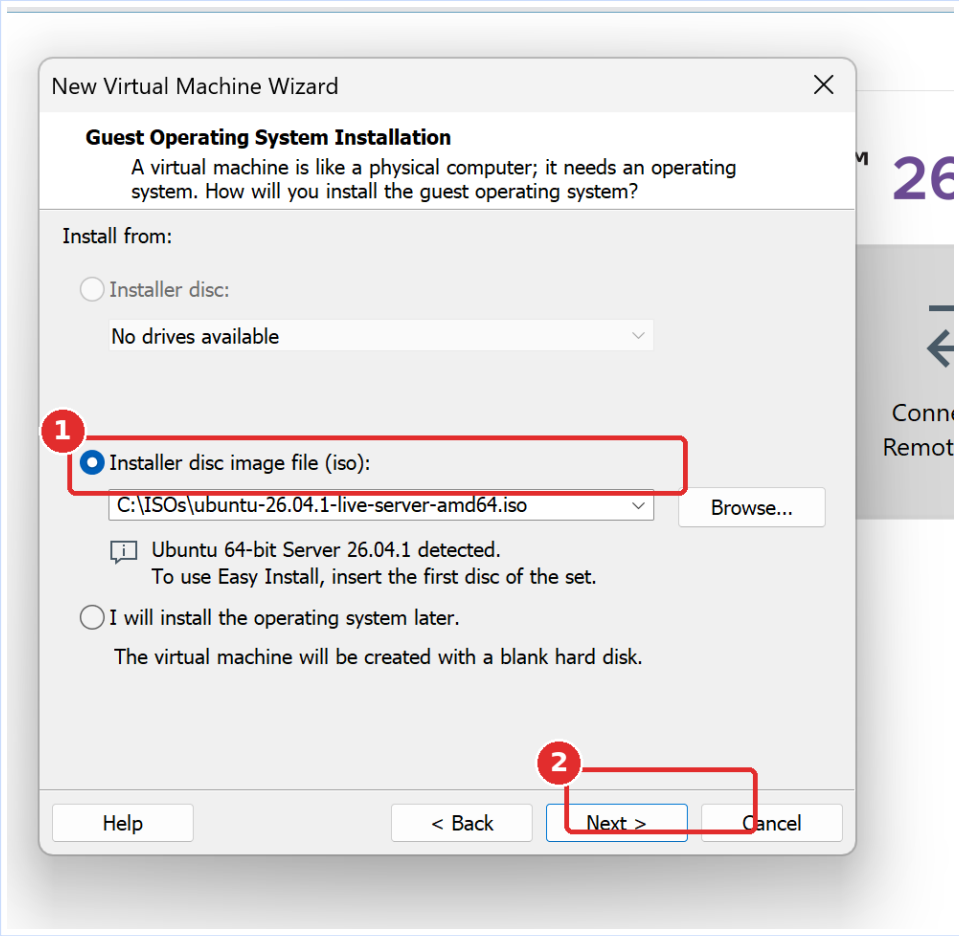
1. Press Browse and open C:\ISOs, then choose ubuntu-26.04.1-live-server-amd64.iso. You can also type the path in the File name box.

The path is filled in



1 The File name box shows the full path of the ISO. Press Open.

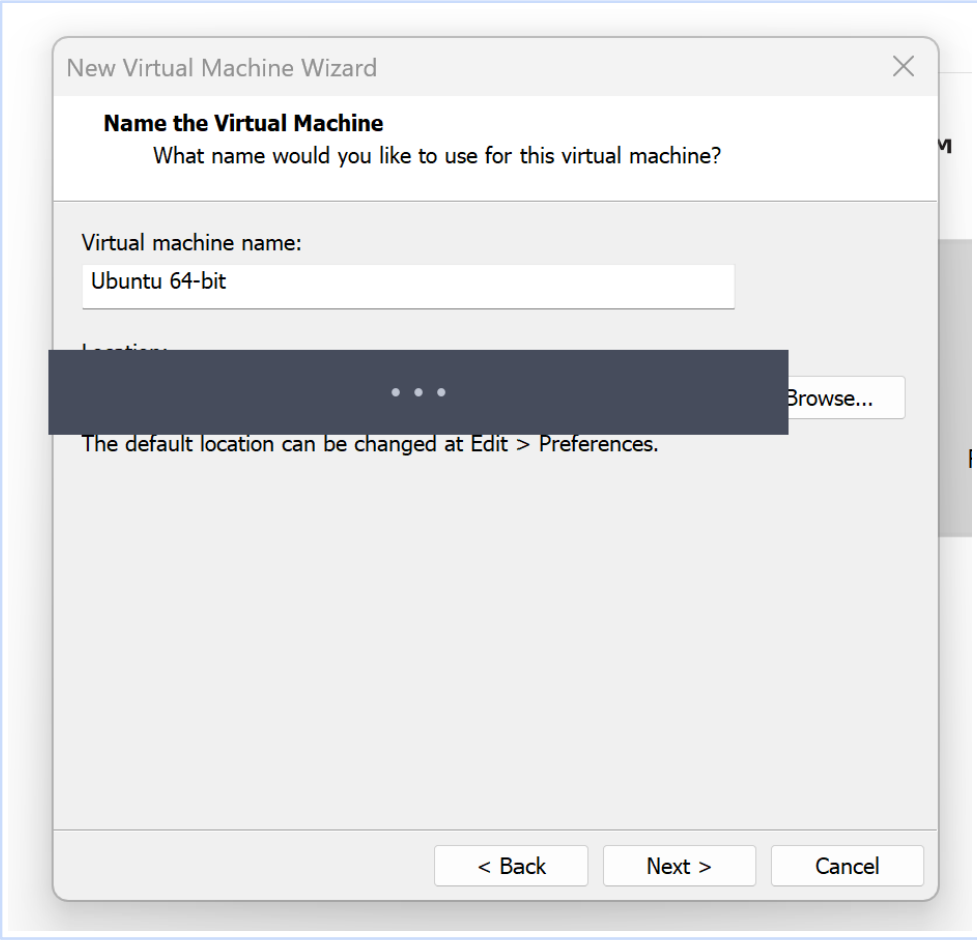
VMware detects the guest OS



1. VMware detected “Ubuntu 64-bit Server 26.04.1”. It cannot use Easy Install with this ISO, so we install Ubuntu manually in Part 5.

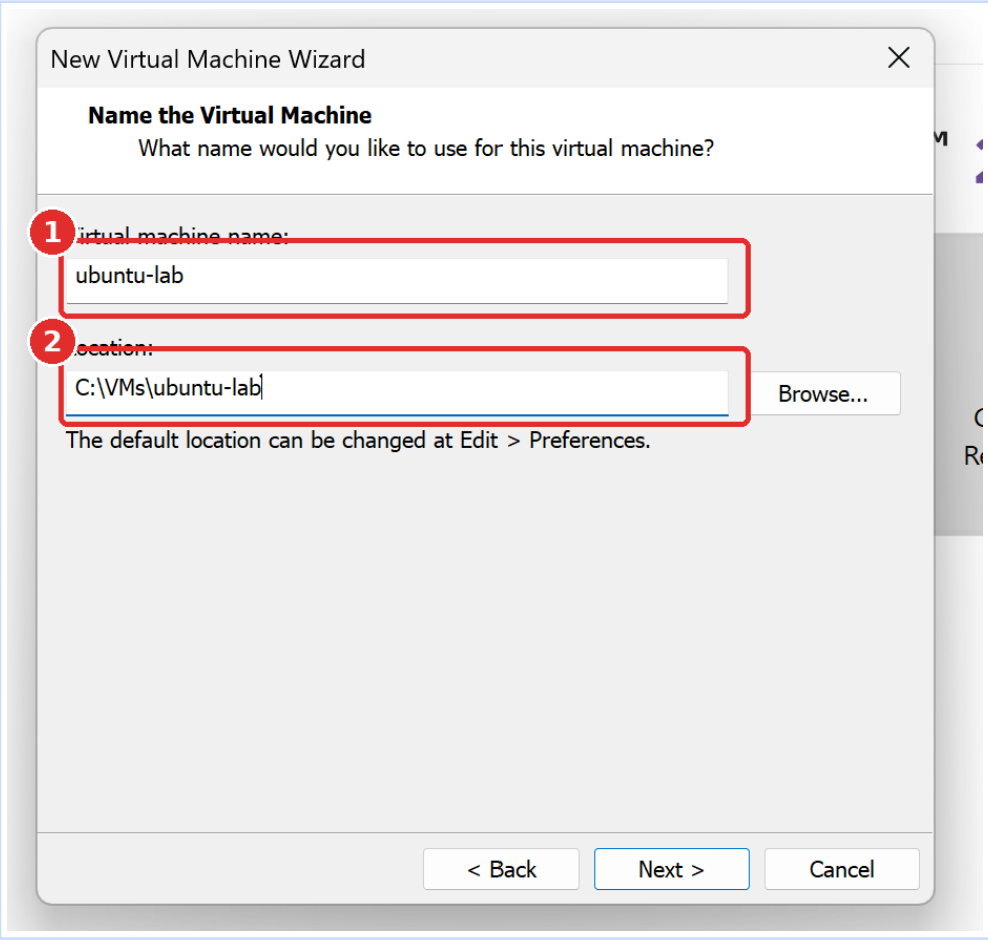
2. Press Next.

Default name and location



- By default VMware names the VM “Ubuntu 64-bit” and stores it under your Documents folder, which includes your Windows user name (hidden here). We change both.

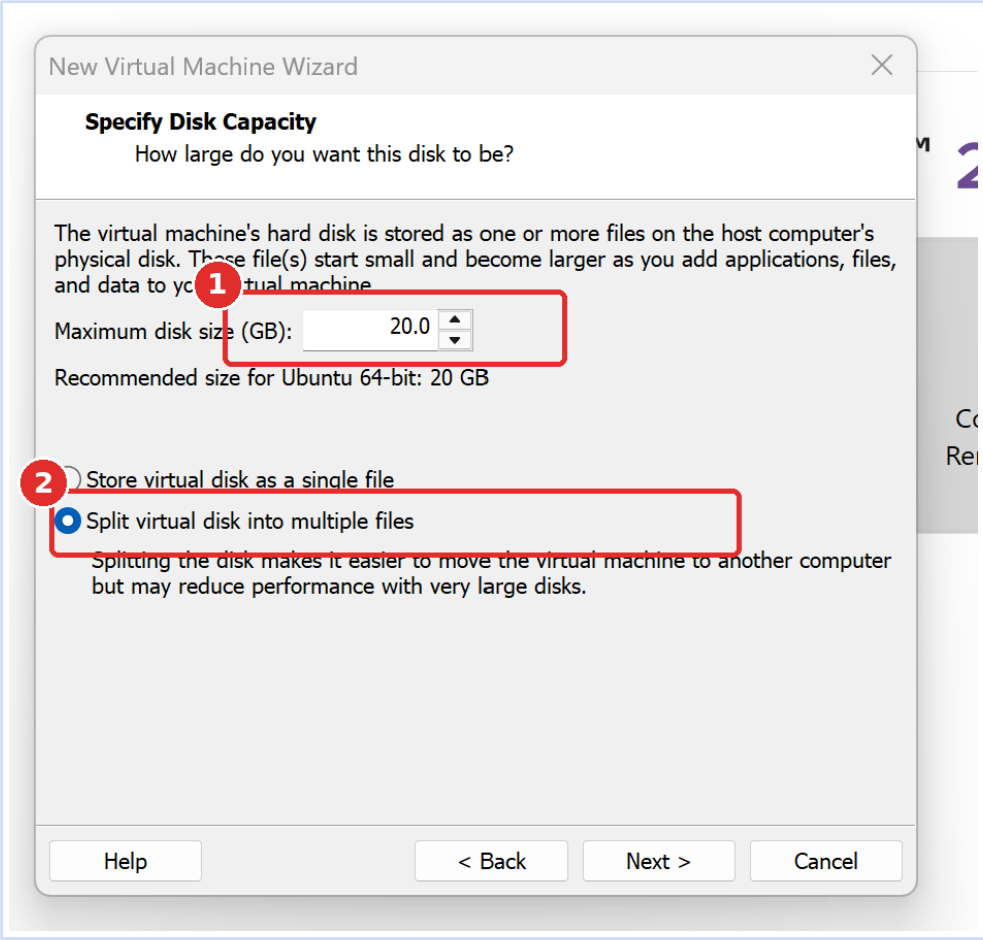
Name it and choose a short location



1. Virtual machine name: ubuntu-lab

2. Location: C:\VMs\ubuntu-lab. VM files are large, so use a drive with free space.

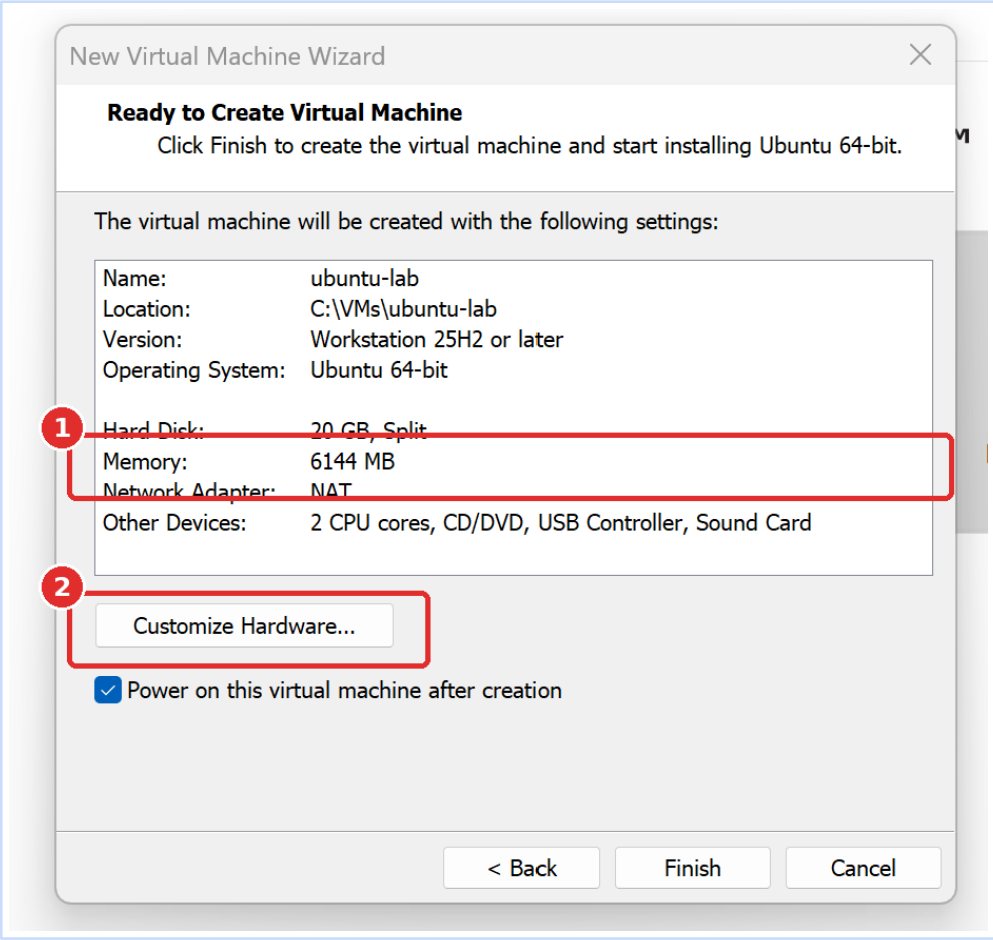
Disk capacity



1. Maximum disk size: 20 GB. The file starts small and grows as needed.

2. Choose “Split virtual disk into multiple files”. It makes the VM easier to copy and move.

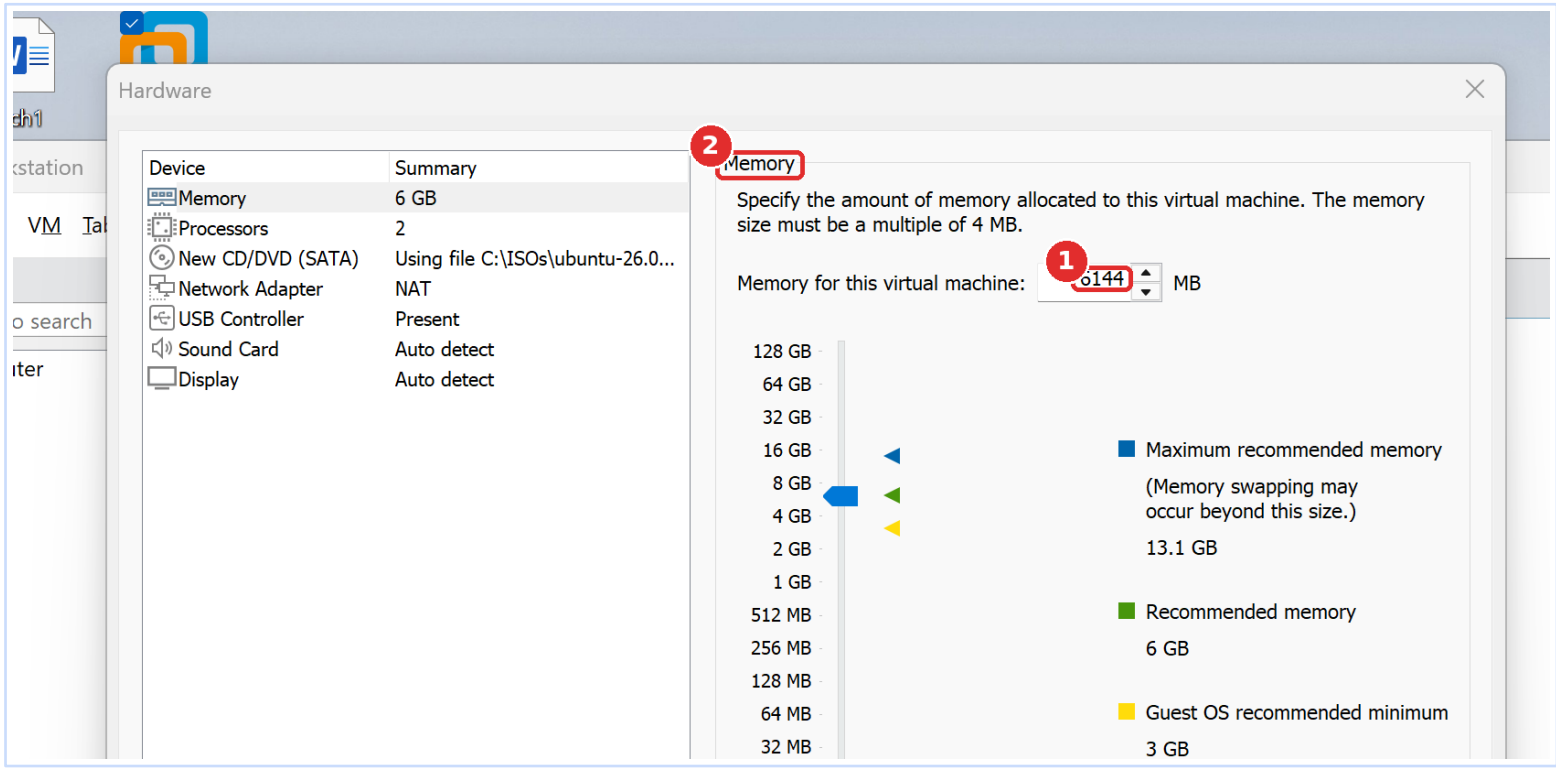
Ready to create



1. The summary lists the defaults. VMware suggests 6144 MB memory, which we will lower.

2. Press Customize Hardware.

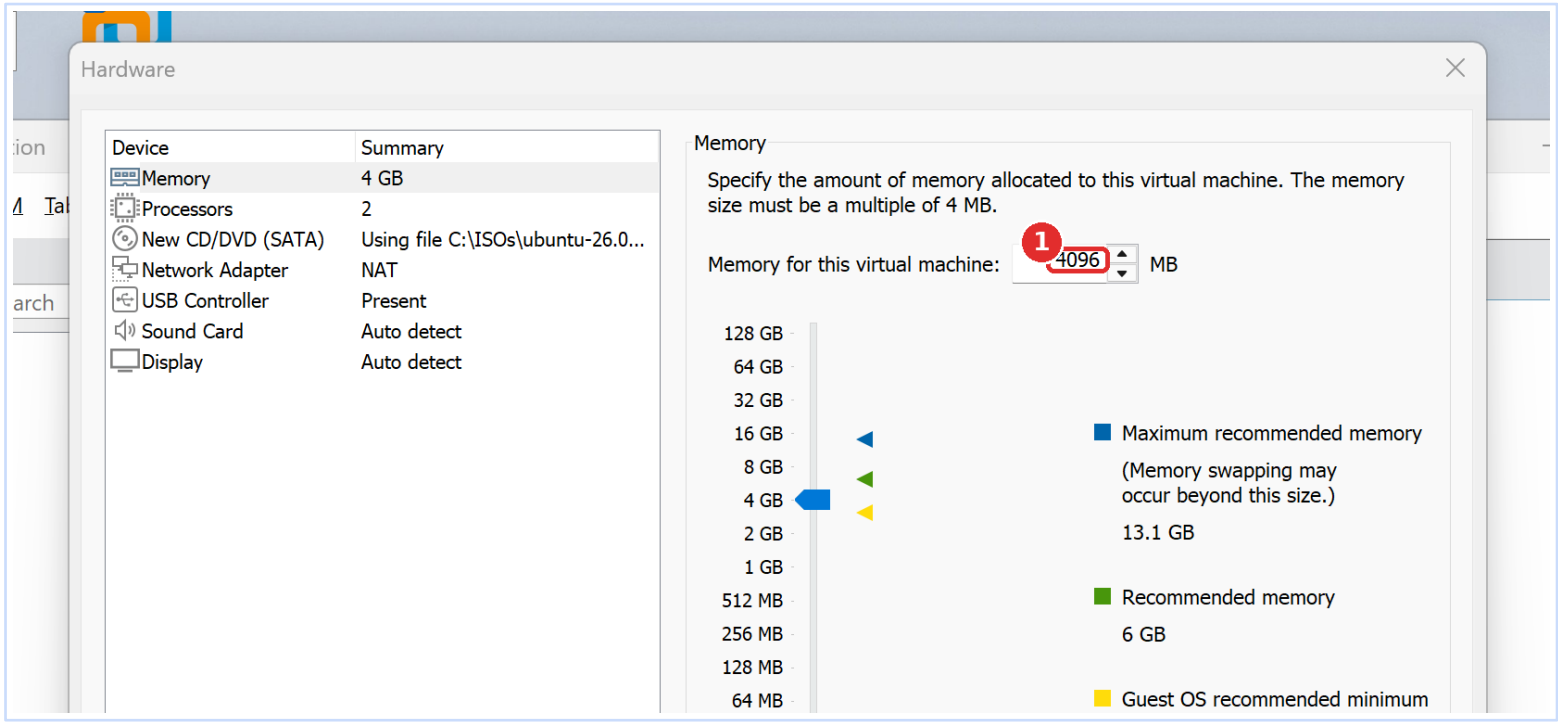
Hardware: memory



1. The default memory is 6144 MB (6 GB).

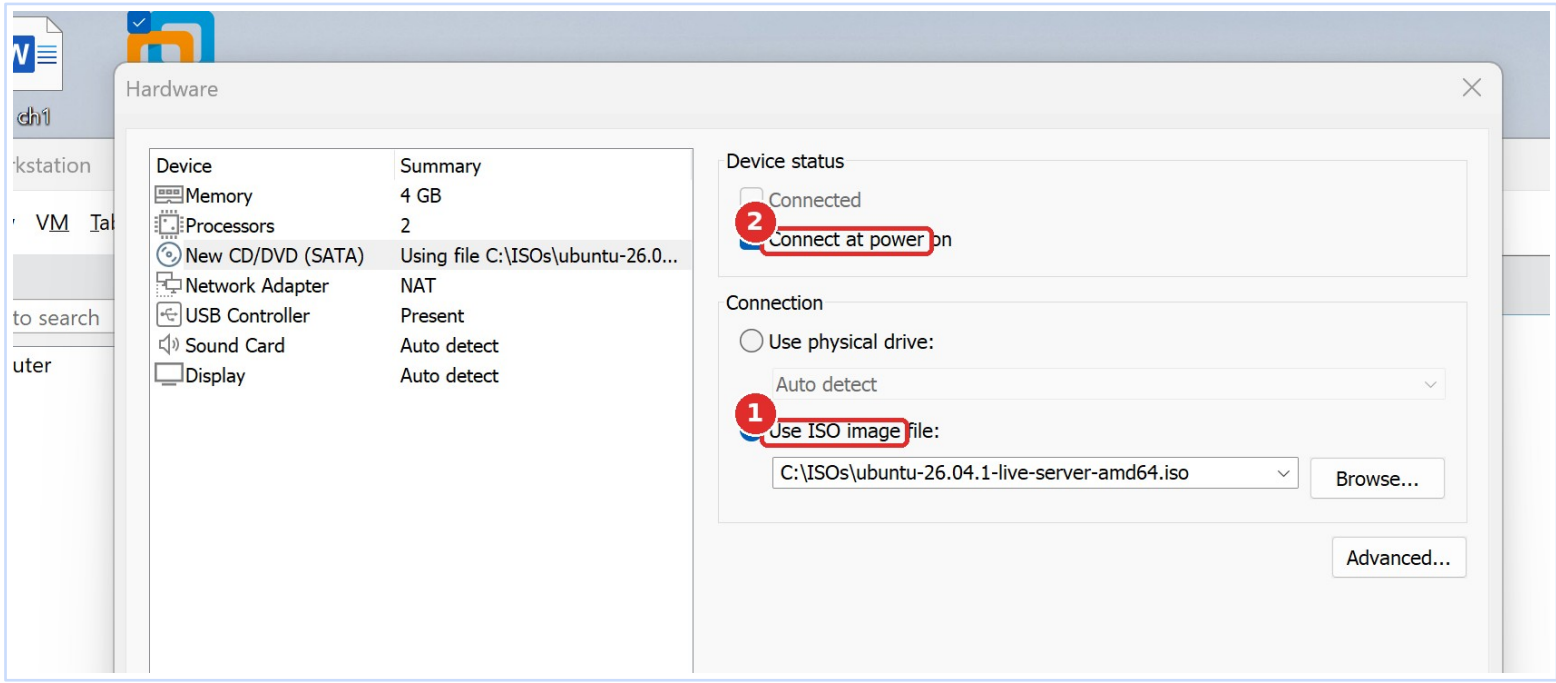
2. Select Memory in the device list on the left.

Set memory to 4096 MB



1. Type 4096. A server guest runs well in 4 GB and leaves enough RAM for Windows.

Hardware: CD/DVD

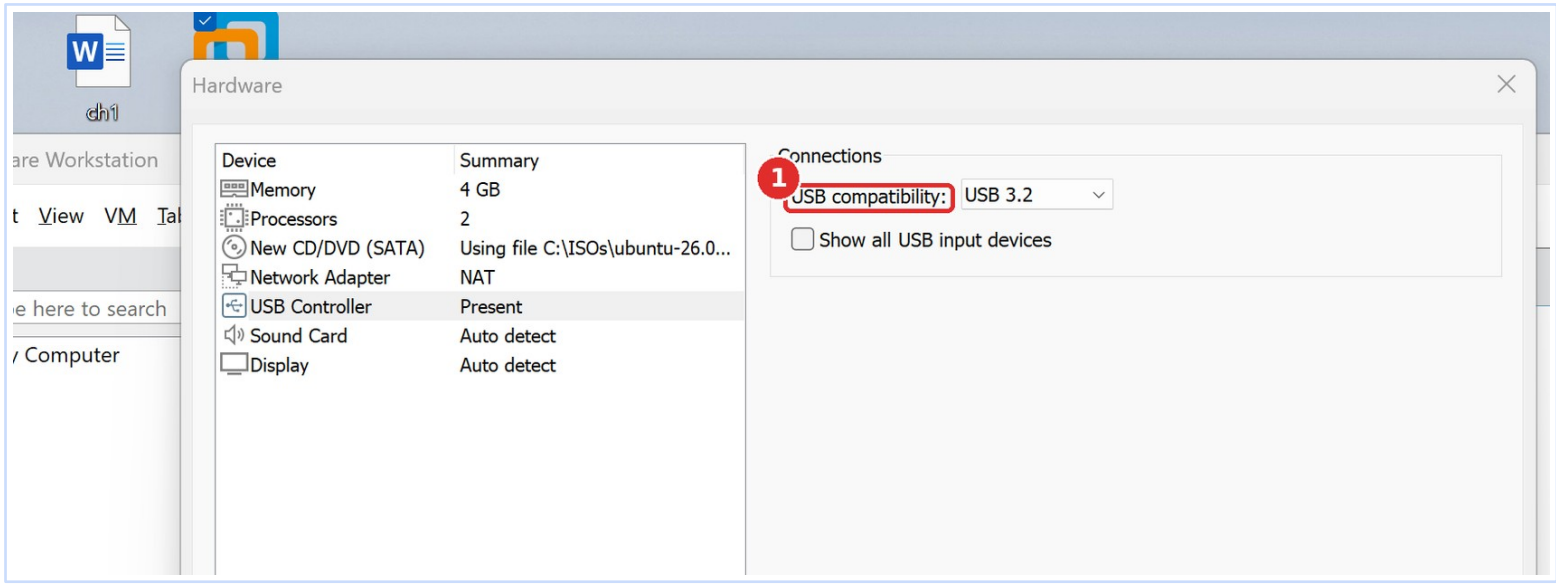


1. “Use ISO image file” points to the Ubuntu ISO.

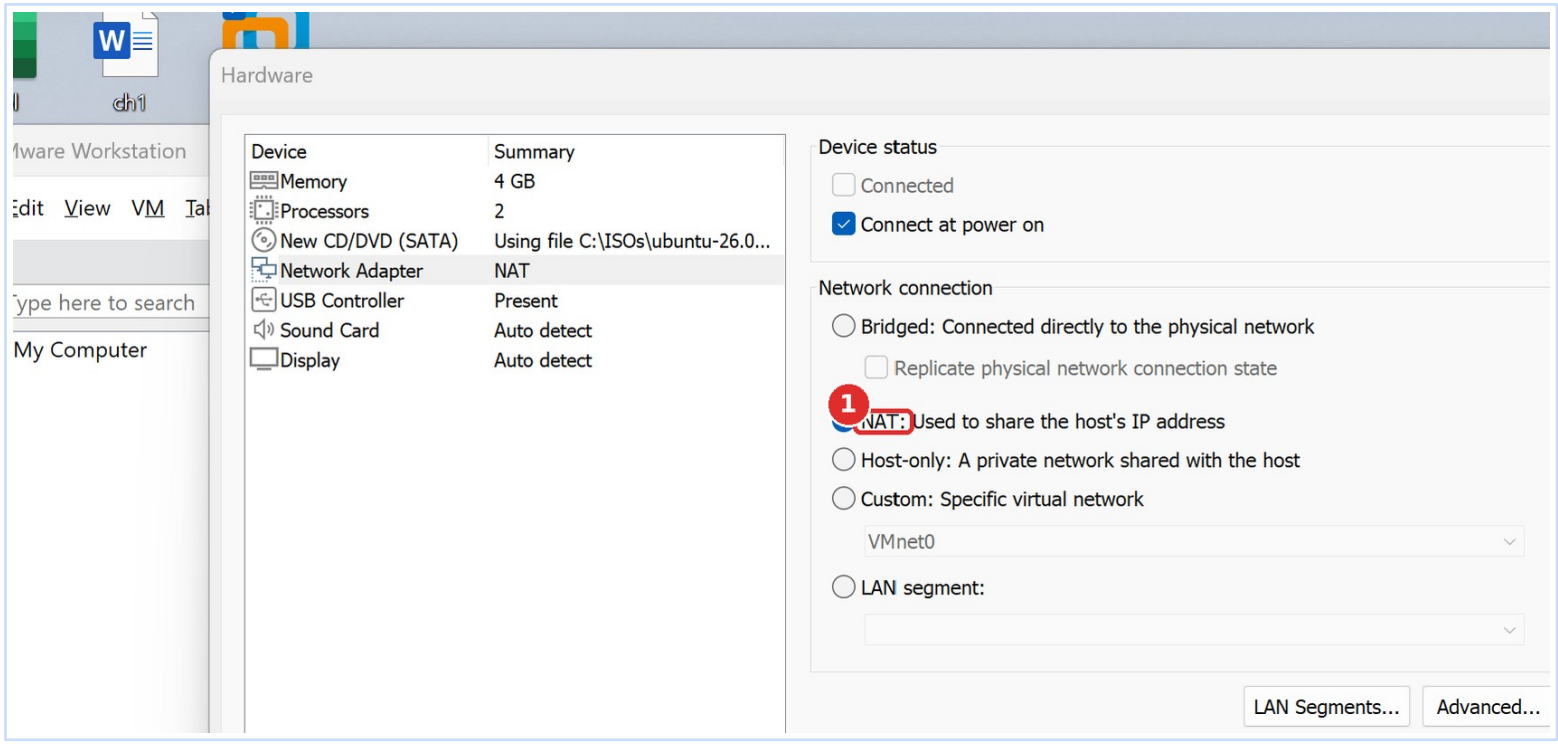
2. “Connect at power on” must be ticked so the VM boots from the ISO.

Hardware: USB

1 Keep USB compatibility on USB 3.2 (default).

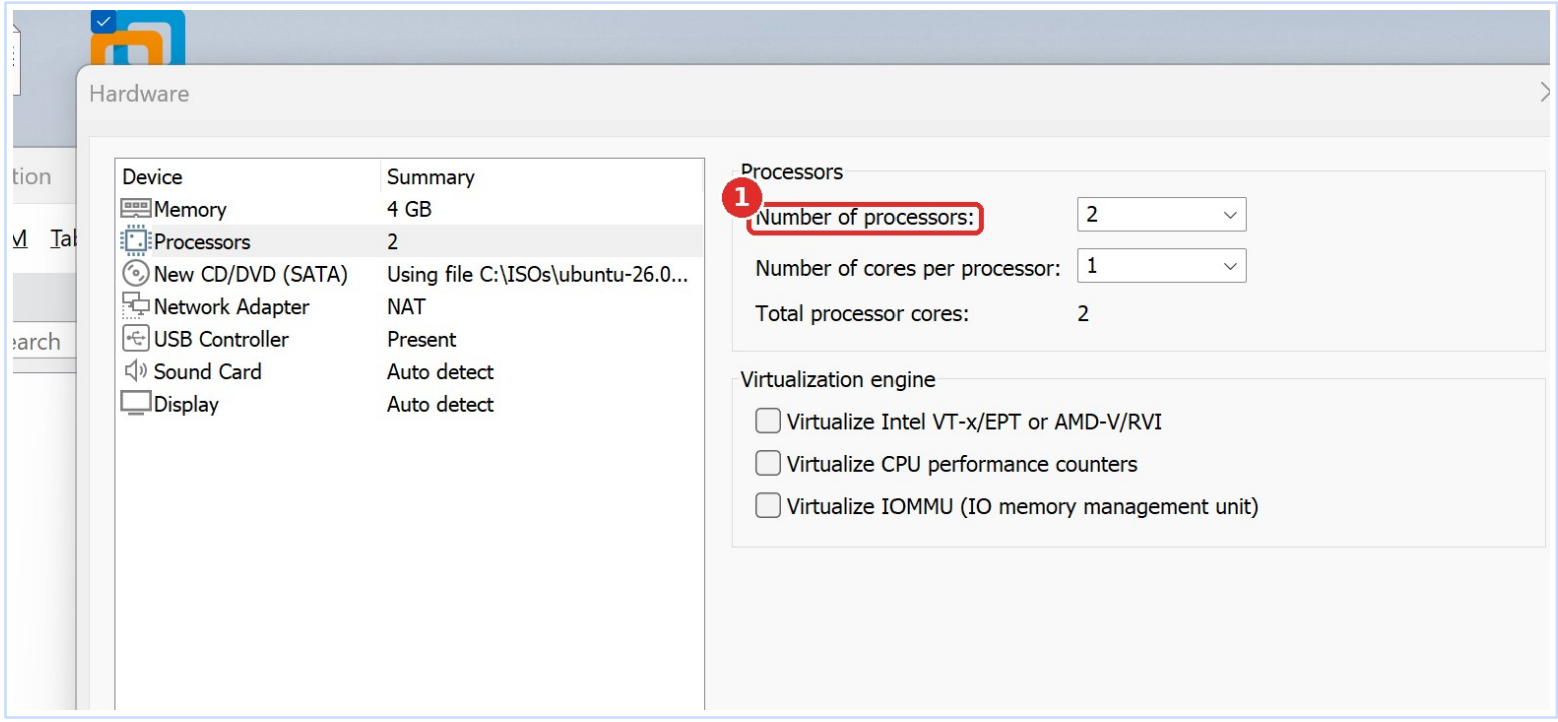


Hardware: network



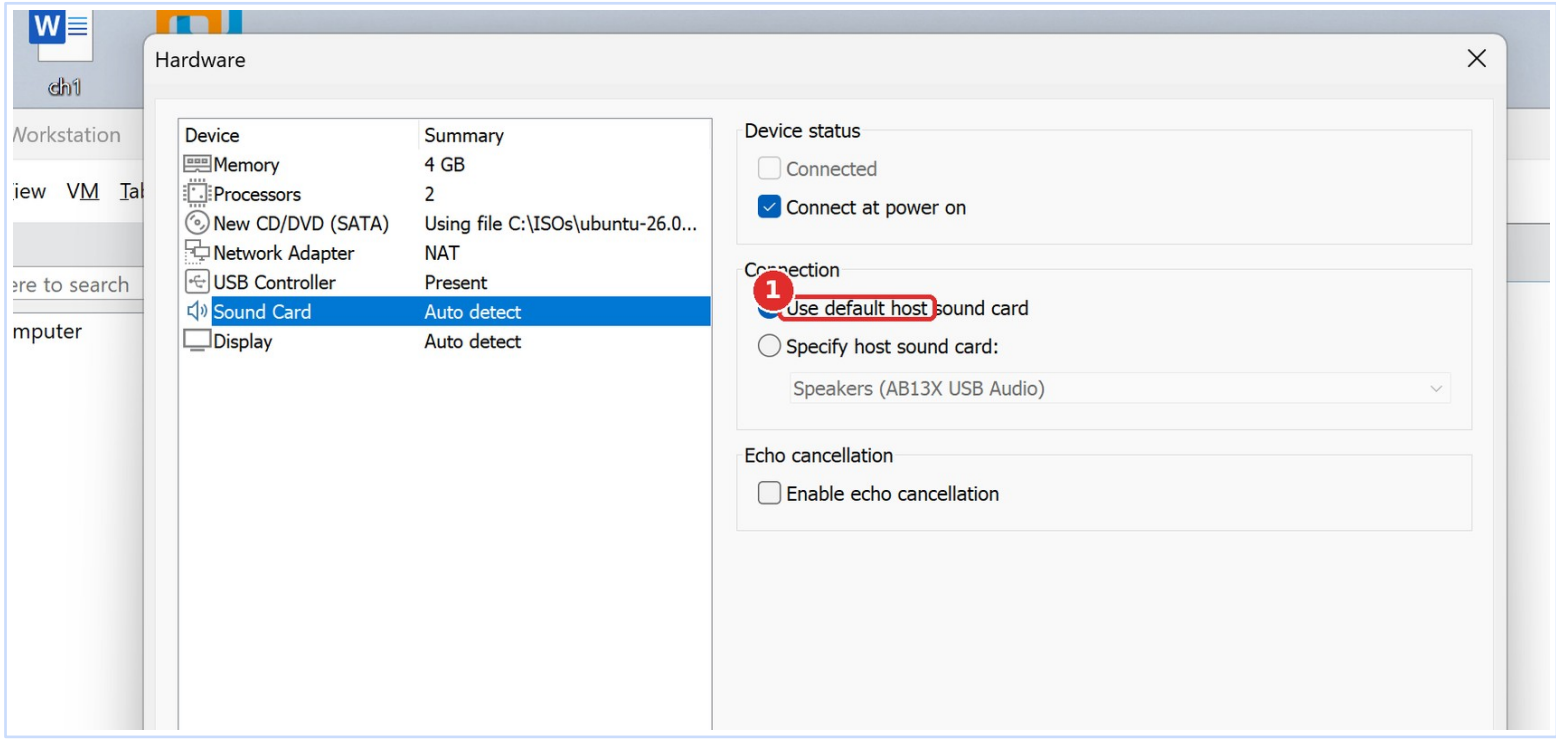
1 Keep NAT. The VM shares the host's internet connection and has a private address. This is the safest default for a lab.

Hardware: processors



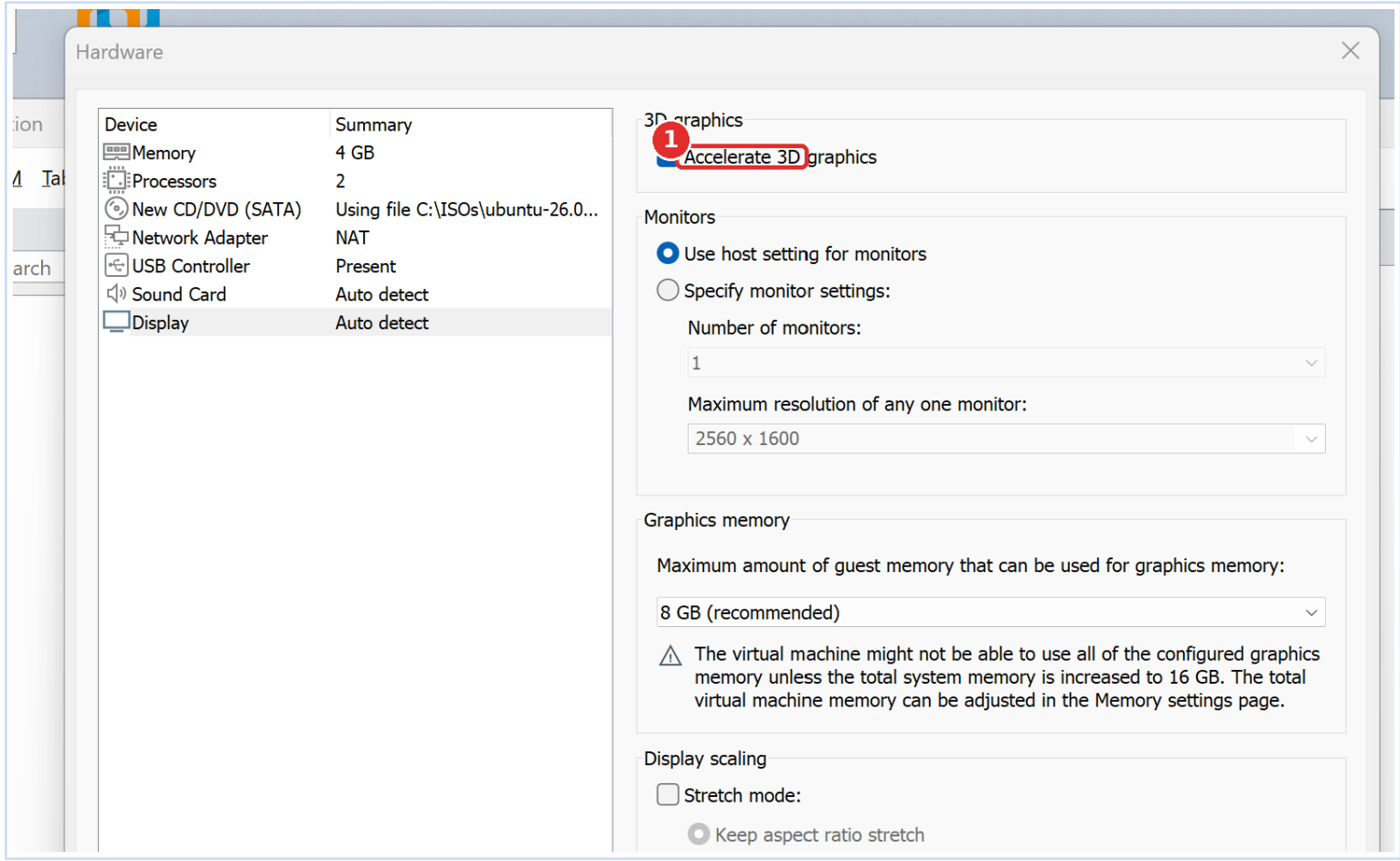
1 Use 2 processors with 1 core each (2 virtual CPUs).

Hardware: sound card



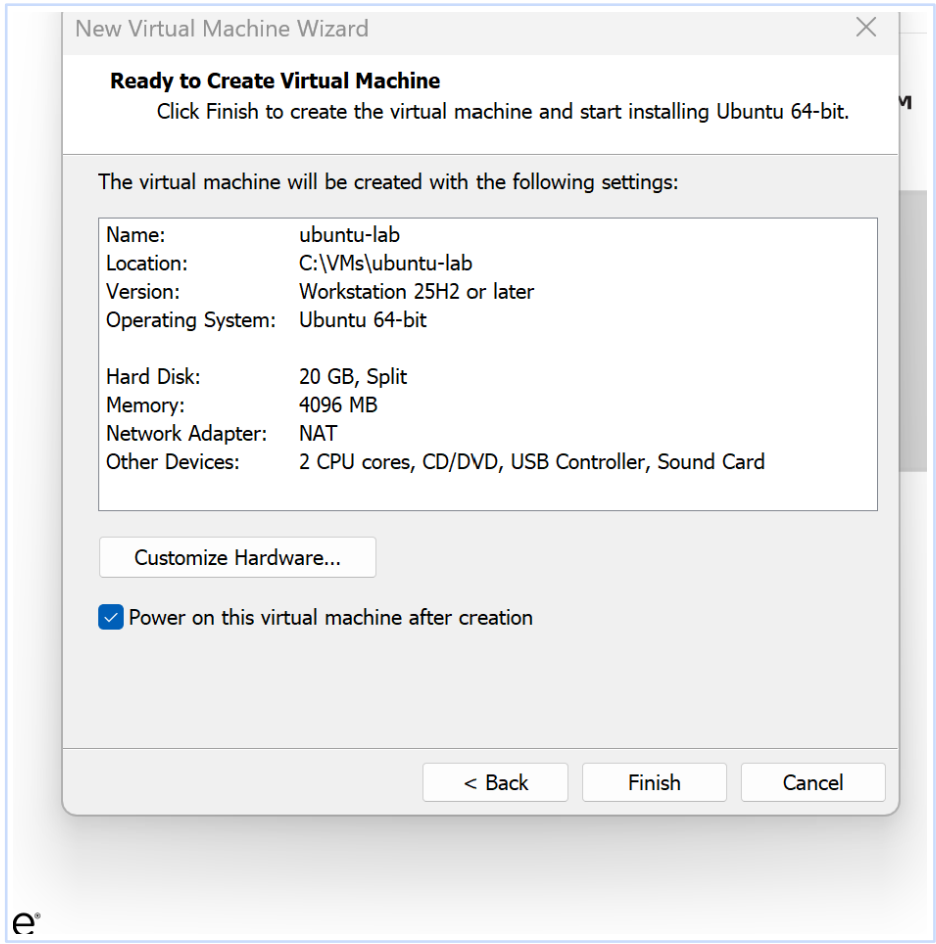
1 The sound card is not needed for a server. Leave the default; it does no harm.

Hardware: display



1 Make sure “Accelerate 3D graphics” is off. A text-only server does not need it.

Final summary



- Check: name ubuntu-lab, location C:\VMs\ubuntu-lab, 20 GB split disk, memory 4096 MB, NAT, 2 CPU cores.
- Leave “Power on this virtual machine after creation” ticked and press Finish. The VM starts and boots from the ISO.

Checkpoint

- ✓ VM named ubuntu-lab with 20 GB disk
- ✓ 4096 MB memory, 2 CPUs, NAT network
- ✓ CD/DVD connected to the Ubuntu ISO

Next: Part 5 — Install Ubuntu Server

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Part 5 of 6

Install Ubuntu Server 26.04 LTS

Follow every screen of the text-based installer, using only the keyboard

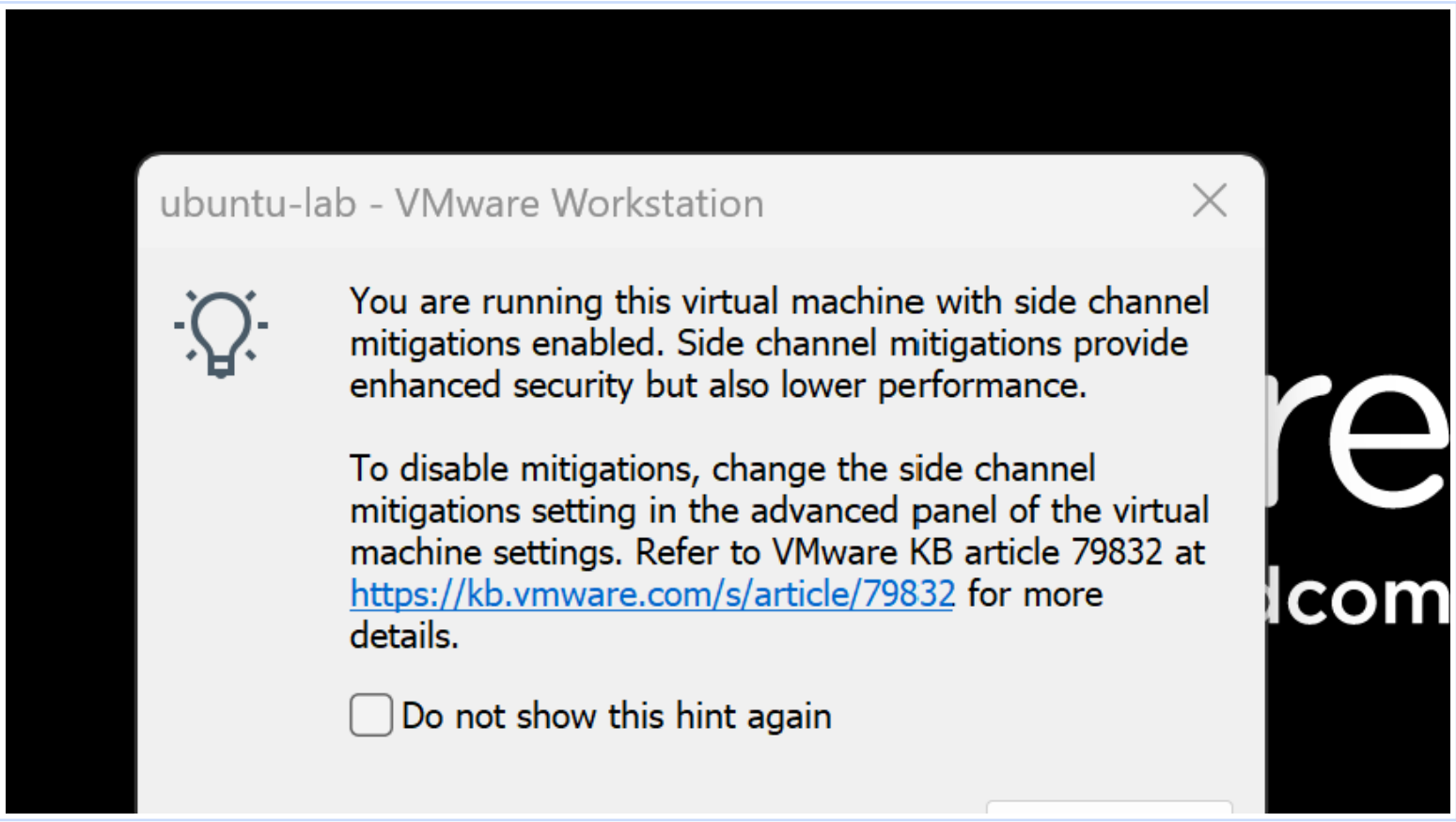
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PART 5 OF 6

How the installer works

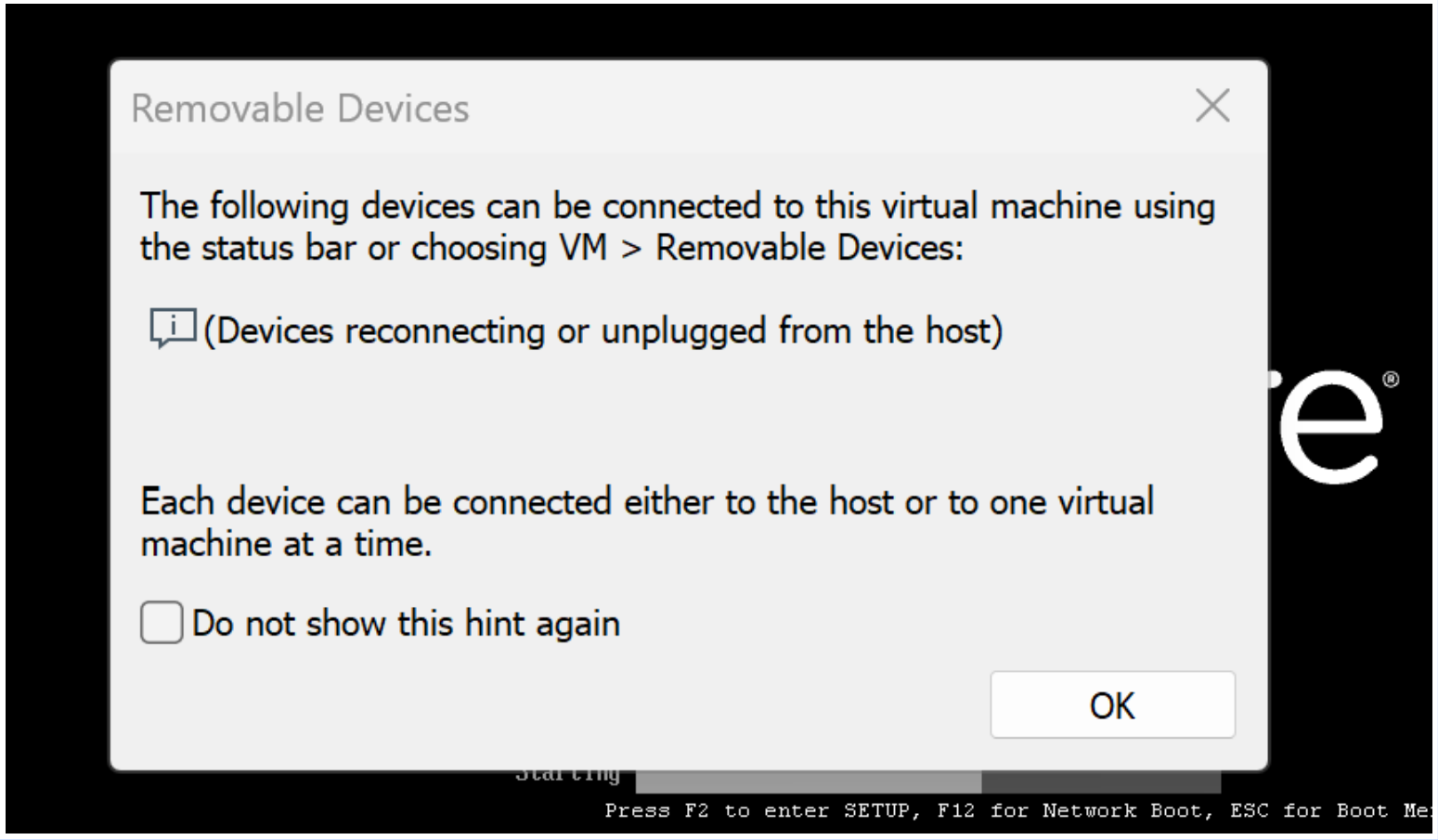
- The Ubuntu Server installer is text-based: the mouse does nothing inside it.
- Click inside the VM window once to give it your keyboard. Press Ctrl+Alt to get the mouse back.
- Use Up/Down arrows to move, Enter to select, Space to tick a box, Tab to jump between areas.
- The whole install takes about 15 minutes. The virtual disk is erased, never your Windows disk.
- The username and password you choose are temporary lab values. Never reuse a real password.

First boot: side-channel hint



- VMware shows a hint about side-channel mitigations (security protections against CPU attacks). It only informs you; it is not an error. Press OK.

Removable devices hint



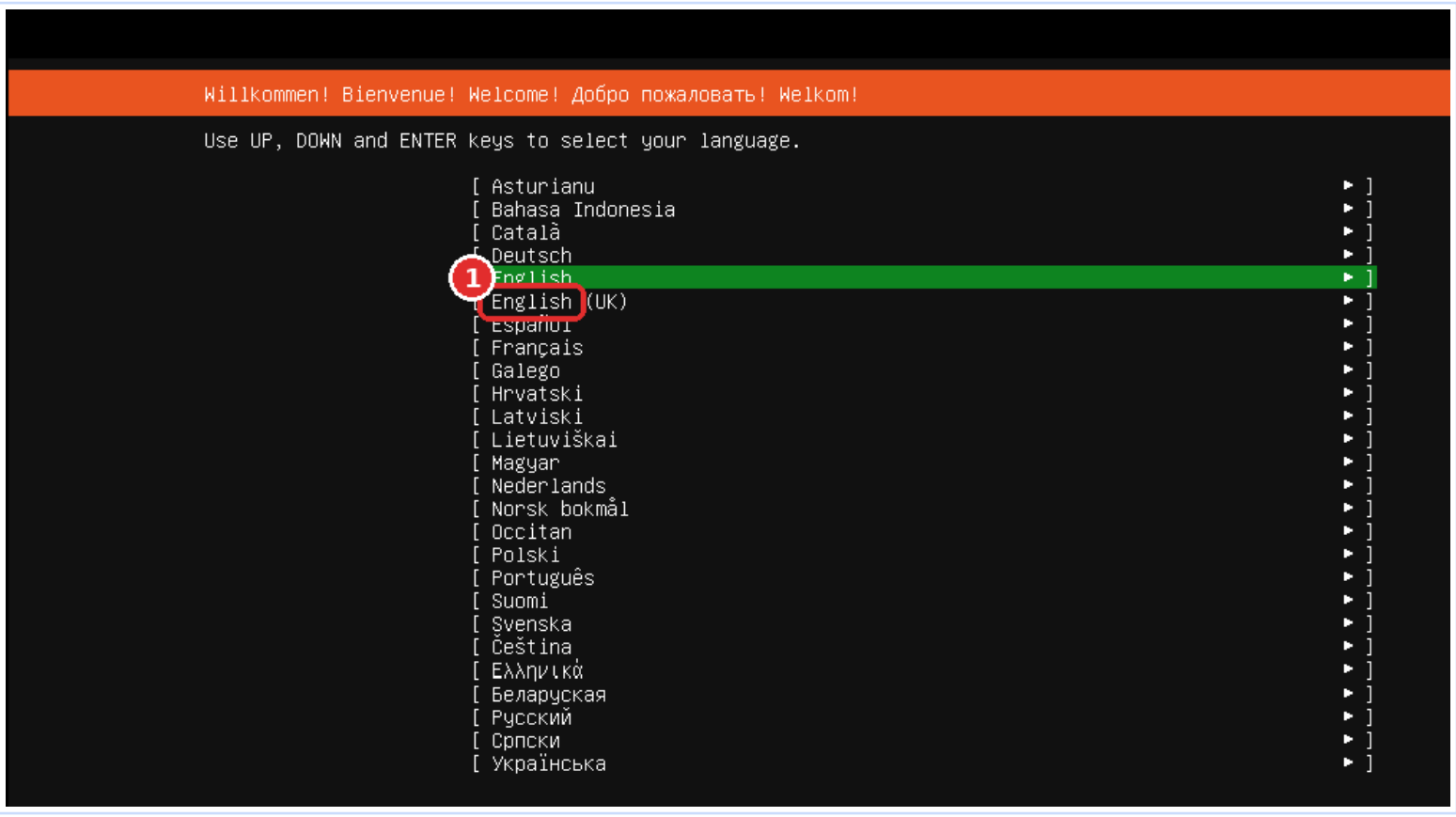
- This tells you how to attach USB or Bluetooth devices to the VM. Tick “Do not show this hint again” if you like, then press OK.

The installer boots



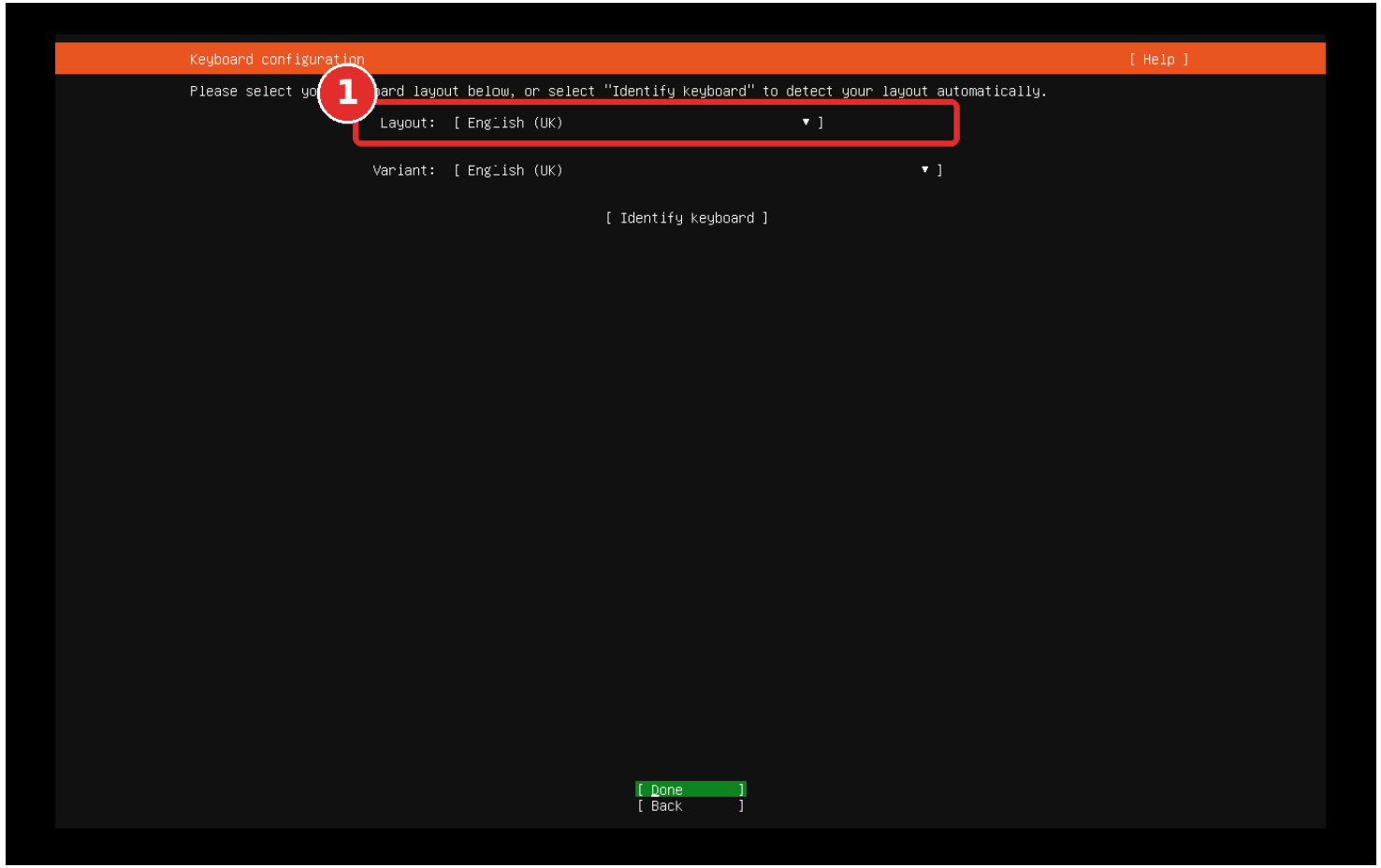
- Boot messages scroll by while the installer environment starts. Wait without pressing keys.

Language



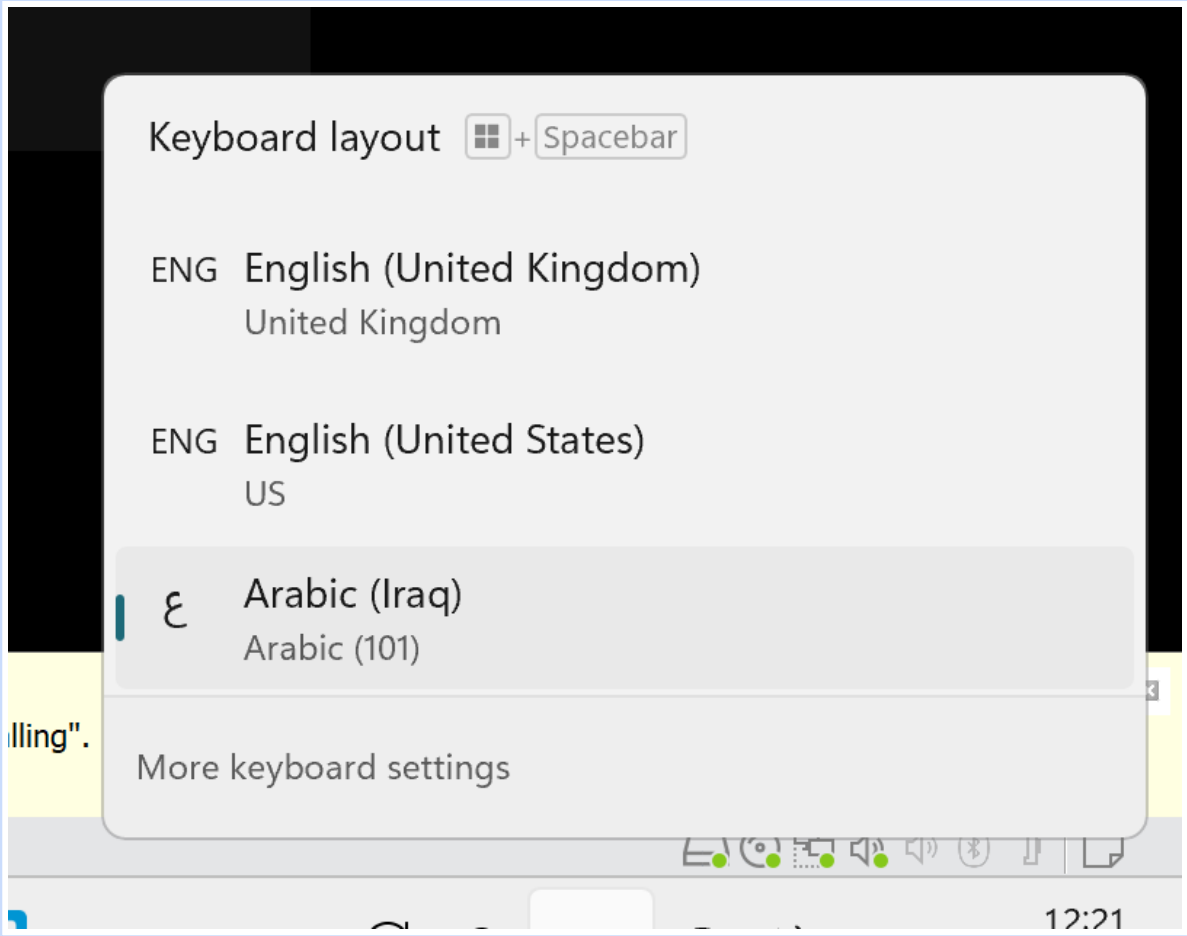
1 Select English with the arrow keys and press Enter.

Keyboard configuration



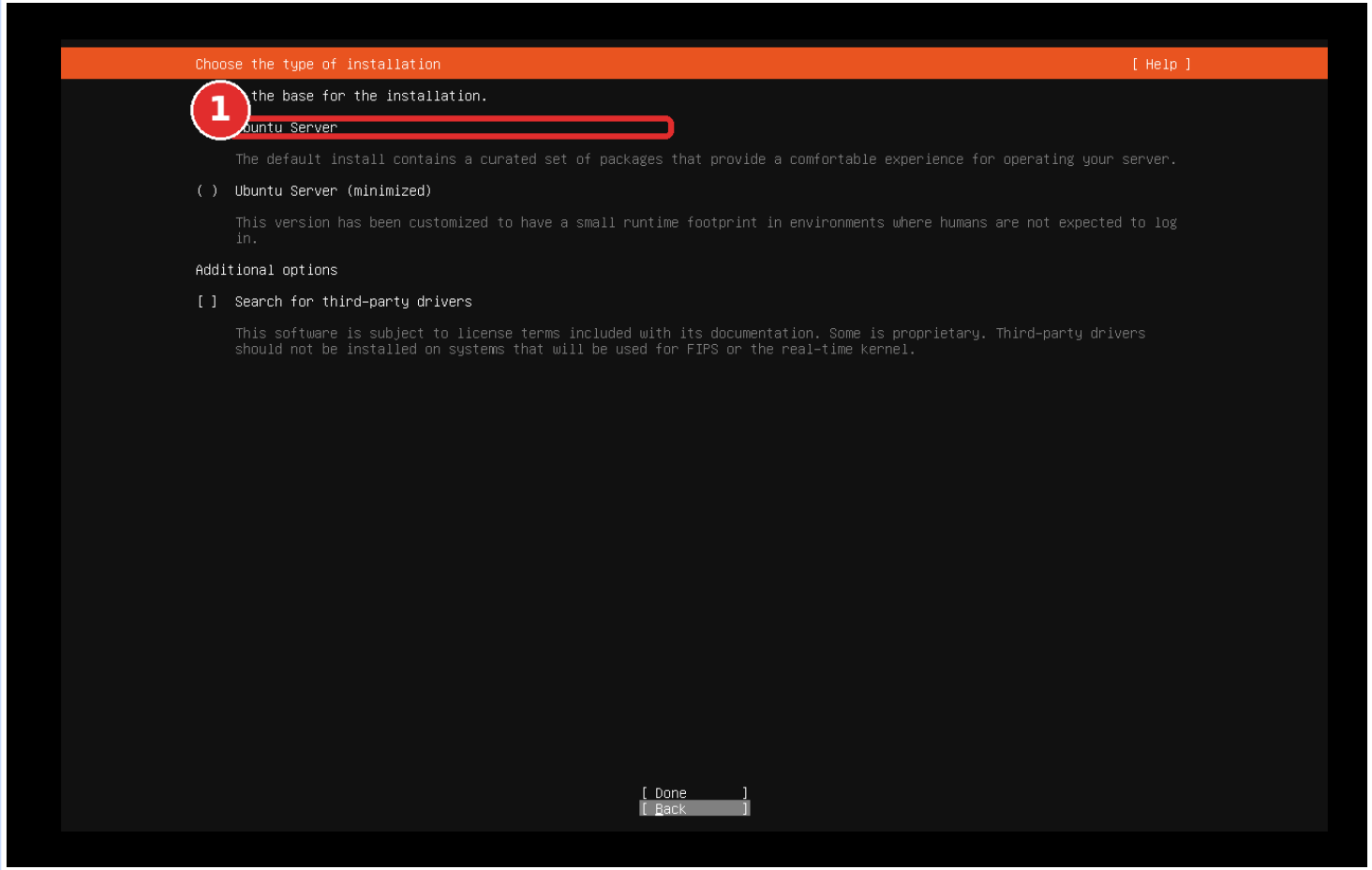
1 The layout and variant are shown. Make sure they match the keyboard you really type on, because passwords depend on it. Select Done.

Pitfall: the Windows input language



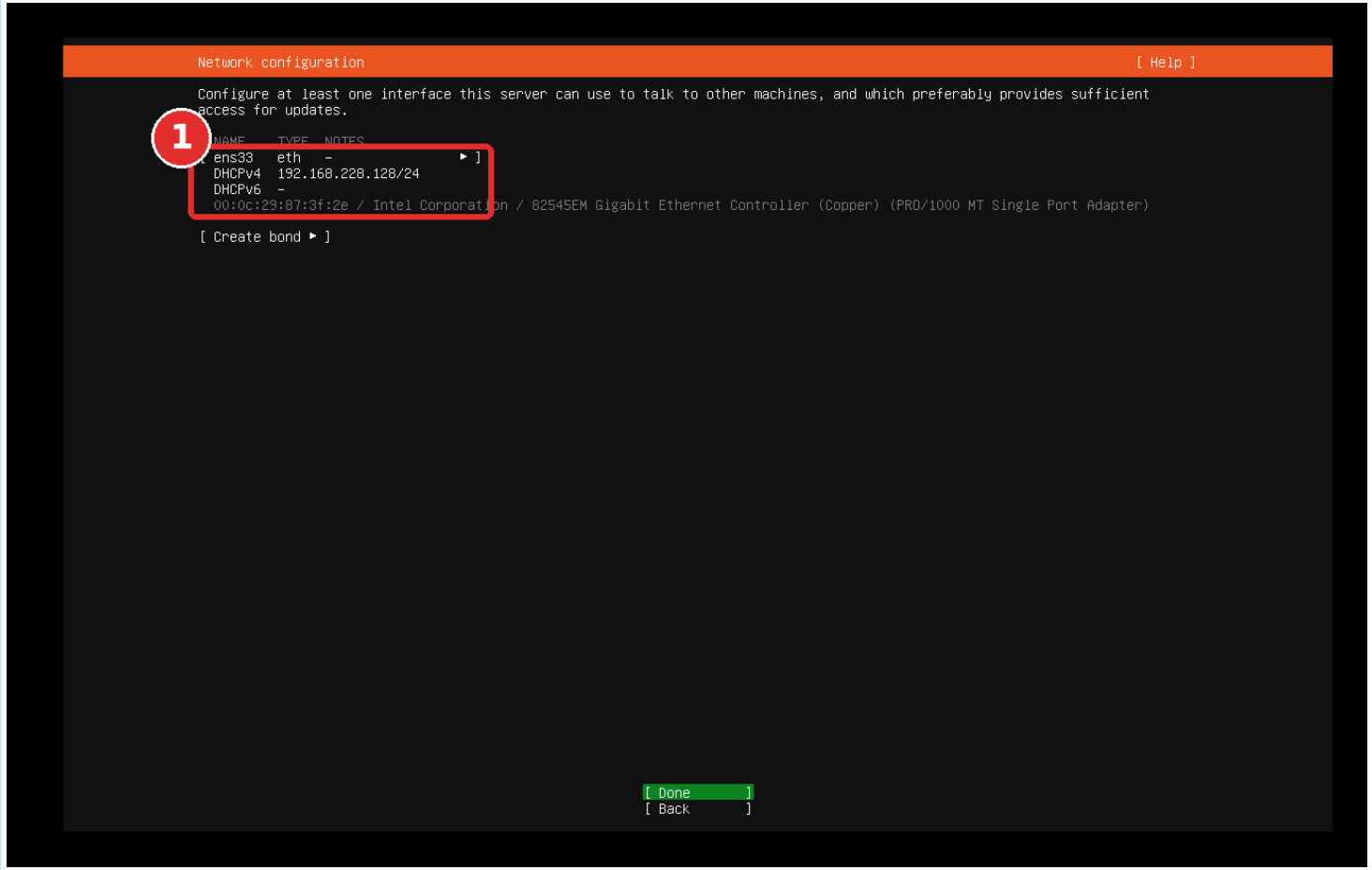
- The Windows language indicator (ENG UK, ENG US, Arabic) changes what the VM receives from your keyboard.
- Switch with Win+Space or Alt+Shift. In the lab we use English (US) for both Windows and Ubuntu.

Type of install



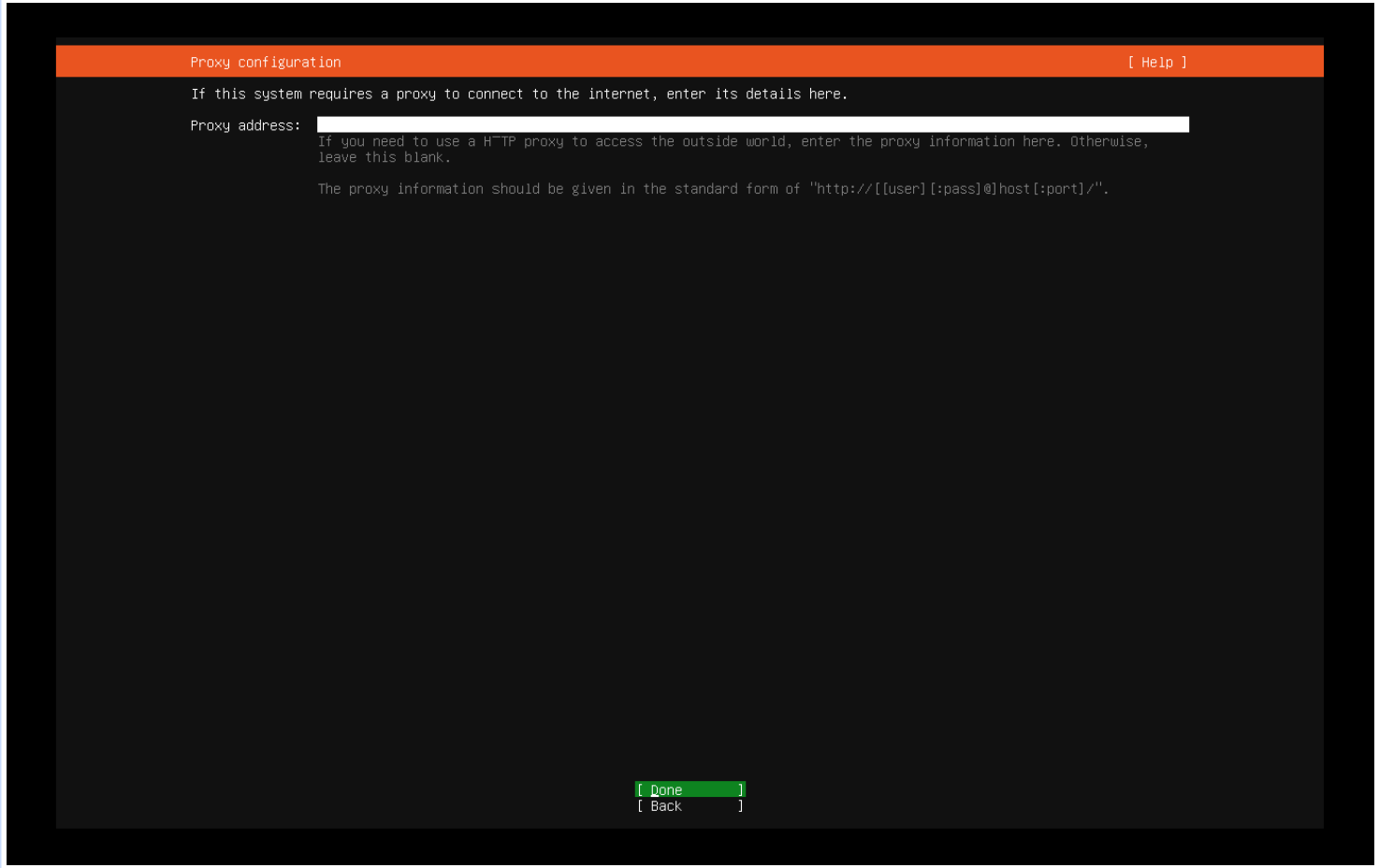
1. Choose “Ubuntu Server” (the standard option), not the minimized one, and do not tick third-party drivers.

Network connection



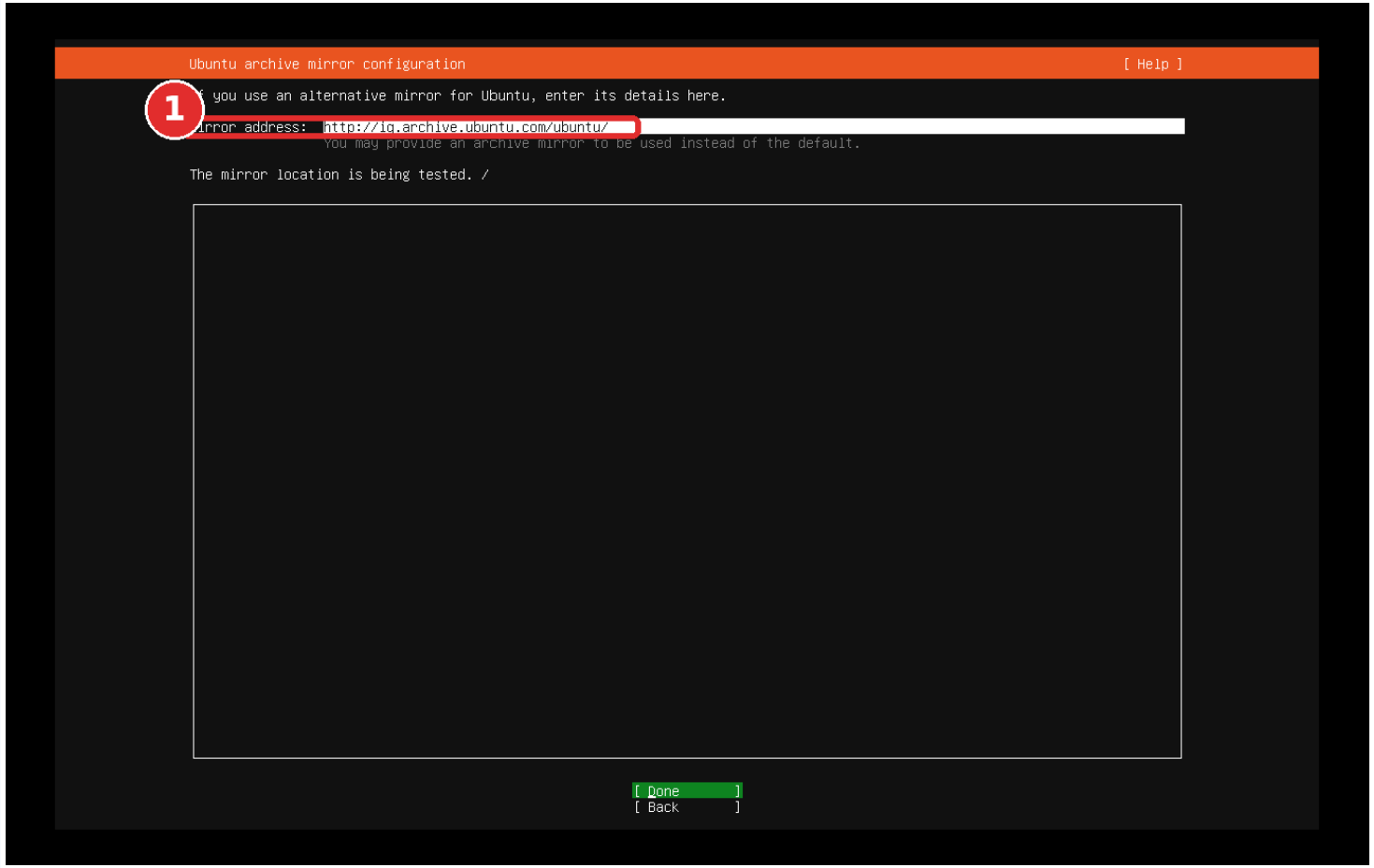
1 The adapter ens33 got an address by DHCP from the VMware NAT network. Nothing to change. Select Done.

Proxy



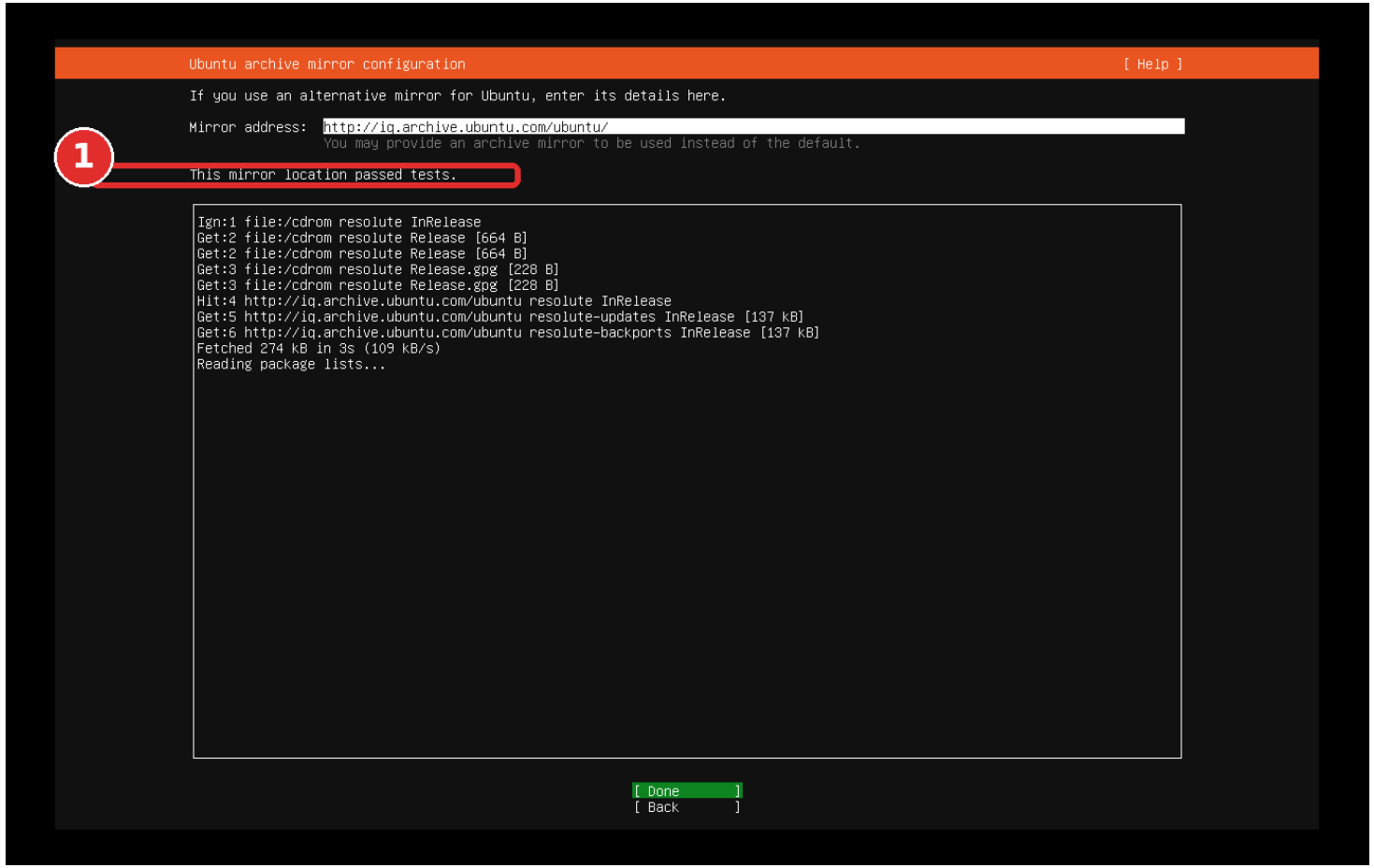
- Leave the proxy address empty (we have no proxy). Select Done.

Ubuntu archive mirror



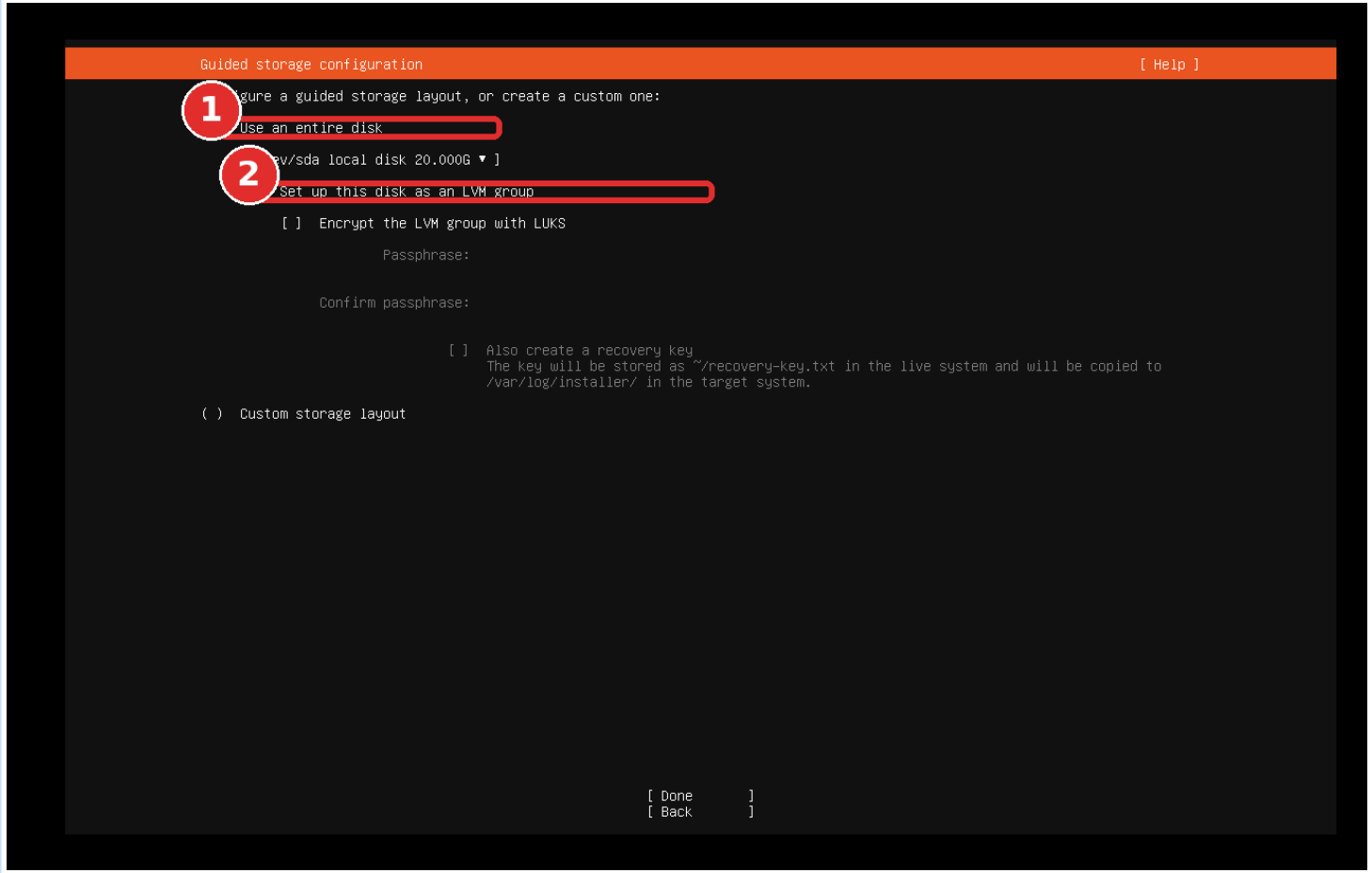
1. The installer picks a nearby mirror for downloading packages. Here it chose iq.archive.ubuntu.com.

The mirror passed its test



1. “This mirror location passed tests” means it is reachable. Select Done.

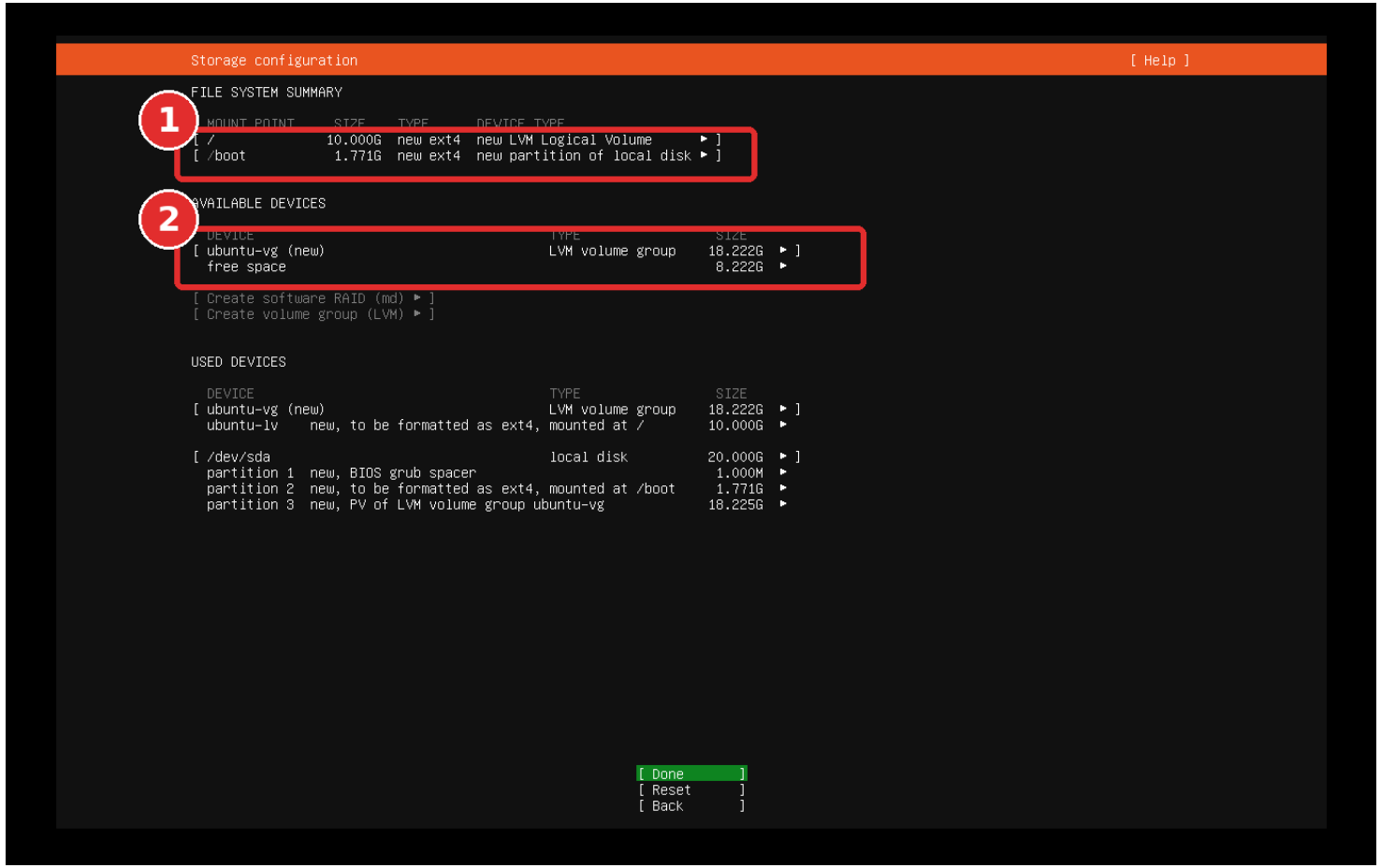
Guided storage configuration



1. Keep “Use an entire disk”. It uses the 20 GB virtual disk only.

2. Keep “Set up this disk as an LVM group” ticked, leave encryption (LUKS) off, and select Done.

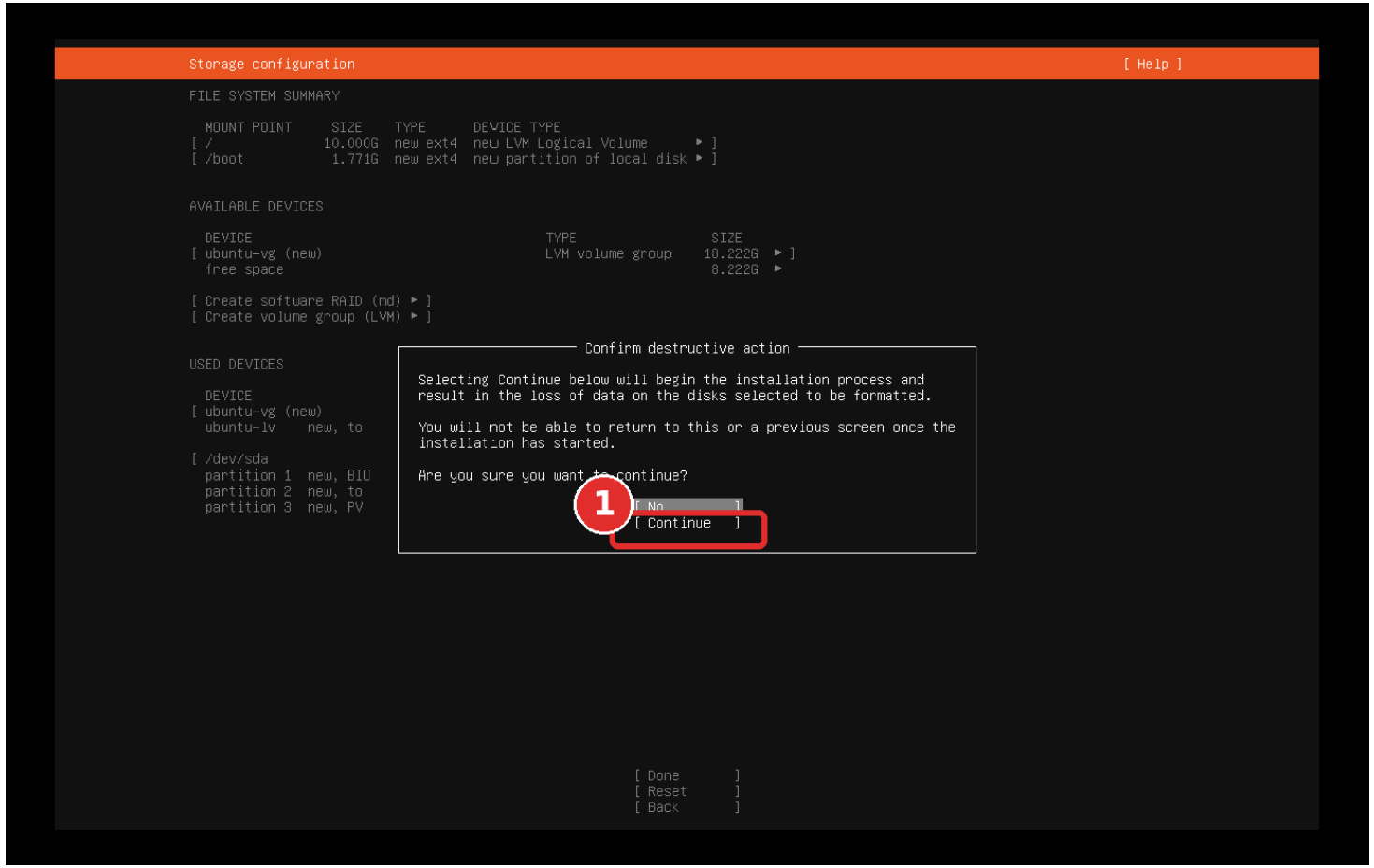
Storage summary



1. The root file system “/” gets 10 GB (ext4 on an LVM logical volume) and /boot about 1.8 GB.

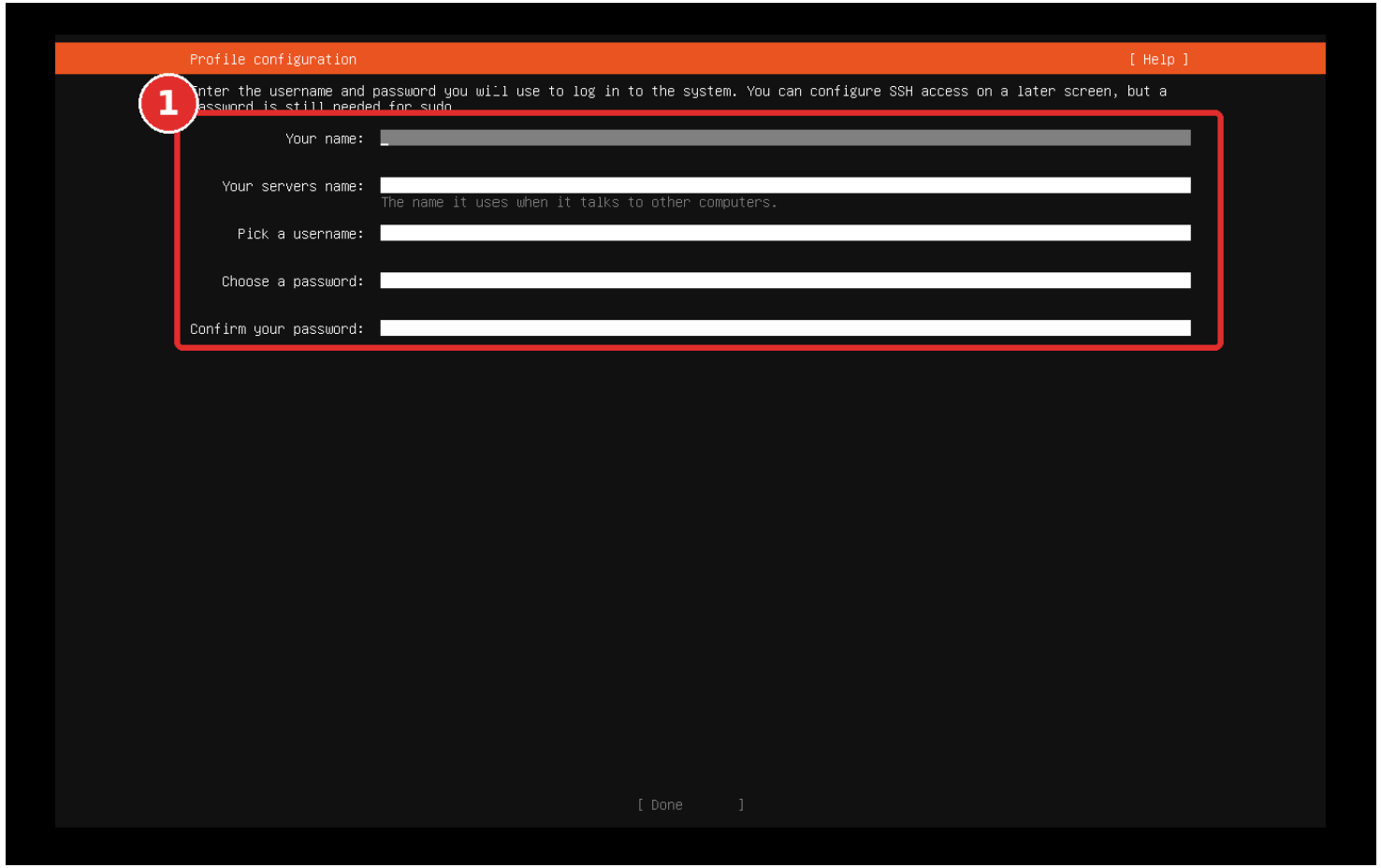
2. The volume group ubuntu-vg is about 18 GB, so about 8 GB stay free. In Lecture 4 you can grow the root volume with lvextend.

Confirm the destructive action



1. The warning says the disk will be formatted. This is only the virtual disk. Select Continue.

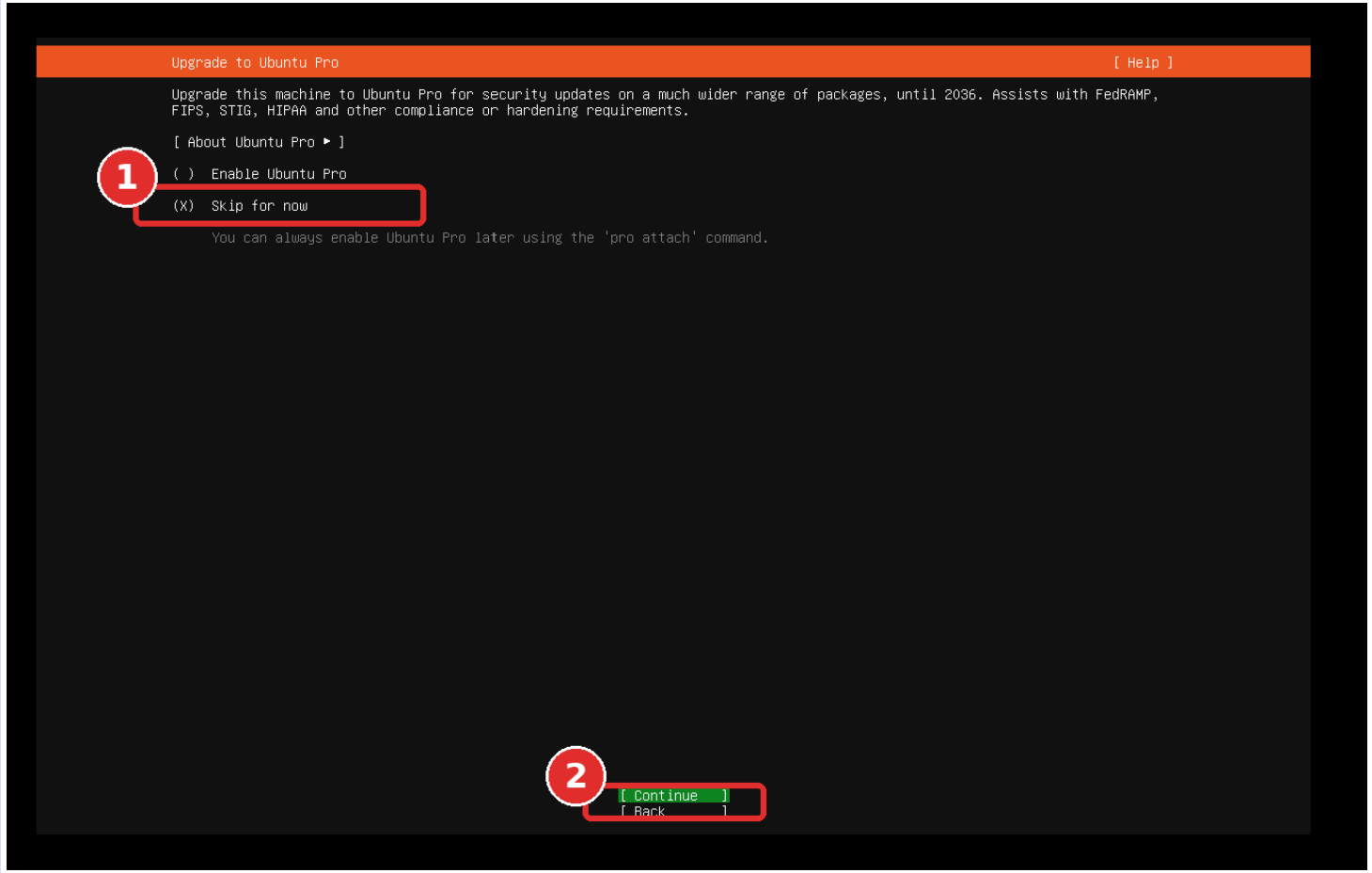
Profile configuration



1. Fill in: your name, server name ubuntu-lab, username student, and a password twice. The boxes are empty in this screenshot on purpose.

💡 Passwords are never echoed on a Linux console. Type carefully; nothing appears.

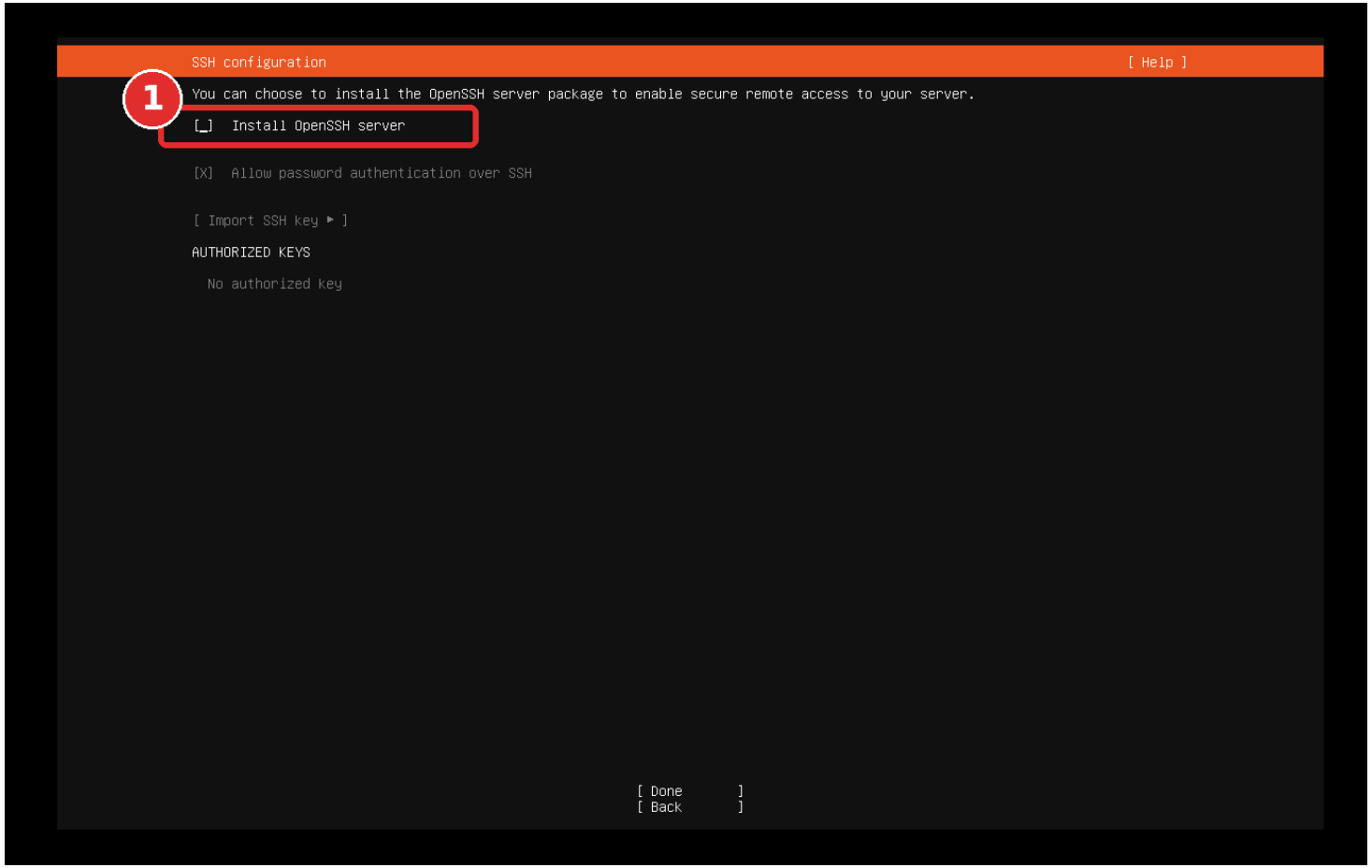
Upgrade to Ubuntu Pro



1. Choose “Skip for now”. Pro is a paid support option we do not need.

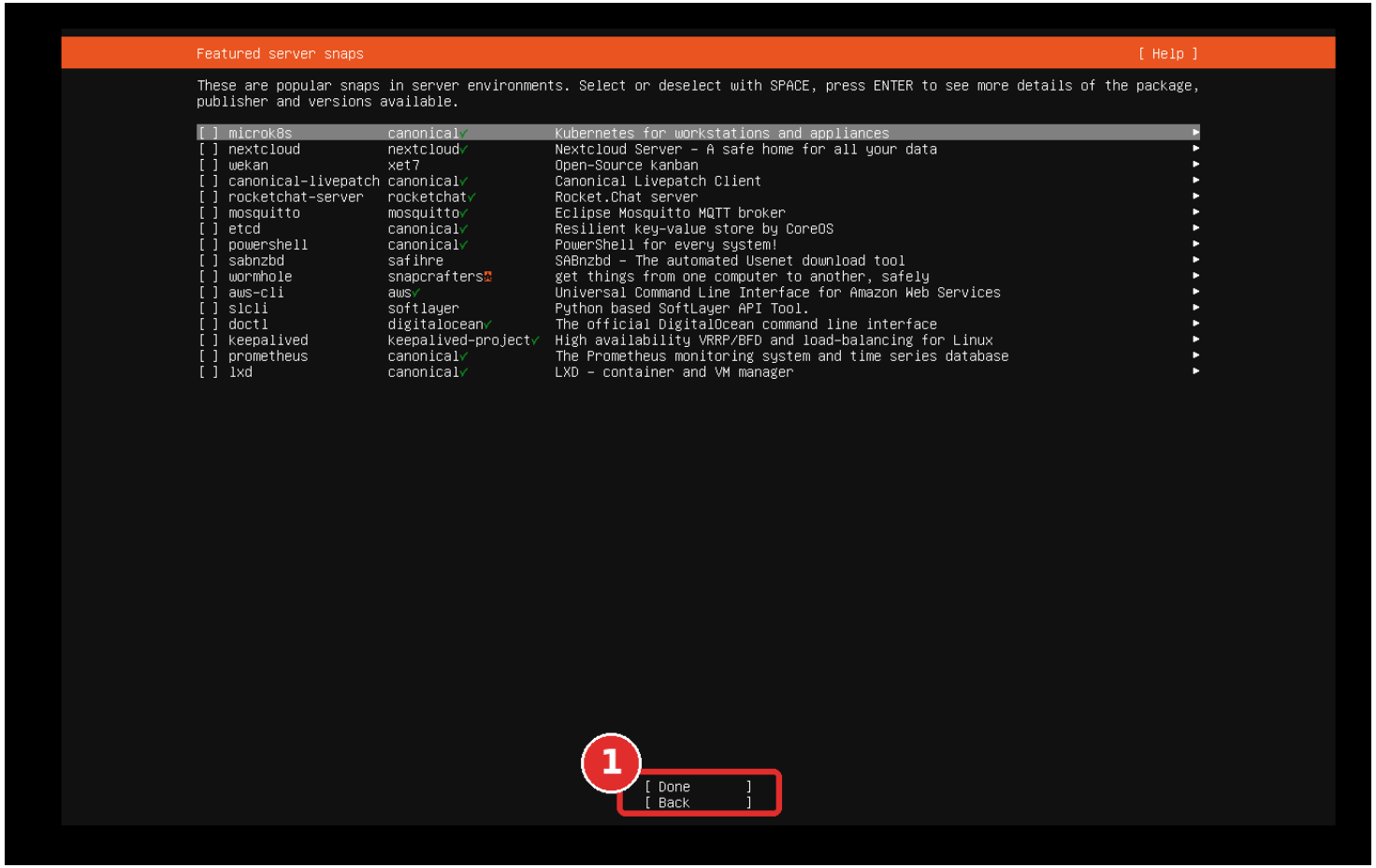
2. Select Continue.

SSH configuration



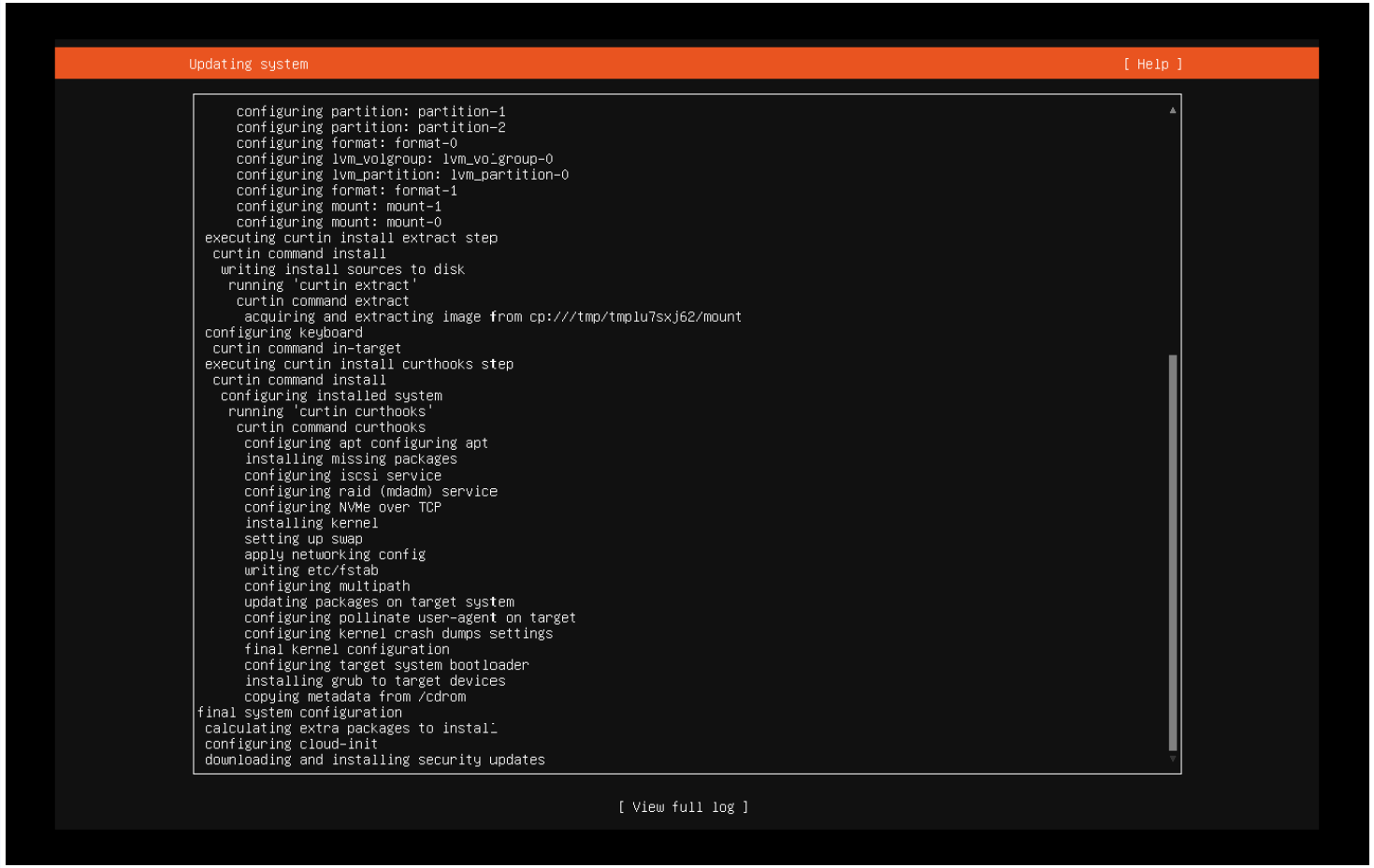
1. Leave “Install OpenSSH server” unticked for now. We use the VMware console in this lab.

Featured server snaps



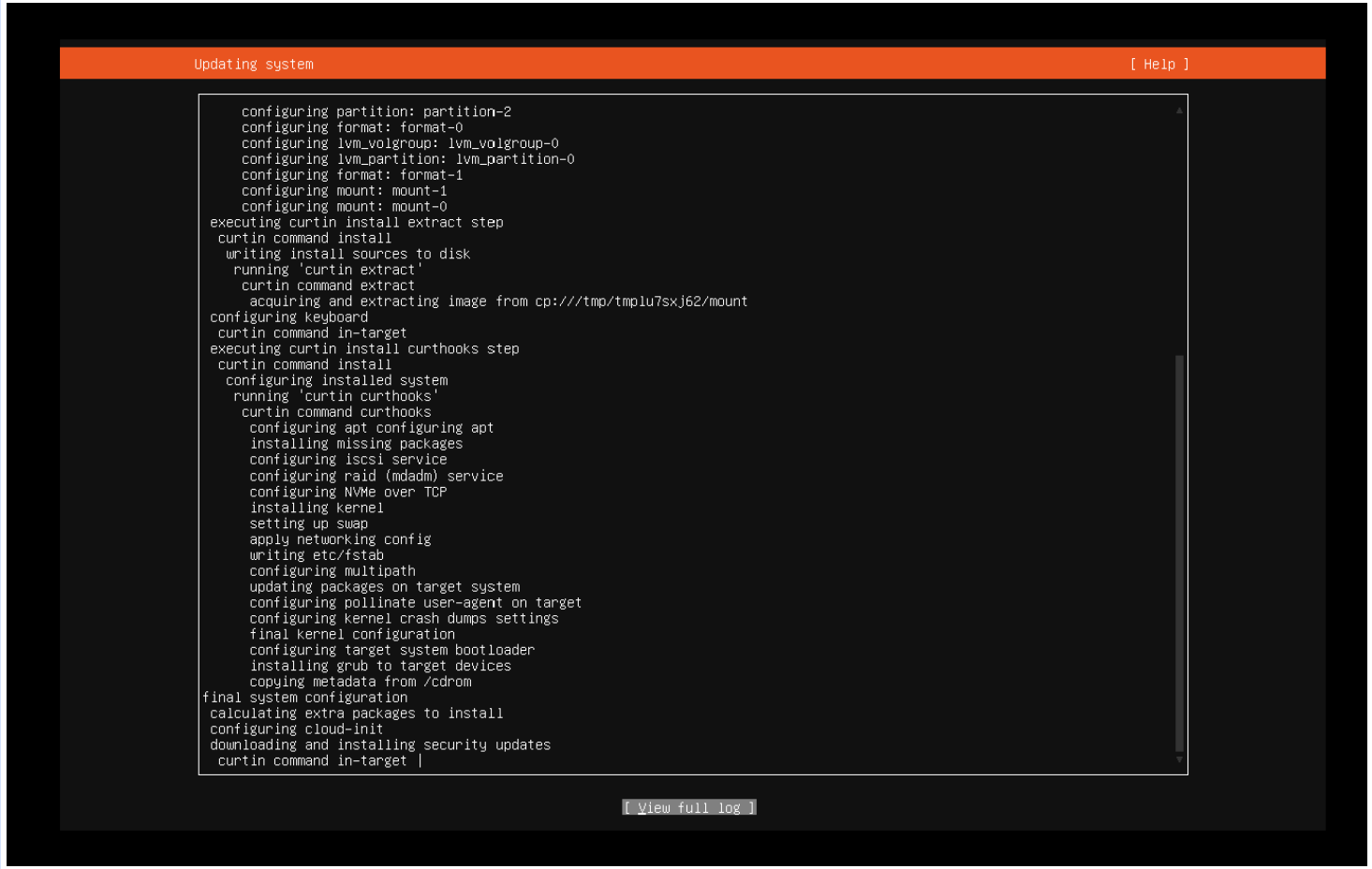
1. Select nothing. Move to Done and press Enter.

Installation begins



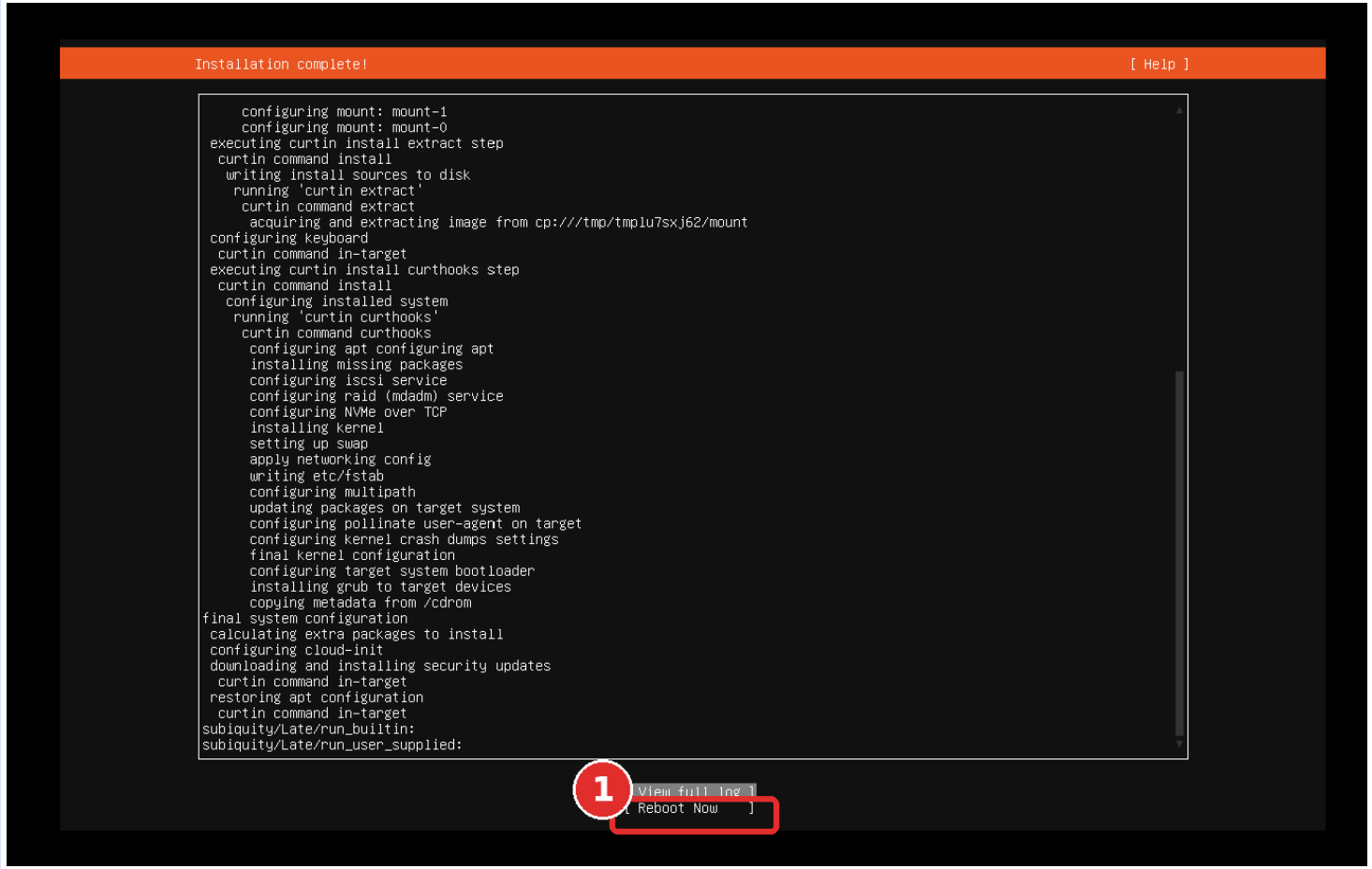
- The installer partitions the disk, copies the system and configures it. Watch the progress; it takes about 3 minutes.

Updating the system



- After copying files, it installs security updates when the internet is reachable. You can open “View full log” to see details.

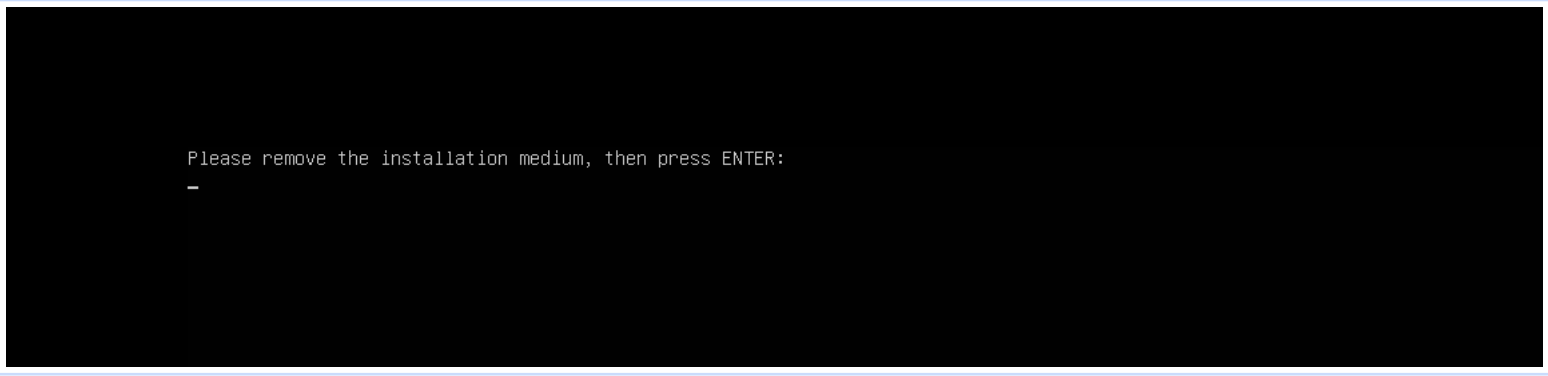
Installation complete



1. When the title says “Installation complete!”, select “Reboot Now” and press Enter.

Remove the installation medium

- “Please remove the installation medium, then press ENTER.” VMware ejects the ISO by itself, so just press Enter.



💡 If the installer starts again instead, disconnect the CD/DVD in VM settings.

Checkpoint

- ✓ Installer finished with “Installation complete!”
- ✓ Server name ubuntu-lab, user student
- ✓ VM rebooted without the ISO

Next: Part 6 — First login, verification and snapshot

CSTE4103 — Cloud Computing • Lab 3 Tutorial

6

Part 6 of 6

First Login, Verification and Snapshot

Log in, prove the machine is virtual, then save a restore point

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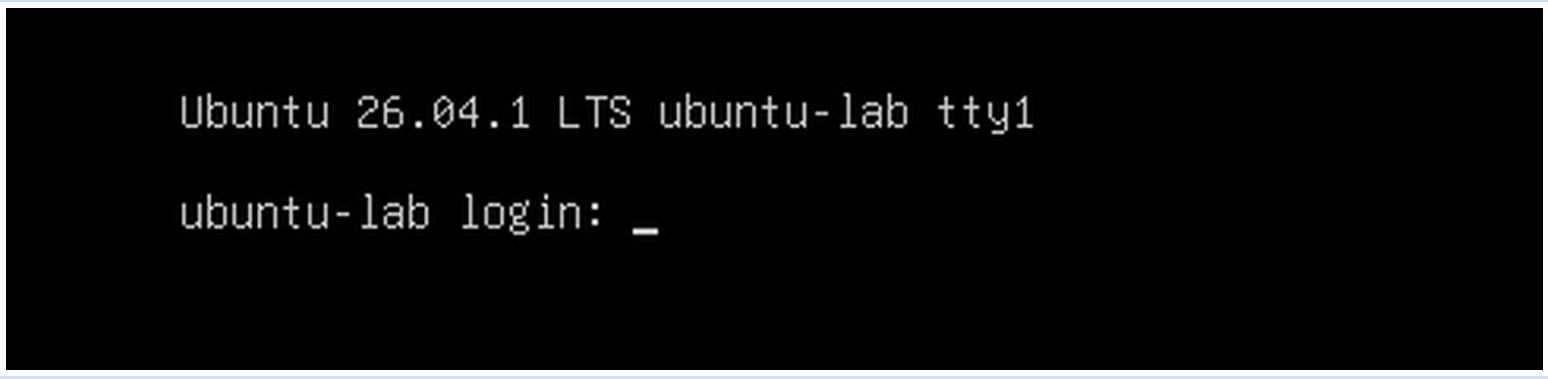
PART 6 OF 6

What this part proves

- You can log in to the new server and read basic system information.
- Commands show the hypervisor, CPUs, memory, disk and network that VMware gave the guest.
- A snapshot saves the clean state so you can return to it after any mistake.
- Commands are in the same order as the report screenshots.

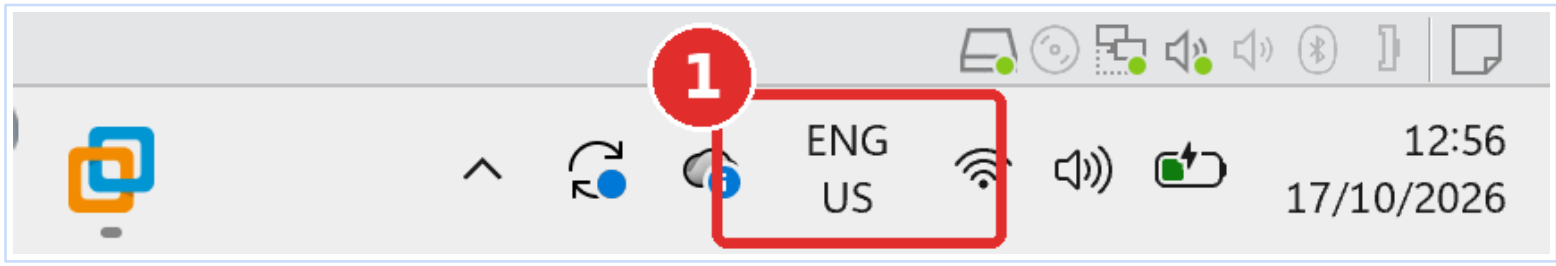
The login prompt

- After the reboot, the console shows “ubuntu-lab login:”. Click inside the VM so it receives your keyboard.



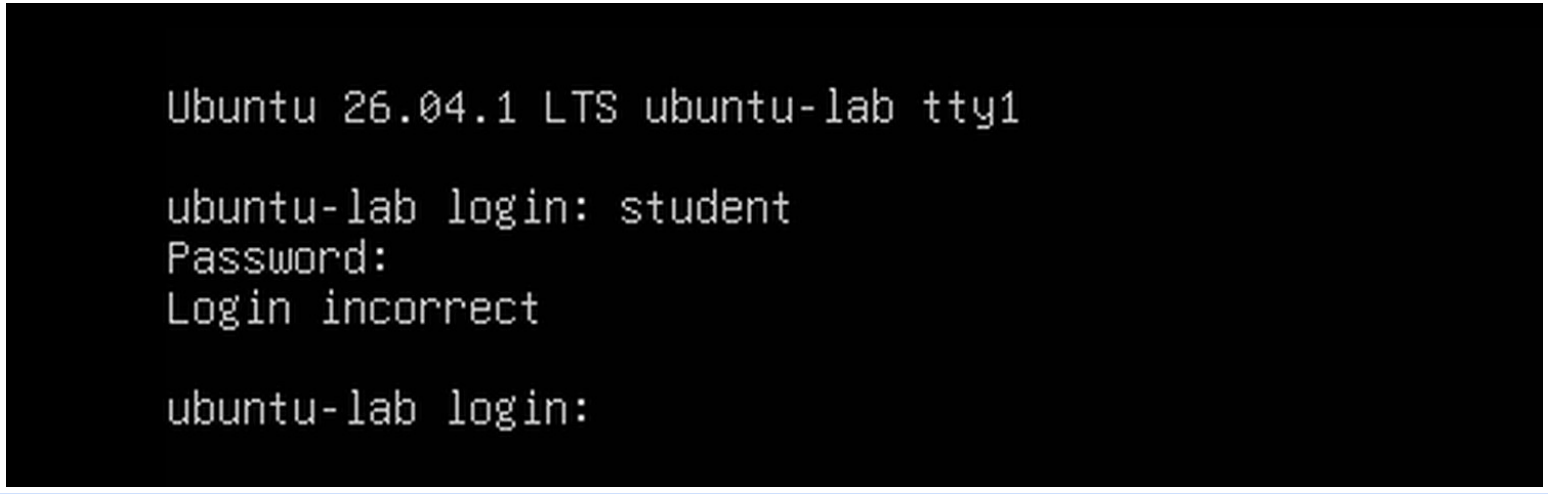
💡 Type student, press Enter, then the password. Nothing appears while typing the password.

Check the Windows input language



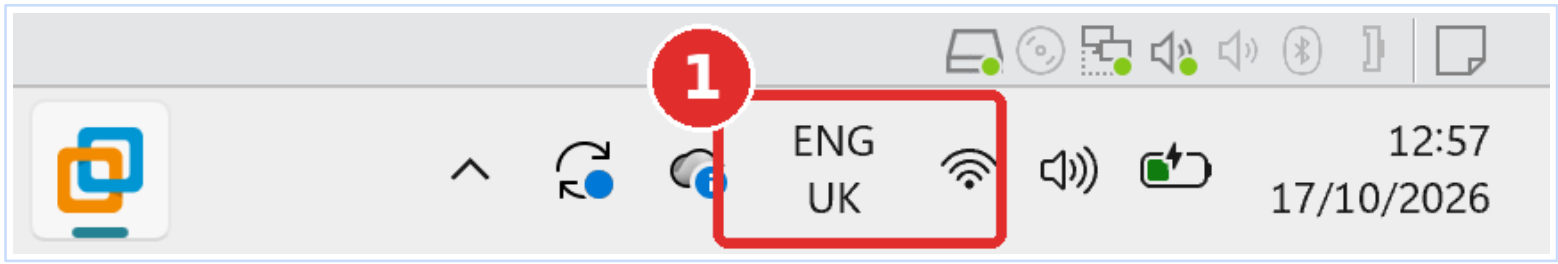
1 The taskbar shows ENG US. With this layout the password was accepted.

Pitfall: “Login incorrect”



- A correct password was rejected because the typing layout was different. Linux shows only “Login incorrect” and asks again.

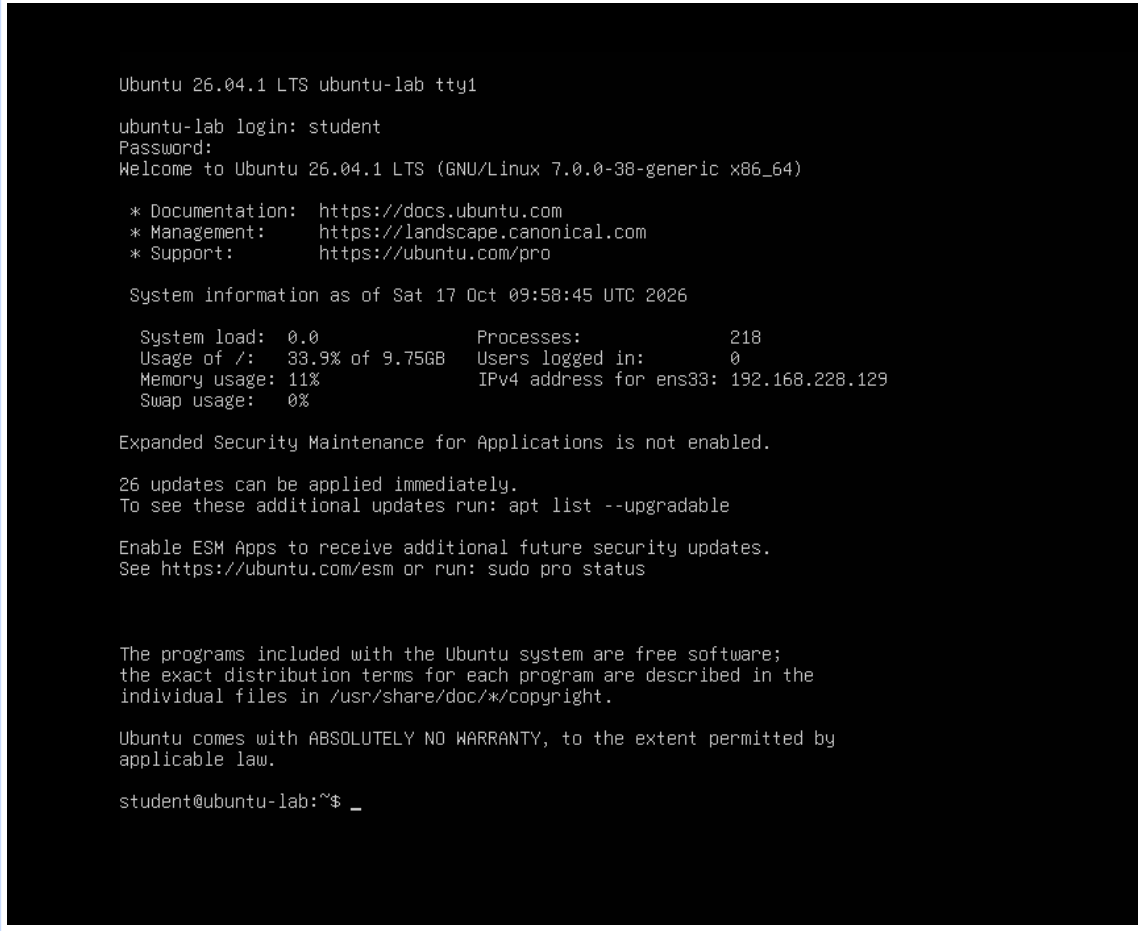
The cause: ENG UK



1 The taskbar showed ENG UK at that moment. Switch to ENG US (Win+Space) and type the username and password again.

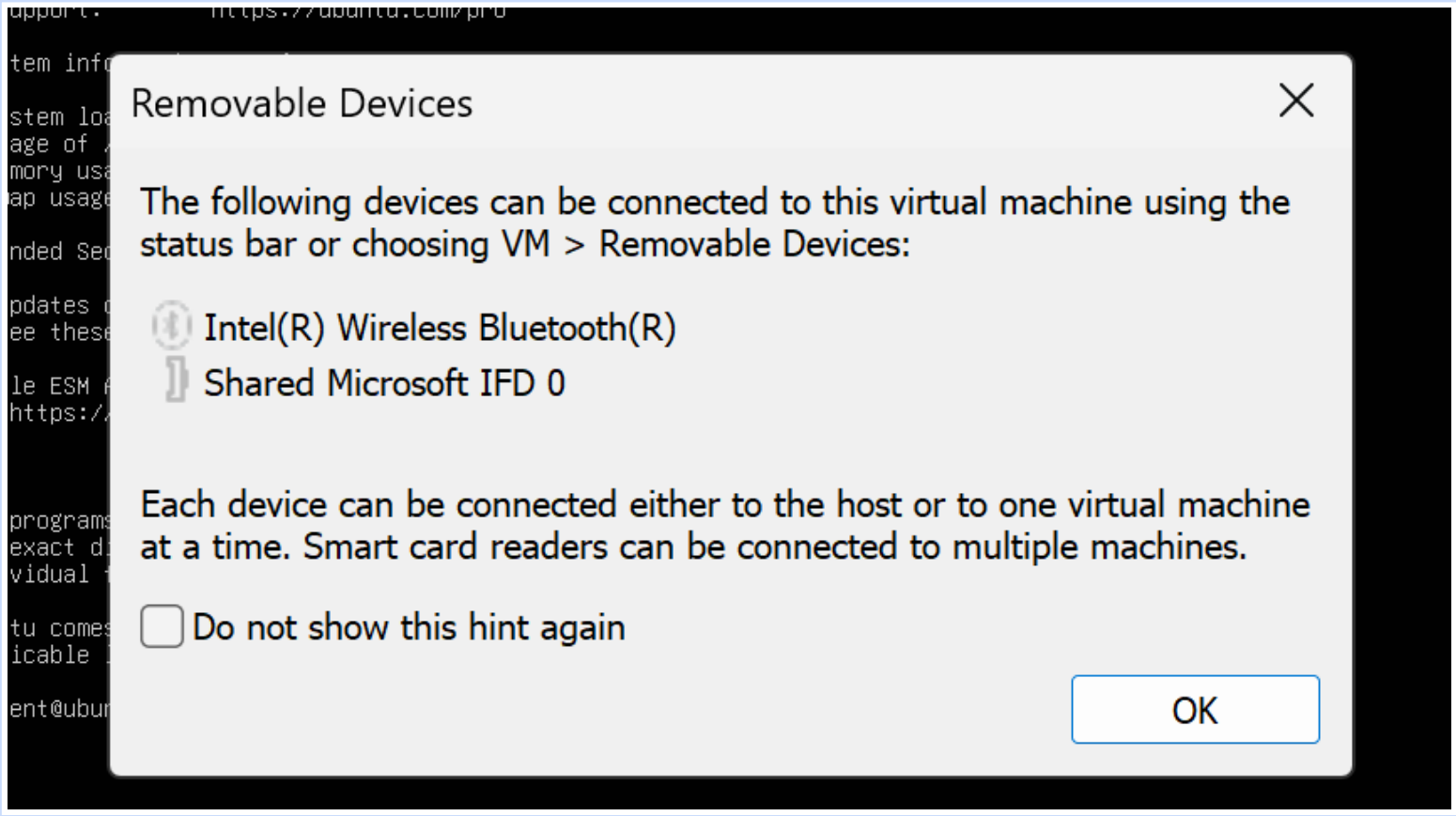
💡 If you mistyped the username, just press Enter twice and retry. Nothing breaks.

Logged in



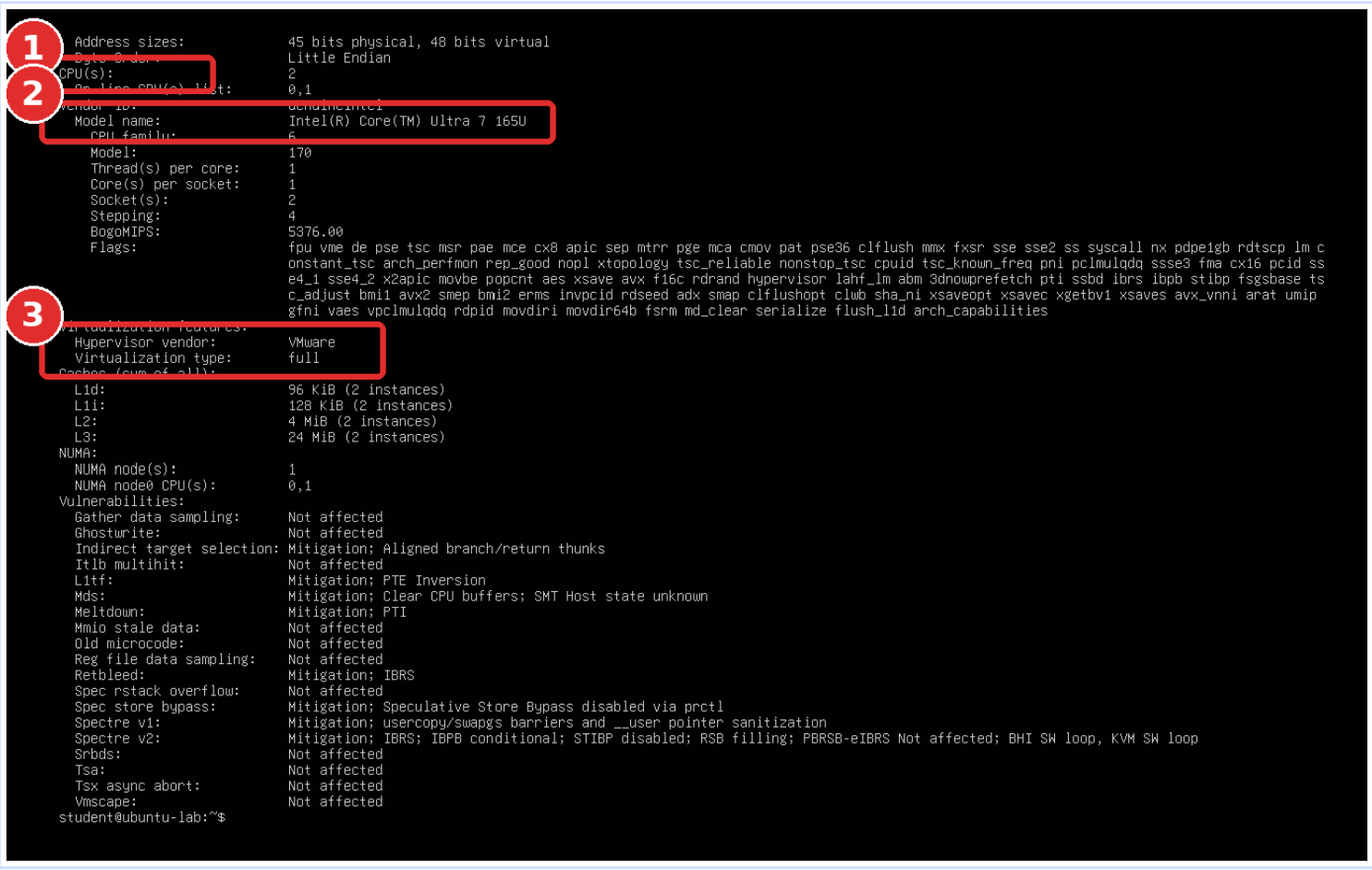
- The welcome message shows the Ubuntu version (26.04.1 LTS) and the kernel. The prompt student@ubuntu-lab:~$ means you are in.

Removable devices hint again



- VMware may show this hint again after a restart. It is harmless; press OK.

Command 1: lscpu

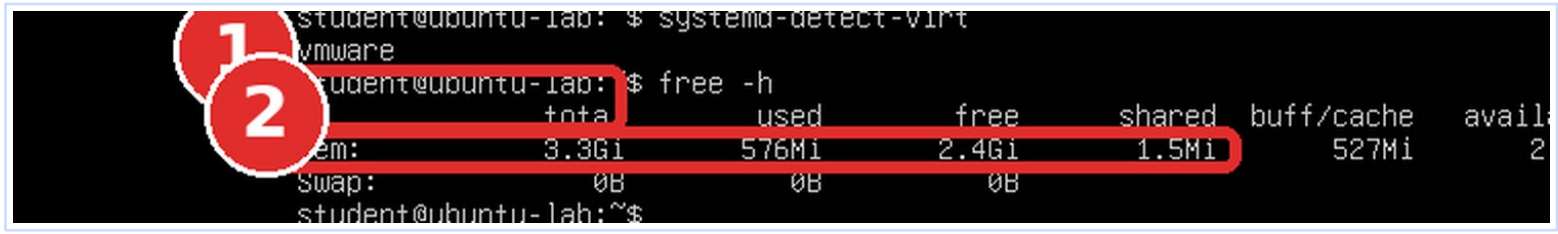


1. CPU(s): 2. These are the two virtual CPUs you assigned in Part 4.

2. Model name: the physical processor of your host PC, exposed to the guest.

3. Hypervisor vendor: VMware, Virtualization type: full. This is proof the OS runs on a virtual machine.

Commands 2 and 3: systemd-detect-virt and free -h



1. systemd-detect-virt prints a single word. “vmware” confirms the hypervisor.

2. free -h: total memory is about 3.3 GiB, i.e. the 4096 MB you gave the VM minus what the kernel reserves. Swap is 0.

Command 4: lsblk

```
student@ubuntu-lab:~$ lsblk
NAME                      MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
loop0                       7:0    0 66.8M  1 loop /snap/core24/1643
loop1                       7:1    0  1.5M  1 loop /snap/hwctl/123
loop2                       7:2    0 50.1M  1 loop /snap/snapd/27710
sda                         8:0    0   20G  0 disk
├─sda1                      8:1    0    1M  0 part
├─sda2                      8:2    0  1.8G  0 part /boot
└─sda3                      8:3    0 18.2G  0 part
  └─ubuntu--vg-ubuntu--lv 252:0    0   10G  0 lvm  /
sr0                        11:0    1 1024M  0 rom
student@ubuntu-lab:~$ ip a
```

- Disk sda is 20 GB. sda2 (about 1.8 GB) is /boot and sda3 holds the LVM volume with root “/” of 10 GB. sr0 is the virtual CD drive.

Command 5: ip a

```
student@ubuntu-lab:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:87:3f:2e brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx000c29873f2e
    inet 192.168.228.129/24 metric 100 brd 192.168.228.255 scope global dynamic ens33
       valid_lft 1018sec preferred_lft 1018sec
    inet6 fe80::20c:29ff:fe87:3f2e/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
student@ubuntu-lab:~$
```

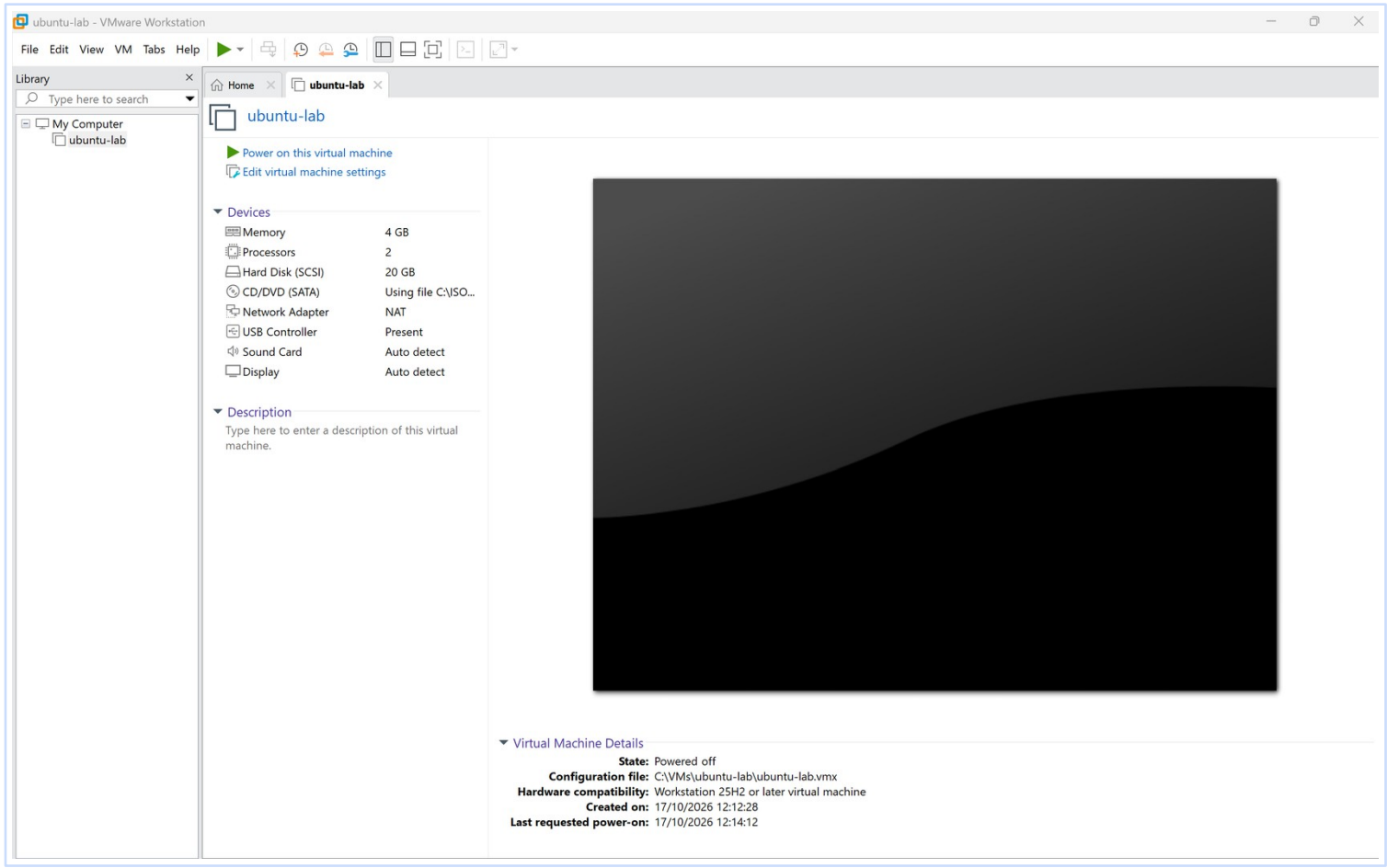
- The interface ens33 is UP with a private 192.168.x.x address from VMware's NAT network. The VM can reach the internet through the host.

PART 6 OF 6

Power off cleanly before the snapshot

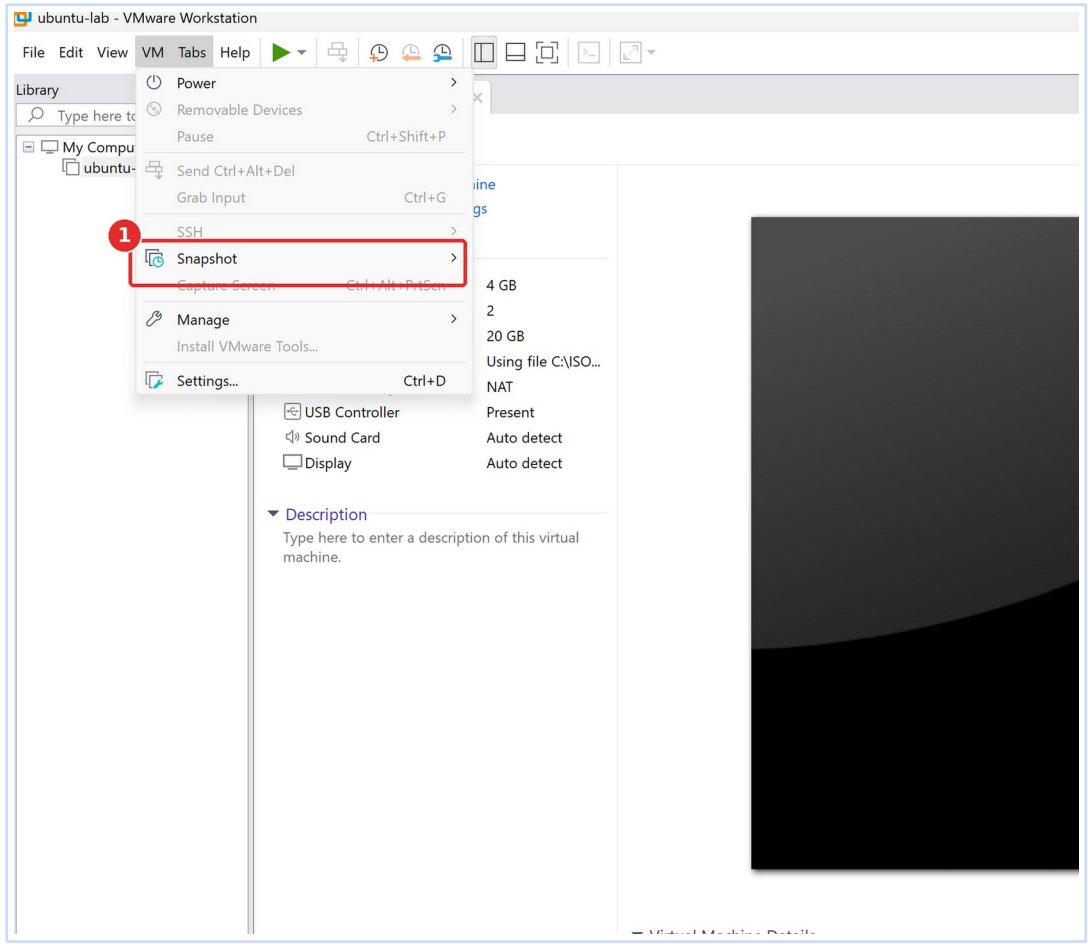
- Run: sudo poweroff, then type the student password.
- Wait until the VM shows “Powered off”. A snapshot of a powered-off VM is small and consistent.
- If you did not make the change in the VM yet, take this snapshot before experimenting.

The VM is powered off



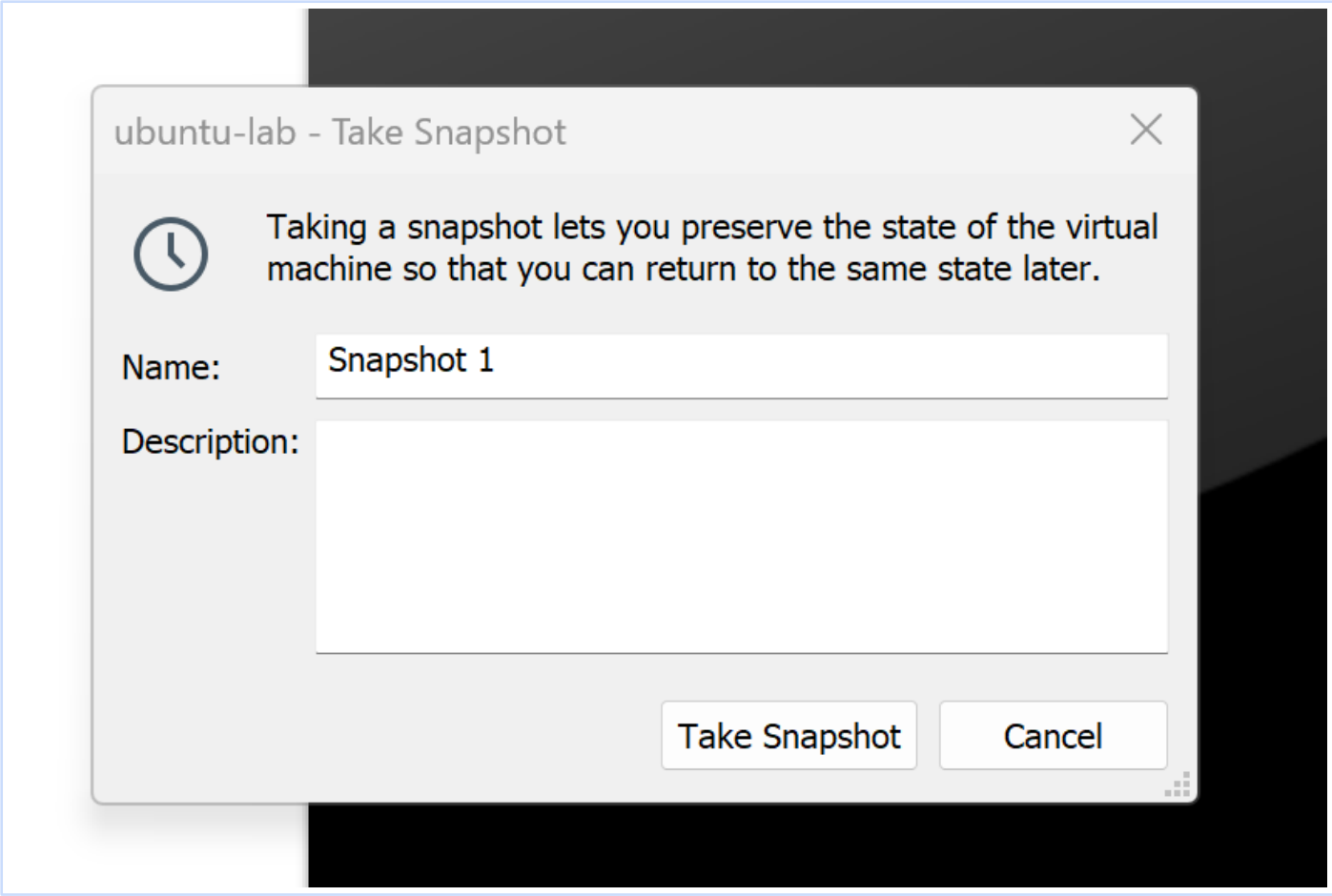
- The VM page shows State: Powered off, with 4 GB memory, 2 processors and 20 GB disk.

Open the Snapshot menu



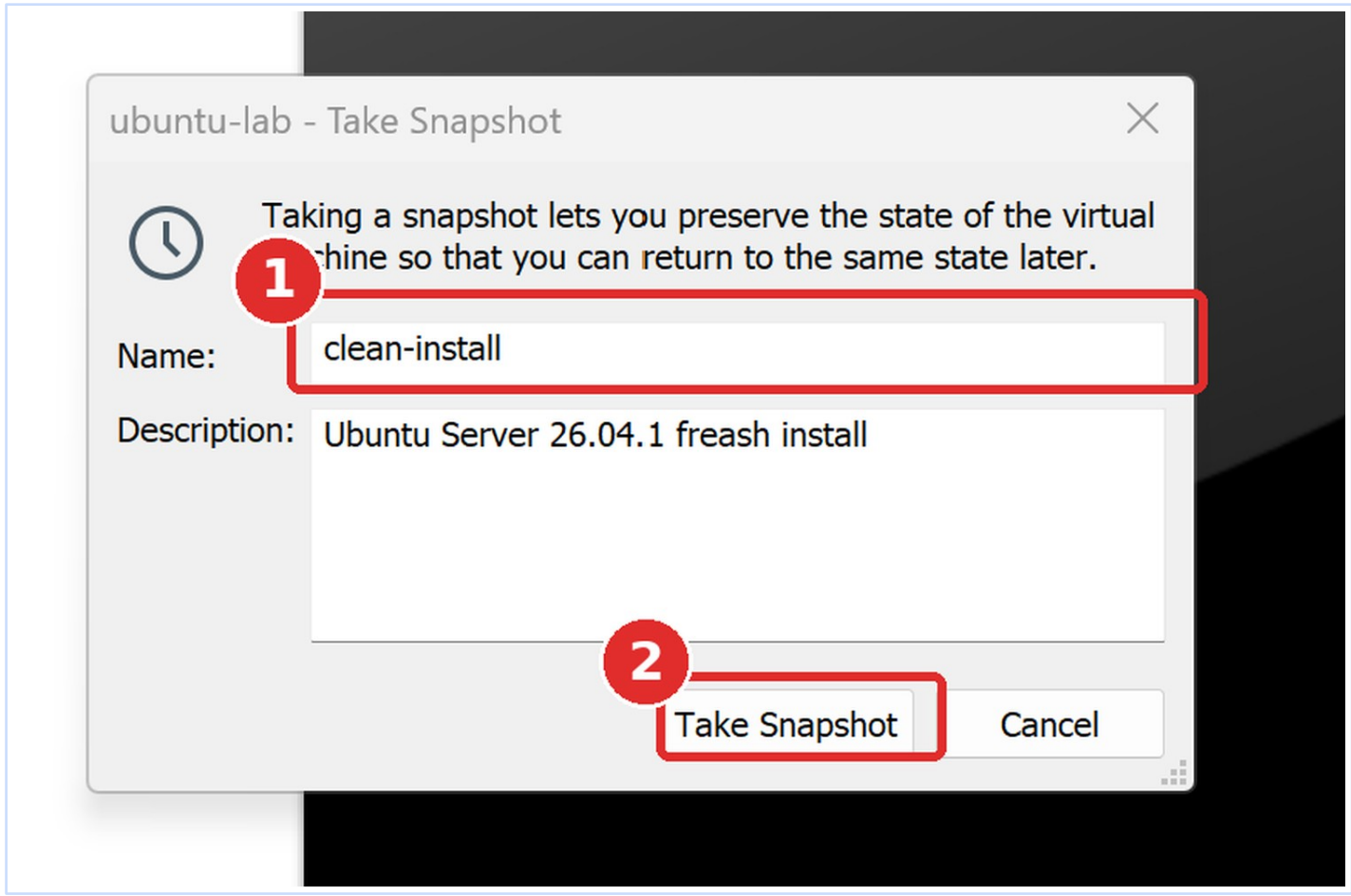
1 Choose VM > Snapshot from the VMware menu bar.

Take Snapshot



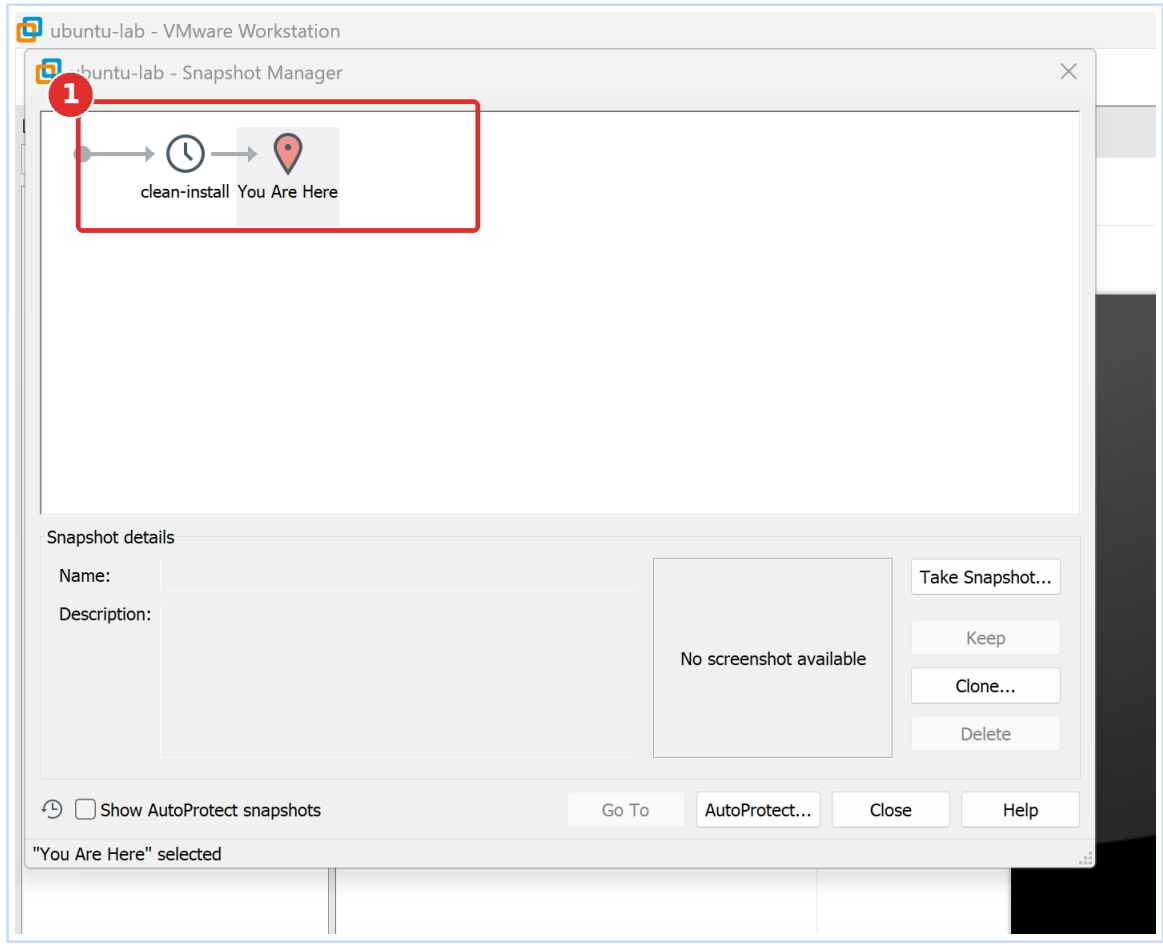
- Choose Take Snapshot. A dialog proposes “Snapshot 1” as the name.

Name the snapshot



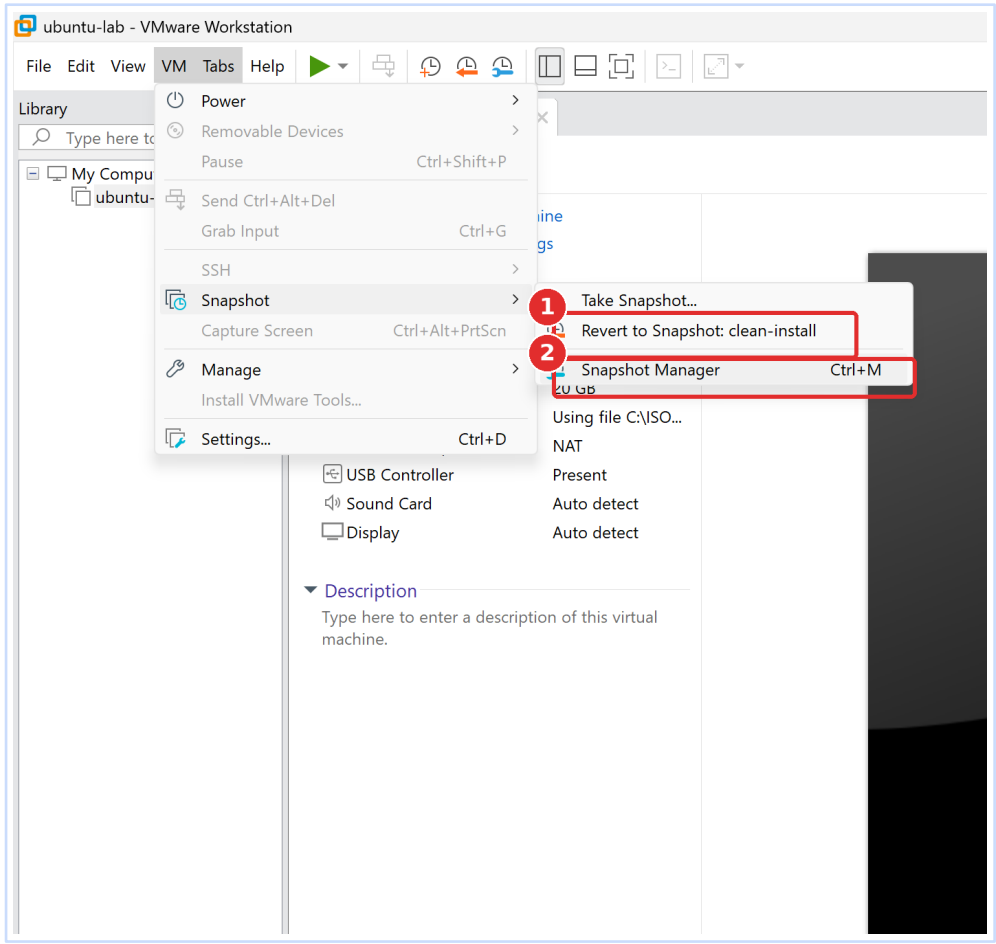
1. Name: clean-install. Add a short description such as “Ubuntu Server 26.04.1 fresh install”.
2. Press Take Snapshot.

Snapshot Manager



1. Open VM > Snapshot > Snapshot Manager. The clean-install snapshot is listed, and “You Are Here” marks the current state.

Restore when needed



1. “Revert to Snapshot: clean-install” returns the VM to this clean state and throws away later changes.

2. Snapshot Manager lists all snapshots.

PART 6 OF 6

Forgot the password? Recovery mode

- Not shown on screen in this lab. Steps in text only.
- Restart the VM and hold Shift (or press Esc) at boot to open GRUB. Choose Advanced options, then a recovery mode entry.
- Choose root (drop to root shell prompt), then run: mount -o remount,rw /
- Run: passwd student, type a new password twice, then run: reboot
- Or simply revert to the clean-install snapshot.

PART 6 OF 6

Lab report screenshots

- 1. Task Manager showing Virtualization: Enabled (Part 1)
- 2. VM Settings or final summary showing your VM size (Part 4)
- 3. Logged-in Ubuntu welcome screen (Part 6)
- 4. Output of lscpu showing Hypervisor vendor: VMware
- 5. Output of free -h, lsblk and ip a
- 6. Snapshot Manager showing clean-install

PART 6 OF 6

Checkpoint

- ✓ Logged in as student on ubuntu-lab
- ✓ lscpu and systemd-detect-virt show VMware
- ✓ Snapshot clean-install exists

Series complete. Next: Lab 3 assignment and Lecture 4